

Photosystem II Function under Extreme Low Light and Low Temperatures; Polar vs. Temperate Phytoplankton

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Abstract

Polar phytoplankton achieve slow, but ecologically significant, productivity under exceptionally low light, where the sequential arrivals of photons at Photosystem II complexes are widely spaced in time. We hypothesized that maintaining photosynthesis under such low growth light involves suppression of energetically wasteful photochemical ‘misses’ in Photosystem II. Under full darkness Photosystem II relaxes predominantly to the S1 oxidation state of the oxygen evolving complex. After such dark relaxation we applied series of single turnover, saturating, flashes to advance Photosystem II through sequential photochemical charge separations, progressively oxidizing the oxygen evolving complex through sequential increments through the S-States, resulting in four-step changes in Photosystem II chlorophyll fluorescence yield. We tracked these oscillations in chlorophyll fluorescence to measure the persistence of synchronized S-States across the Photosystem II populations in diatoms and green algae, in response to changing measurement temperatures and extremely low effective light levels, which we emulated by increasing the spacing between photon deliveries. Prolonged synchronous S-State cycling indicates fewer desynchronizations from wasteful photochemical misses and, thus, potentially more efficient photosynthetic energy conversion under low growth light. We observed that colder temperatures result in more sustained cycling at longer photon spacings, within taxa. Further, polar taxa maintained synchronized cycles for longer than did temperate taxa, even under comparable measurement conditions. Our findings support our hypothesis that diverse lineages of polar phytoplankton can suppress energetically wasteful photochemical misses under low temperature, to help sustain photosynthesis even under extremely low growth light. Therefore, ambient temperature interacts with the lower-light limitation on photosynthesis.

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Introduction

Phytoplankton are a functional grouping of photosynthetic microorganisms, of diverse lineages and ecologies (Pierella Karlusich et al., 2020). Phytoplankters share photolithotrophic growth, using a set of highly conserved protein complexes (Falkowski et al., 2008) to harness energy from Photosynthetically Active Radiation (PAR), to drive reduction and assimilation of inorganic carbon, nitrogen, phosphorus, sulphur, and other essential micronutrients (Raven et al., 2000). Phytoplankton taxa use diverse metabolic and regulatory responses (Croteau et al., 2024; Ananyev et al., 2017; Zournas et al., 2023), but all rely upon PAR, which decreases with water depth, as light passing through the water column is scattered and absorbed (Kirk, 2011). Therefore, with photons as an energy input, phytoplankton growth is constrained to the photic zone, the region of water receiving sufficient light for photosynthesis, was in past operationally defined as extending down to a limit receiving 1% of surface irradiance, or roughly 2-20 $\mu\text{mol photons m}^{-2} \text{ s}^{-1}$ (Raven et al., 2000; Falkowski, 1994).

Light availability is further constrained in polar aquatic habitats, with light penetration of the water limited by seasonally low solar angles, ice cover, and snow cover (Cohen et al., 2020; Leu et al., 2015; Hancke et al., 2018), thereby presenting challenges for phytoplankton growth (Hancke et al., 2018; Randelhoff et al., 2020; Hoppe et al., 2024). Indeed, the lower limit of the photic zone was reconsidered when benthic microalgae in the Antarctic were reported as photosynthetically active at light levels less than 1 $\mu\text{mol photons m}^{-2} \text{ s}^{-1}$ (Robinson et al., 1995; McMinn et al., 2003). Some psychrophilic phytoplankton further demonstrate slow, but ecologically significant, photosynthetic productivity during the polar winter at PAR below 0.17 $\mu\text{mol photons m}^{-2} \text{ s}^{-1}$ (Randelhoff et al., 2020; Hancke et

al., 2018; Morgan-Kiss et al., 2016; Hoppe et al., 2024), with implications for the responses of polar phytoplankton communities to climate change (Ardyna and Arrigo, 2020).

This slow, but potentially significant, phytoplankton photosynthesis during polar winters underscores the ability of psychrophilic phytoplankton to maintain functional photosystems throughout the polar night (Morgan-Kiss et al., 2016; Kvernvik et al., 2018; Randelhoff et al., 2020; Hoppe et al., 2024), and supports a theoretical minimum light level for phytoplankton photosynthesis of $0.01 \mu\text{mol photons m}^{-2} \text{ s}^{-1}$ (Raven et al., 2000). Such low light phytoplankton photosynthesis could mitigate cell mortality in the extended darkness of winter, maintaining seeding populations for the spring bloom (Randelhoff et al., 2020). These spring blooms, in turn, are a major source of annual net primary production in polar regions (Ardyna and Arrigo, 2020). In parallel with seasonally low light, psychrophile phytoplankton contend with cold temperature constraints on water viscosity, solute diffusion rates, membrane fluidity, enzyme kinetics and macromolecule interactions (Lyon and Mock, 2014).

Salinities vary across polar habitats (Morgan-Kiss et al., 2016; Smoot and Hopcroft, 2017; Hoppmann et al., 2022; Li et al., 2025), and can fluctuate annually, driven by freezing and thawing of ice (Carton et al., 2015). Such high, or variable, salinities further impact photosynthesis. The polar green algae *Chlamydomonas priscui* and *Chlamydomonas* ICE-MDV are naturally found in Antarctic saline waters, where acclimation to high salinity leads to the formation of stable PSI supercomplexes associated with increased rates of cyclic electron flow (CEF) and a higher capacity for nonphotochemical quenching (NPQ), predicted to have a photoprotective role in their natural environment (Kalra et al., 2020;

Kalra et al., 2023; Stahl-Rommel et al., 2022). This adaptation may be independent of low temperature, as long term exposure to high salinity also led to restructuring of PSI supercomplexes and increased rates of CEF and NPQ in *Chlamydomonas reinhardtii*, a temperate species naturally found in low salinities (Kalra et al., 2023).

Such extreme abiotic constraints probably act as powerful ecological filters, by imposing niche-driven speciation processes among phytoplankton lineages adapted to polar environments (Ibarbalz et al., 2023). Some psychrophilic phytoplankton indeed exhibit high genetic divergence from related temperate species. The polar diatom *Fragilariopsis cylindrus* shows divergent alleles compared to their mesophilic relatives (Mock et al., 2017), with changes in protein structure, including amino acid substitutions, H-bonds, and salt bridges (Lyon and Mock, 2014), cold shock proteins (Lyon and Mock, 2014) and anti-freeze proteins (Bayer-Giraldi et al., 2011). Similarly, the polar green algae *Chlamydomonas priscui* and *Chlamydomonas* sp. ICE-L harbour large genomes enriched in gene families encoding ice-binding proteins, molecular chaperones, light harvesting proteins and core histones (Zhang et al., 2020). Polar microbes use cellular-compatible solutes, including sugars, polyols, amino acids, betaine, and DMSP, which lower intracellular freezing points and maintain enzyme hydration spheres, stabilizing catalytic activity (Lyon and Mock, 2014; Zhang et al., 2020). Additionally, they exhibit high levels of polyunsaturated fatty acids (PUFAs) in their lipid membranes (Lyon and Mock, 2014; Cvetkovska et al., 2022) contributing to maintenance of membrane fluidity at cold temperatures.

Oxygenic photosynthesis in eukaryotic phytoplankton occurs in chloroplasts, bounded by a two- to four-membrane envelope, depending upon taxa (Kirk, 2011). Thylakoid membranes, containing pigments and electron carriers, traverse the chloroplast stroma. Photosystem II (PSII), a multi-subunit protein complex embedded in the thylakoid membranes (Shevela et al., 2023), catalyzes the photooxidation of water, releasing O_2 and H^+ , and passing reductant into the photosynthetic electron transport chain.

Photons are absorbed by diverse light harvesting complexes, which vary across taxa (Kirk, 2011), initiating a transition from the ground state to an electronically excited state of a pigment. Energy transfers among multiple pigments before eventually reaching the highly conserved photochemical reaction center of PSII, P_{680} (Kirk, 2011), composed of a Chl a heterodimer (Shen, 2021; Vass, 2011). The excitation energy is then distributed, in variable proportions, to i) photochemistry; ii) dissipation as heat; or iii) re-emission as fluorescence (ChlF) (Schuback et al., 2021). When P_{680} is raised to its excited state, P_{680}^* , it shifts to a stronger (more negative) reduction potential, allowing photooxidation to P_{680}^+ through transfer of an electron to a phaeophytin (Phe) initial acceptor molecule (Vass, 2011; Shevela et al., 2023). In productive electron transport the electron from reduced Phe- is transferred to plastoquinone A (Q_A), followed by transfer to plastoquinone B (Q_B) (Shen, 2021). Q_B , once fully reduced after receiving two electrons, is released, carrying the electrons into the mobile plastoquinone pool in the lipid phase of the thylakoid membrane (Kirk, 2011). P_{680}^+ is reduced back to P_{680} by taking an electron from a tyrosine residue D1-Tyr-161 (Yz). Yz, in turn, extracts an electron from a manganese-oxo cluster (Mn_4CaO_5) on the thylakoid lumenal side of PSII, where water is oxidized (Shen, 2021; Mukhopadhyay et al., 2004; Gates et al., 2020). Four consecutive charge separations at P_{680} induce four

increasingly oxidized states of the Oxygen Evolving Complex (OEC) Mn cluster, known as S-States (Kok et al., 1970; Zaharieva and Dau, 2019), denoted, from most reduced to most oxidized, as S0, S1, S2, and S3, followed by a transient S4 state, which rapidly decays to S0. Once the OEC has progressively lost four electrons, it oxidizes two water molecules, releasing one molecule of O₂. Therefore, a complete water oxidation cycle during oxygenic photosynthesis requires four sequential photon excitations of a single PSII, with the progressive accumulation of four oxidizing equivalents in the OEC (Dau and Haumann, 2007; Gates et al., 2020; Zaharieva and Dau, 2019), and the reduction of two Q_B which enter linear electron transport.

During photosynthesis, electron transfers thus sequentially stabilize the photochemical charge separations (Vass, 2011). These charge separations are, however, reversible through recombination reactions (Vass and Cser, 2009; Han et al., 2012), which are futile cycles of photochemistry (Rappaport et al., 2002; Rappaport et al., 2005; Laloi et al., 2006) forming one category of photochemical ‘miss’ (Han et al., 2012, 2022). Recombinations can generate chlorophyll triplets which react with ground-state molecular oxygen to produce singlet oxygen (¹O₂), a highly damaging, photoinhibitory reactive oxygen species (ROS) (Rappaport et al., 2005). Conversely, harmless non-radiative charge recombinations act as a mechanism of photoprotection under excess excitation (Vass, 2011), enabling direct recombination from the singlet P₆₈₀+Phe⁻ or P₆₈₀+Q_A⁻ states (Rutherford et al., 2012), to help suppress the formation of harmful ROS (Vass and Cser, 2009). Since these recombinations have characteristic activation energies (Rappaport et al., 2002), the probabilities of recombinations vary with temperature, (Ivanov et al., 2006; Han et al., 2008; Han et al., 2012, 2022). Both plants and cyanobacteria furthermore show

acclimatory modulations (Quigg et al., 2003) of the reduction potentials of electron transfer intermediates, thereby changing the probabilities and temperature dependencies of recombinations (Ivanov et al., 2006; Sane et al., 2003; Sane et al., 2002; Ivanov et al., 2003).

Beyond their roles in photodamage and photoprotection, charge recombination reactions are a category of wasteful 'miss' processes that compete with linear electron transport (Rappaport et al., 2005; Han et al., 2012; Ananyev et al., 2017; Zournas et al., 2023). Shifting reduction potentials of downstream electron acceptors, leading to changes in energy gaps for recombinations, may represent evolutionary adaptations aimed at maximizing photoprotection and minimizing losses through back-reactions under light-limited conditions (Vinyard et al., 2013).

Under light limitation, sequential arrivals of excitation of P_{680} in each PSII are widely spaced in time. Thus, in order to maintain productive electron transport, the electron transfer intermediates must persist for long periods of time, elevating the probabilities of energetically wasteful recombination reactions, which represent a step backward in the S-State cycle (Keren et al., 1997; de Wijn and van Gorkom, 2002; Han et al., 2008). Furthermore, the probabilities of charge recombinations decrease as temperature drops below the activation energies for recombination paths, 21 - 29 °C for recombination from Q_A^- to S2, or 35 - 41 °C for recombination from Q_B^- to S2 (Ivanov et al., 2006). Stable and progressive S-State cycling under low light, in contrast, sustains electron flow for ATP and NADPH production, while minimizing risks of low-light photodamage to PSII (Rappaport et al., 2005; Keren et al., 1997).

S-State cycling in a phytoplankton sample can be tracked by applications of sequences of short, single-turnover saturating light flashes, bright enough to deliver a photon to each PSII. In darkness the PSII population relaxes primarily to S1 (Han et al., 2008). As sequential light flashes are applied, the population of PSII complexes is driven through the S-State cycle (Dau and Haumann, 2007). The S-States have different intrinsic PSII fluorescence yields (Gates et al., 2020; Zaharieva and Dau, 2019), so in an idealized sample, the four sequential S-States will be reflected by an ongoing periodic oscillation in ChlF. Recombination reactions, however, contribute to wasteful misses in the S-State cycling of individual PSII (Han et al., 2012). As misses occur, they progressively desynchronize S-State cycling across the PSII population, dampening the observed ChlF oscillation (de Wijn and van Gorkom, 2002) towards an averaged ChlF for the PSII population distributed across the S-States. An organism exhibiting synchronized S-State cycling sustained over more flash cycles indicates fewer photochemical misses, and thus higher conversion of sequentially spaced excitations to productive photosynthetic electron transport.

We hypothesized that, since psychrophilic phytoplankton are able to grow at low temperatures, they show suppression of temperature dependent recombinations, thereby lowering the probability of photochemical misses. Suppressing photochemical misses then improves the potential to exploit extreme low light and low temperatures, for productive photosynthesis in psychrophilic diatoms and green algae. We further hypothesized that growth acclimation to changing salinity might interact with PSII function under extreme low light and low temperatures. Therefore we compared S-State cycling over flash cycles, over ranges of measurement temperatures, for psychrophilic and temperate phytoplankton taxa, after growth across ranges of ecophysiological relevant salinities.

Materials and Methods

Study Strains and Culturing Conditions

The nine study taxa, including polar and temperate strains of diatoms and green algae, and their respective culturing conditions, are summarized in Table 1. All tested cultures were grown at their respective near-optimal temperatures and light levels, to provoke relatively consistent growth, and limit subsequent measurements to healthy cultures. The polar diatom *Fragilariopsis cylindricus* was grown at two temperatures and two light levels. We did not impose different growth light or temperature levels on other individual taxa, although both growth light and temperature are potentially relevant factors for future research. Previous research showed impacts of salinity on photosynthetic performance in some psychrophile strains (Kalra et al., 2020; Kalra et al., 2023; Stahl-Rommel et al., 2022), so we chose a range of polar and temperate taxa from both freshwater and saline environments, grown at their respective near-optimal salinities. Since polar *Chlamydomonas* taxa grow under both low temperature and high salinity, we used a range of growth-permissive salinities for some *Chlamydomonas*, to detect whether growth salinity would influence our results. Most taxa and growth combinations were grown in two culture replicates, measured on different days. Some *Chlamydomonas* cultures under non-optimal salinities were grown only once, although sampled for measurements over multiple days.

Table 1: Study taxa and culture growth conditions, including media salinity (NaCl (mM)); approximate optimal salinity for the taxa (NaCl (opt)); Photosynthetically Active Radiation for growth, $\mu\text{mol photons m}^{-2} \text{ s}^{-1}$, (PAR (uE)); and growth media.

Taxa	Line age	Origin	Growt h °C	NaCl (mM)	NaCl (opt)	PAR (μE)	Photoperiod (h)	Medi a
<i>F. cylindrus</i>	Diato m	Polar	0	500	500	10	24	F2
<i>F. cylindrus</i>	Diato m	Polar	0	500	500	1	24	F2
<i>F. cylindrus</i>	Diato m	Polar	6	500	500	10	24	F2
<i>F. cylindrus</i>	Diato m	Polar	6	500	500	1	24	F2
<i>T. pseudonana</i>	Diato m	Tempe rate	22	500	500	50	12	F2
<i>C. ICEMDV</i>	Gree n	Polar	4	70	500	10	24	BBM
<i>C. klinobasis</i>	Gree n	Polar	4	0	0	10	24	BBM
<i>C. klinobasis</i>	Gree n	Polar	4	10	0	10	24	BBM
<i>C. klinobasis</i>	Gree n	Polar	4	30	0	10	24	BBM
<i>C. malina</i>	Gree n	Polar	4	30	500	10	24	BBM
<i>C. malina</i>	Gree n	Polar	4	70	500	10	24	BBM
<i>C. malina</i>	Gree n	Polar	4	500	500	10	24	BBM
<i>C. priscui</i>	Gree n	Polar	4	30	500	10	24	BBM
<i>C. priscui</i>	Gree n	Polar	4	70	500	10	24	BBM
<i>C. priscui</i>	Gree n	Polar	4	500	500	10	24	BBM
<i>C. priscui</i>	Gree n	Polar	4	700	500	10	24	BBM
<i>C. euryale</i>	Gree n	Tempe rate	24	30	500	10	24	BBM _NH4
<i>C. euryale</i>	Gree n	Tempe rate	24	70	500	10	24	BBM _NH4

Taxa	Line age	Origin	Growth h °C	NaCl (mM)	NaCl (opt)	PAR (μE)	Photoperiod (h)	Media
<i>C. euryale</i>	Green	Temperature	24	500	500	10	24	BBM _NH4
<i>C. reinhardtii</i>	Green	Temperature	24	0	0	10	24	BBM
<i>C. reinhardtii</i>	Green	Temperature	24	10	0	10	24	BBM
<i>C. reinhardtii</i>	Green	Temperature	24	30	0	10	24	BBM
<i>C. vulgaris</i>	Green	Temperature	22	0	0	50	12	BG1 1
<i>C. vulgaris</i>	Green	Temperature	22	500	0	50	12	F2

Fragilariopsis cylindrus is a psychrophilic (Fiala and Oriol, 1990) pennate diatom measuring 15-55 μm which forms large blooms in the high salinity and subzero temperatures of Arctic and Antarctic sea-ice systems (Otte et al., 2023; Cefarelli et al., 2010), and is a keystone species for polar ecosystems (von Quillfeldt, 2004; Otte et al., 2023; Kang and Fryxell, 1992). Sequenced in 2017, *Fragilariopsis cylindrus* has since become the model psychrophilic diatom species for investigating molecular and physiological mechanisms of cold adaptation (Mock et al., 2017). Conversely, *Thalassiosira pseudonana* (*Cyclotella nana*) is a centric diatom (2.5-15 μm) found in diverse freshwater, coastal, brackish, and marine habitats (Poulsen and Kröger, 2023), across a range of salinities and temperatures (4–25°C) (Poulsen and Kröger, 2023). *Chlamydomonas* sp. ICE-MDV and *Chlamydomonas priscui* are halotolerant algae isolated from the perennially ice-covered hypersaline Lake Bonney, in McMurdo Dry Valleys, Antarctica (Cook et al., 2019; Stahl-Rommel et al., 2022). With 15 to 20 μm biflagellate cells, *Chlamydomonas* sp. ICE-MDV dominates the shallow photic zone, where it experiences relatively higher irradiance, extreme nutrient limitation, and lower salinity (Cook et al., 2019; Li et al., 2016). The smaller *Chlamydomonas priscui* dominates the deep photic zone, characterized by permanent low temperatures, low irradiance, and high salinity (Cvetkovska et al., 2022; Hüner et al., 2023). *Chlamydomonas malina* is a marine microalga isolated from the Arctic Beaufort Sea, around 10 μm in length and 5 μm in width, growing optimally at 4°C (Balzano et al., 2012; Morales-Sánchez et al., 2020). *Chlamydomonas klinobasis* strain 041-99 is a freshwater microalga isolated from an ephemeral snowfield on Svalbard Island, Norway, of around 12 x 6 μm (P. Osmer, unpub), growing from 2-6°C (Hulatt et al., 2017). The temperate *Chlamydomonas reinhardtii* is a model green alga approximately 10 μm in

diameter, found in freshwater soil and aquatic environments with an optimal temperature range of 20-32°C (Sasso et al., 2018; Xie et al., 2013). The temperate marine *Chlamydomonas euryale* is approximately 4 µm in diameter, isolated from various locations within the northern Atlantic and Pacific, including the Yellow Sea, with a known growth temperature range of 18-26°C Carrasco Flores et al. (2021). *Chlorella vulgaris*, ranging from 2 µm to 10 µm in size, is primarily found in freshwater environments and grows optimally at 27°C (Wiel et al., 2017; Leyva et al., 2014).

Thalassiosira pseudonana and *Chlorella vulgaris* were cultured by Naaman Omar (Mount Allison University); *Chlamydomonas* strains were cultured by Mackenzie Poirier (Cvetkovska Lab, University of Ottawa) as reported previously (Poirier et al., 2025); while *Fragilariopsis cylindrus* was cultured by Sébastien Guérin (Takuvik International Research Laboratory, Université Laval).

Single Turnover Variable Chlorophyll Fluorescence

We used series of single turnover, saturating, flashes to induce variable chlorophyll fluorescence (ChlF) (Kolber, 1998), and evaluate progressive desynchronization of the S-State cycle in several polar and temperate phytoplankton species acclimated to a range of growth temperatures. We loaded a 3 mL sample of culture into a temperature-controlled water-jacketed quartz cuvette (Cafferty's Scientific Glassblowing, Anola, Manitoba, Canada), placed within the measurement chamber of a Soliense fluorometer (Version LIFT-REM 1.0, Soliense Inc., Shoreham NY, USA). The apparatus was covered to block out incident light, for a minimum of 30 s prior to measurements. In the dark PSII reaction centres relax to ground state, and are open for photochemistry upon receipt of an absorbed photon, with a low

fluorescence yield (Schuback et al., 2021). In parallel the OEC relaxes primarily to S1 (Han et al., 2008), so the S-States of the population of PSII largely synchronize. We did not systematically test the effect of duration of dark relaxation on subsequent results, but our experience is that 30 s gives full relaxation, even for psychrophile strains at low temperature. The sample was then exposed to a series of 20 to 32 short, high irradiance evenly spaced flashes centred at 445 nm, with the number of flashes always sufficient to provide at least 10 flashes after the sample reached full damping of S-State oscillations. Excitation at 445 nm preferentially excites chlorophyll a, thereby limiting the influence of different light harvesting complexes across taxa. Each flash comprised a rapid series of 50-70 sub-saturating flashlets of 1.6 μ s and $\sim 30000 \mu\text{mol photons m}^{-2} \text{s}^{-1}$, delivered every 4.1 μ s, over a total 205 - 287 μ s per flash (Schuback et al., 2021). These flashlets cumulatively deliver a photon to each PSII in the population, with each PSII ideally undergoing a single round of photochemistry and passing an electron downstream to Q_A , closing PSII for photochemistry for $\sim 1000 \mu$ s, the lifetime (τ) for re-opening by downstream electron transport (Dau and Haumann, 2007; Schuback et al., 2021). Closing the photochemistry pathway transiently redirects a greater proportion of additional incoming excitation to ChlF, driving ChlF progressively from minimum (F_0) towards a maximum (F_M) as the PSII centres close (Schuback et al., 2021). For each flash-induced fluorescence rise, F_0 and F_M were extracted using a fitting model (Solisense LIFT software version 22.11.11, Solisense Inc) (Kolber, 1998). F_0 and F_M were then used to derive the maximum quantum yield of photochemistry for open PSII (Schuback et al., 2021), a secondary ChlF parameter calculated as:

$$\frac{F_V}{F_M} = \frac{F_M - F_O}{F_M} (1)$$

For each culture sample we adjusted the number and excitation power of flashlets per flash to achieve closure of ~ 0.95 of PSII complexes, as estimated by the fitting of the chlorophyll fluorescence induction (Kolber, 1998; Schuback et al., 2021). Ideally the saturating flash would achieve complete closure of the PSII population, but concomitant re-opening of PSII centres by downstream electron transport results in a steady state ratio of mainly closed PSII to a fraction of open PSII, even at F_M . The flash power required to drive the PSII population to closure depends largely upon the effective absorption cross-section of photosystem II (σ_{PSII}), itself estimated from the ChlF induction curve during the saturation phase (Kolber, 1998). σ_{PSII} represents the probability of light capture by the PSII antenna bed associated with the dark-adapted PSII (Schuback et al., 2021; Xu et al., 2018). When σ_{PSII} for a sample is large, the higher probability of excitation delivery to PSII means fewer, or weaker, flashlets are required to sequentially saturate the population of PSII, while limiting excitation stress or double turnovers of PSII associated with excess light (Xu et al., 2018). Conversely, a sample with a smaller σ_{PSII} requires more numerous, or brighter, flashlets, to drive the proportion of closed reaction centres progressively towards saturation. Because we used excitation at 445 nm, delivered through PSII-associated chlorophyll a, our σ_{PSII} were quite consistent across taxa (Xu et al., 2018), even though the light harvesting complexes differ between the green and diatom taxa. If the brightness and number of sub-saturating flashlets are appropriate for the culture, the fluorescence will reach a plateau after approximately 70% of the flashlets are applied (Kolber, 1998).

We kept the single turnover excitation flash train as short as feasible, to avoid excessive excitation, and to limit the probability of ‘double hits’ (Han et al., 2012, 2022); re-openings of PSII, followed by re-closures of PSII complexes by a second round of photochemistry, which contribute to desynchronization of the progression of the PSII complexes through S-States. Furthermore we did not use probe flashlets to track re-opening of PSII after closure by each saturating flash (Kolber, 1998), to limit differential re-openings and re-excitations across the PSII population. We thereby forewent analyses of PSII re-opening kinetics.

As sequential flashes are applied to the culture, each individual PSII is ideally driven through the four S-States (Fig 1) (Dau and Haumann, 2007). As the OEC of PSII moves between S-States, it alters the system kinetics and free energy (Vinyard et al., 2013), so the yield of ChlF, and thus F_V/F_M , varies between S-States (Gates et al., 2020), and we can use F_V/F_M to monitor S-State progression, until the PSII population desynchronizes to a distribution of S-States with an averaged F_V/F_M .

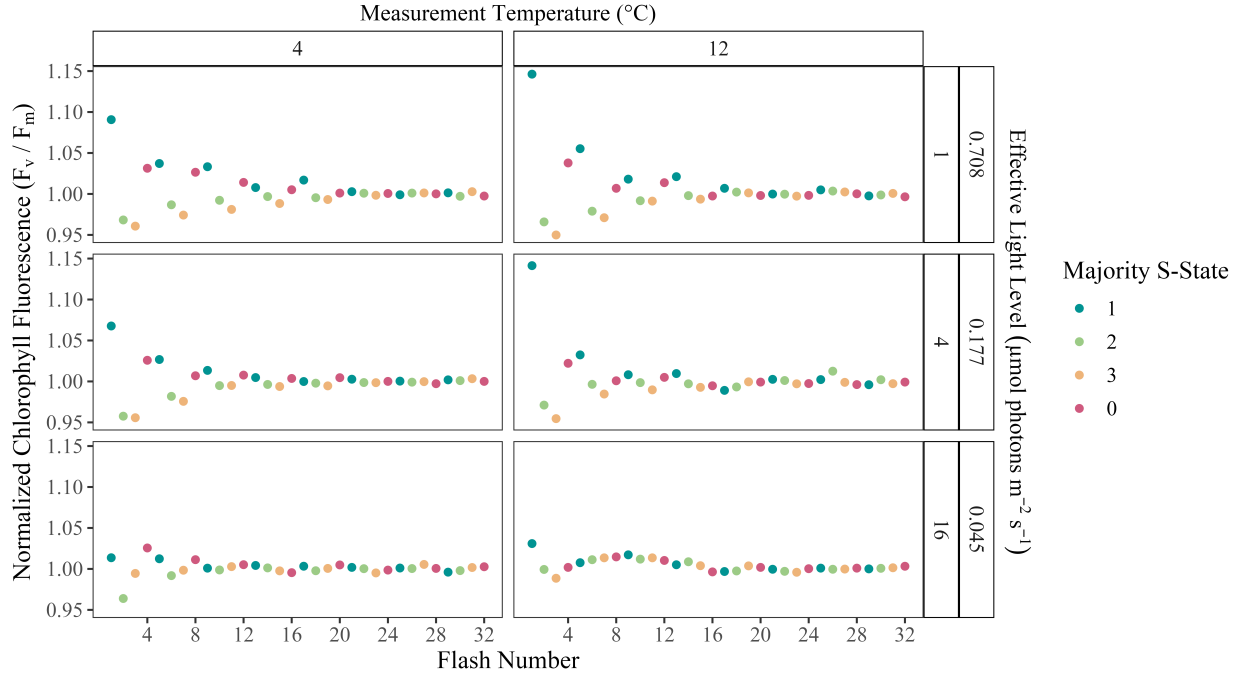


Fig 1: Representative oscillations of normalized maximum quantum yield of PSII photochemistry (F_v/F_m) over a Series of Single Turnover Saturating Flashes applied to Antarctic green algae *Chlamydomonas priscui*. F_v/F_m was normalized to the average value over the series, measured at 4 or 12 °C, with spacings of 1, 4, or 16 s between sequential flashes, equivalent to photon delivery rates to PSII achieved under effective light levels from 0.708, 0.177 or 0.044 $\mu\text{mol photons m}^{-2} \text{s}^{-1}$, estimated from flash spacing (s) and effective absorption cross section of photosystem II, σ_{PSII} (A^2). Symbol colour indicates the inferred majority S-State, based upon assumption of relaxation to S1 during the dark incubation preceding the flash sequence.

Measurements

By evaluating the oscillations in F_v/F_m (Fig 1), we tracked S-State cycling of polar and temperate taxa of diatoms and green algae under a range of measurement temperatures and effective light levels, after growth under a range of salinities (Table 4), to determine if polar taxa maintain synchronized PSII function, as an index of their capacity for electron transport under low light, by suppressing wasteful photochemical misses. We further tested whether change in growth salinity altered the responses, for a sub-set of the

Chlamydomonas strains. Measurement temperatures ranged from 0 to 28°C, depending on the taxa (Table 4).

Increasing the spacing between sequential flashes decreases the repeat rate for excitations arriving at PSII, thereby approximating a decrease in excitations delivered to PSII per second under a decrease in continuous irradiance (Fig 1). Our instrument control limited our minimum flash spacing to 1 s, so cultures were evaluated at flash spacings of 1, 2, 4, 8, and 16 seconds (Table 4). The equivalent effective light levels were then estimated using the σ_{PSII} determined for each culture as follows:

$$\text{Light}(\text{photons m}^{-2}\text{s}^{-1}) = \frac{1}{\text{flashspacing(s)}} \times \frac{1}{\sigma_{PSII}(\text{m}^2\text{photons}^{-1})} \quad (2)$$

$1/\sigma_{PSII}$ is multiplied by flash spacing (s) to convert the probability that an individual incident photon is absorbed by PSII, into an equivalent photon flux density, i.e. to estimate effective equivalent light availability serving PSII photochemistry.

Since the population of PSII starts predominantly from S1 in the dark acclimated state, the population of PSII will initially cycle synchronously through S-States, reflected by an oscillation in chlorophyll fluorescence with a period of four (Fig 1)(de Wijn and van Gorkom, 2002). Photochemical misses, including wasteful recombination reactions causing loss of a charge separation in a PSII, will cause a missed step in the S-State cycling of an individual PSII. As more photochemical misses occur, desynchronization of S-State cycling among the population of PSII will scramble the periodic changes in F_V/F_M , dampening the

observed oscillation of the population-level fluorescence (de Wijn and van Gorkom, 2002; Han et al., 2022). Prolonged four-step synchronous cycling of PSII fluorescence thus reflects a lower probability of desynchronization at each excitation, and thus a lower probability of losses through photochemical misses. Photochemical double hits also contribute to desynchronization (Han et al., 2022; Han et al., 2008), but should not vary strongly across changes in flash spacing nor measurement temperature within a sample.

Analytical Methods

Data was processed using R version 4.3.2 (R Core Team, 2023) and RStudio version 2023.12.0+369 (Posit team, 2025) on the x86_64-apple-darwin20 (64-bit) platform and running under macOS Sonoma 14.3.1. Fluorescence data files generated by LIFT software were imported, tidied and combined with metadata on each culture using the tidyverse (Wickham, 2023), lubridate (Spinu et al., 2024), and googlesheets4 (Bryan, 2023) packages. The doBy (Halekoh and Højsgaard, 2025), and WaveletComp (Roesch and Schmidbauer, 2018) packages were used for wavelet analyses. The mgcv (Wood, 2011) and mgcViz (Fasiolo et al., 2020) packages were used for generalized additive modelling, while the ggplot2 (Wickham, 2016), ggh4x (van den Brand, 2025), and metR (Campitelli, 2021) packages were used for data visualization. Statistical results tables were done using the packages 'broom' (Robinson et al., 2025), 'knitr' (Xie, 2025), and 'kableExtra' (Zhu, 2024).

Wavelet Transformations

The fluorescence data yields a time series of F_V/F_M over 20 to 32 sequentially applied flashes, always sufficient to reach full damping of oscillations, for each flash spacing, and each measurement temperature, for each culture sample. The F_V/F_M time

series were analyzed for each combination of strain, growth conditions, measurement temperature, and flash spacing, using wavelet transformations (Theis and Meyer-Bäse, 2010), as exemplified in (Fig 2). Unlike traditional methods, wavelet analysis does not assume that the statistical properties of a time series are constant. Instead, wavelet transformations locally decompose the signal across different time scales and estimate spectral characteristics as a function of time (Cazelles et al., 2008). By examining the frequency and wavelet power spectra, we can uncover the dominant patterns in the data (Theis and Meyer-Bäse, 2010).

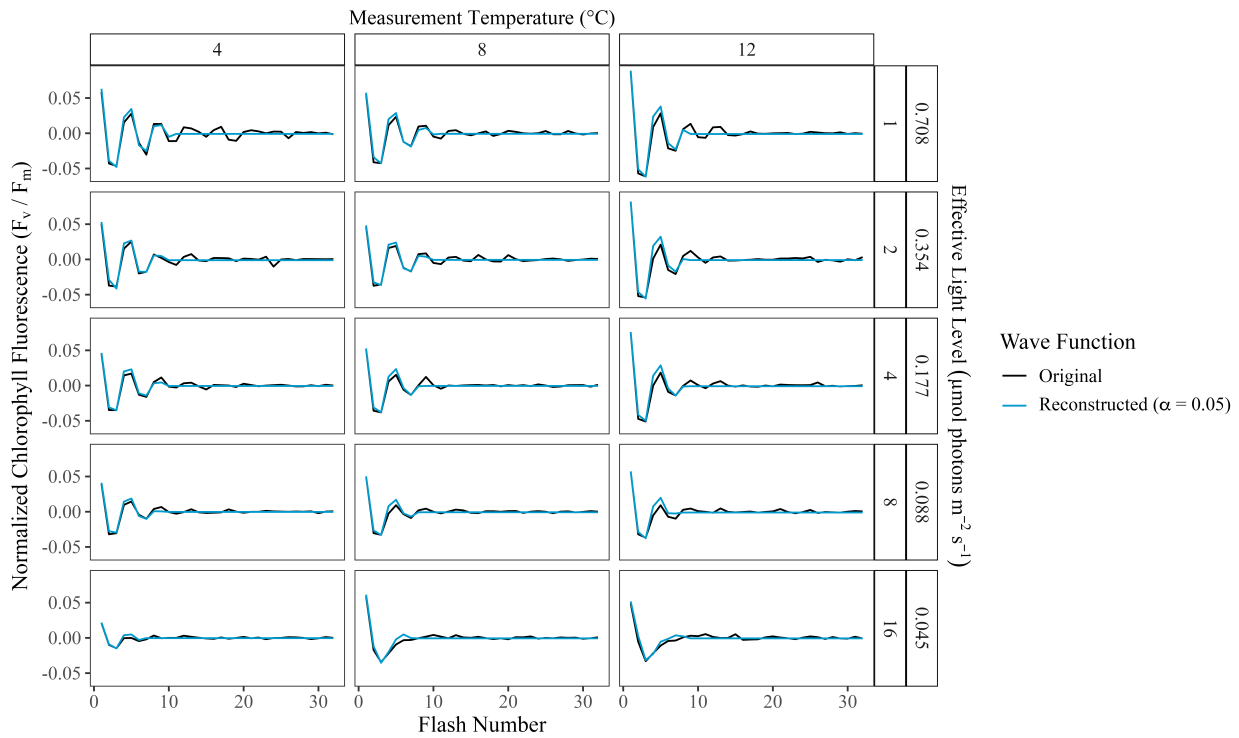


Fig 2: Representative wavelet transformations of normalized maximum quantum yield of PSII photochemistry (F_V/F_M) over a Series of Single Turnover Saturating Flashes applied to Antarctic green algae *Chlamydomonas priscui*. F_V/F_M was normalized to the average value over the series, measured at 4, 8 or 12°C, over 32 consecutive single turnover saturating flashes, applied at spacings of 1, 2, 4, 8, 16 s, equivalent to photon delivery rates to PSII achieved under effective light levels from 0.708 down to 0.044 $\mu\text{mol photons m}^{-2} \text{s}^{-1}$, estimated from flash spacing (s) and effective absorption cross section of photosystem II, σ_{PSII} (A^2).

The wavelet transformation involves computing the wavelet power spectrum of the standardized time series using the Morlet wavelet (Theis and Meyer-Bäse, 2010). The statistical significance of the periodic components in the time series was then calculated using a simulation algorithm. Surrogate time series are generated based on a white noise model, consisting of uncorrelated random values with constant mean and variance. The wavelet transform of the data is compared with the white noise model, to estimate p-values for whether the observed periodic components are statistically significant (Theis and Meyer-Bäse, 2010). Statistical significance of the wavelet power at a periodicity of four indicates the culture is exhibiting periodic oscillations in chlorophyll fluorescence, reflecting synchronous S-State cycling across the PSII population. For wavelets exhibiting detectable S-State cycling, we generated a reconstruction, limited to the sub-set of flashes showing a signal that was statistically different from the average value of F_V/F_M . We then extracted a Damping Index as the number of flashes applied before the reconstructed wavelet dropping below a detection threshold significance of $p=0.05$ above random noise. This Damping Index indicates how many successive photochemical charge separations occur in the PSII population before recombinations, other photochemical misses, or photochemical double hits, desynchronize the S-State cycle to a randomized distribution of S-States across the population.

Generalized Additive Modelling

Patterns in the persistence of S-State cycling within taxa were then modelled using the nonparametric method of Generalized Additive Modelling (GAM). For GAM we used S-State damping indices estimated for each strain, from each measurement temperature and

flash spacing. The damping index response variable is linked to the independent variables using a smoothing function, where many localized polynomials are joined to form a piecewise spline function (Pinilla and Negrín, 2021). GAM thus generated a matrix of predicted damping indices based on a tensor product smooth of the two predictors, measurement temperature, and equivalent effective light level derived from flash spacing. For each strain, GAM were fit to the data using the restricted maximum likelihood method (REML), using the R-package *mgcv* (Wood, 2025). GAM were validated by verifying the choice of basis dimensions (k) and evaluating the residual plots (Schoenig et al., 2023) using the function `gam.check` from the R-package *mgcv* (Wood, 2025) (Supp. Figs. 7-16). The predicted damping indices for combinations of measurement temperature and effective light (Wood, 2017) were represented with contour plots. For GAM plots we only include estimates of S-State damping for which the amplitude of the standard error was smaller than 100% of the fitted estimates, resulting in blank areas on some plots. The resulting GAM plots examine a response surface of S-State damping to the measurement temperature (°C) and the equivalent effective light level ($\mu\text{mol photons m}^{-2} \text{ s}^{-1}$). For GAM we only included data from taxa grown and measured at their respective approximate optimal salinities (Table 1).

Analyses of Variance

Analyses of Variance (ANOVA) were run using the `aov` function (R Core Team, 2023) to detect effects of taxa, measurement effective light levels (derived from flash spacing and σ_{PSII}), measurement temperature, and growth salinity, upon the Damping Index.

Results

Single Turnover Variable Chlorophyll Fluorescence

Exposing phytoplankton cultures to a series of 20-32 successive single turnover, saturating flashes produced the expected oscillations in the maximum quantum yield of photochemistry of PSII, as estimated through the secondary chlorophyll fluorescence parameter F_V/F_M , normalized to 1 for each run by dividing by the averaged value for each measurement run, determined after oscillations damped out (Table 4). Initially, after dark relaxation the majority of the population of PSII is at S1, with a smaller fraction at S0 (Gates et al., 2020; de Wijn and van Gorkom, 2002). As shown in representative data from polar *Chlamydomonas priscui* (Figs. 1, 2), grown at optimal salinity, the time series of F_V/F_M over successive flashes reveals consistent variations in fluorescence yield as the predominant S-States, with different yields for ChlF and thus F_V/F_M , follow each other across the PSII within the population. The amplitudes of the F_V/F_M oscillations, however, decline progressively over successive flashes. The persistence of oscillations decreases with wider spacing between sequential flashes, equivalent to decreasing effective light levels, and at higher measurement temperatures. The full set of conditions under which S-State time courses were captured across taxa and growth salinities are in Supplemental Data (Table 4). The full data sets are available through the GitHub repository (<https://github.com/FundyPhytoPhys/LowLightPhotosynth>) hosting the data & analyses.

σ_{PSII} for each run was determined for the 445 nm blue light flashes, absorbed primarily by the Chla associated with PSII. σ_{PSII} were thus similar (Xu et al., 2018), in spite of variations in light harvesting complexes across taxa and growth conditions (Table 4).

Conversions of flash spacing to equivalent effective light levels, using σ_{PSII} , therefore gave similar ranges of effective light levels ranging from ~ 0.02 to $\sim 1 \mu\text{mol photons m}^{-2}\text{s}^{-1}$, applied to each sample (Table 4). For comparison, full sunlight at the sea surface is $\sim 2000 \mu\text{mol photons m}^{-2}\text{s}^{-1}$, while the threshold for the bottom of the photic zone supporting photosynthetic productivity in the oceans was conventionally defined as $\sim 20 \mu\text{mol photons m}^{-2}\text{s}^{-1}$ (Raven et al., 2000). Therefore, our effective equivalent measurement light ranges are ~ 3 -5 orders of magnitude lower than full sunlight, and ~ 2 -3 orders of magnitude below the traditional minimum photic zone threshold, but nevertheless span light ranges found to support phytoplankton productivity in field studies (Hancke et al., 2018; Randelhoff et al., 2020; Hoppe et al., 2024).

Wavelet Analysis

To detect statistically significant 4-step oscillations in F_V/F_M , reflecting S-State synchronization, across samples and measurements, we computed wavelet transformations for the F_V/F_M time series of each combination of measurement temperature, flash spacing (and thus equivalent effective light), growth temperature, growth salinity, and species or strain. The wavelet transformations only detected 4-step periodicities, consistent with our expectation that S-State synchronization underlies the oscillations in F_V/F_M . Assessing the wavelet power of the 4-step periodicity across conditions, key trends emerged. As exemplified by the Antarctic green alga *Chlamydomonas priscui* (Fig. 3), the average wavelet power declines with increasing flash spacings, equivalent to decreasing effective light levels, and also declines with increasing measurement temperatures.

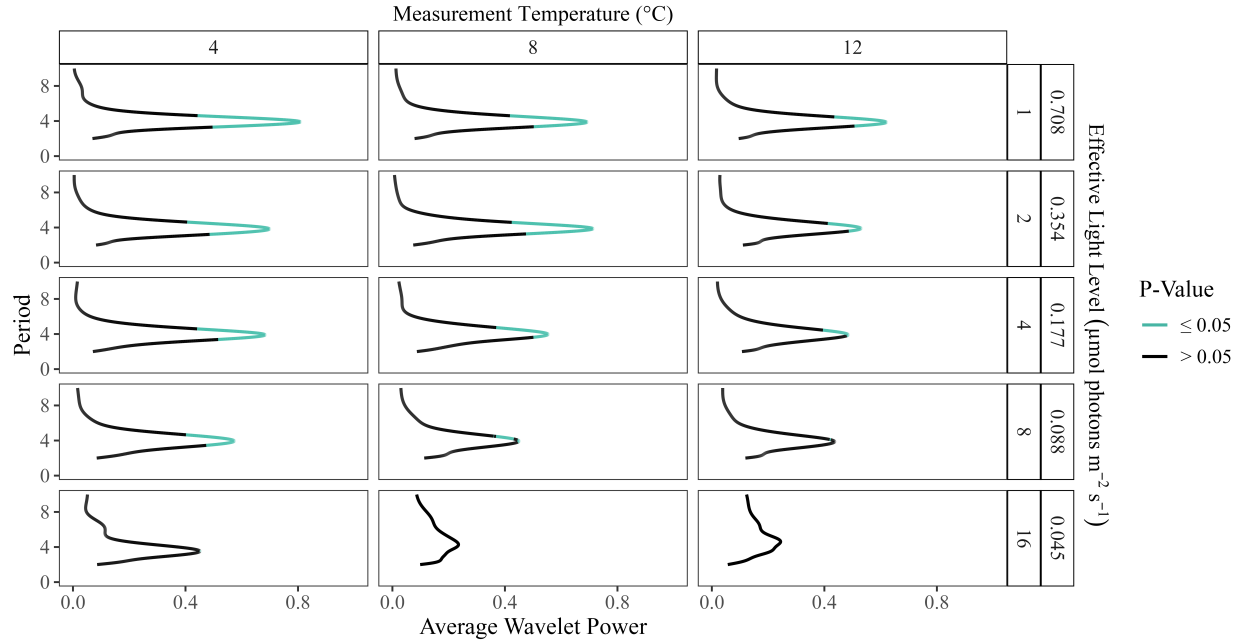


Fig 3: Sample plot of wavelet powers by period of oscillations in the maximum quantum yield of photochemistry in the Antarctic green algae *Chlamydomonas priscui*, grown at 4°C and 500 mM NaCl, and measured across a range of measurement temperatures and flash spacings, with the equivalent effective light levels ($\mu\text{mol photons m}^{-2} \text{s}^{-1}$), estimated from flash spacing (s) and effective absorption cross section of photosystem II, σ_{PSII} (A^2). Regions where wavelet power reached statistical significance ($p < 0.05$) are shaded in blue.

In contrast, in the temperate green algae *Chlamydomonas reinhardtii* (Fig. 4) both grown, and measured, at warmer temperatures, the wavelet power is consistently lower, showing a weaker 4-step periodicity of F_V/F_M in the temperate taxa, which only reaches statistical significance at shorter flash spacings, with higher equivalent effective light levels. Note that at a common measurement temperature of 12°C *Chlamydomonas reinhardtii* and *Chlamydomonas priscui* show similar wavelet power patterns, suggested the sustained PSII synchronization in *Chlamydomonas priscui* reflects the ability to grow at lower temperatures which stabilize PSII synchronization, rather than an intrinsic difference in PSII from *Chlamydomonas reinhardtii*.

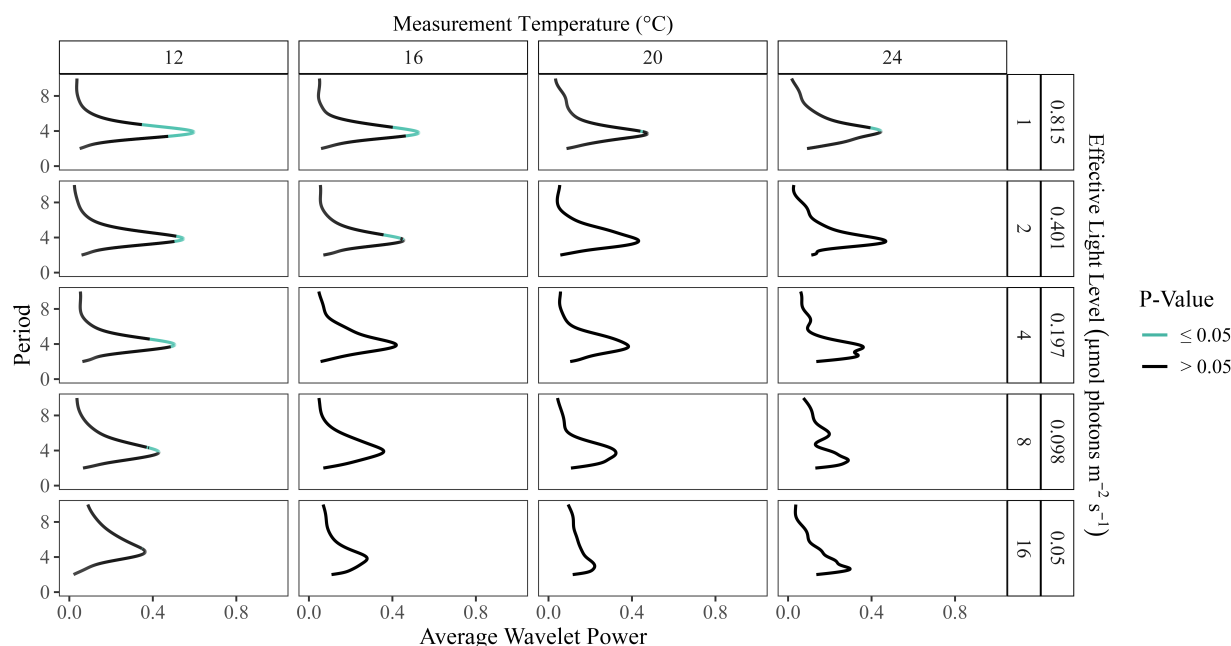


Fig 4: Sample plot of wavelet powers by period of oscillations in the maximum quantum yield of photochemistry in the temperate green algae *Chlamydomonas reinhardtii*, grown at 24°C and 0.43 mM NaCl, and measured across a range of measurement temperatures and flash spacings, with the equivalent effective light levels (μmol photons m⁻²s⁻¹, estimated from flash spacings (s) and effective absorption cross-section of photosystem II, σ_{PSII} (Å²). Regions where wavelet power reached statistical significance ($p < 0.05$) shaded in blue.

Generalized Additive Modelling of Damping of 4-Step Oscillations By Measurement Temperature

To systematically compare the persistence of synchronized PSII S-State cycling across measurement temperatures, effective equivalent light levels, and taxa, we then extracted a Damping Index, as the number of flashes applied before the reconstructed wavelet describing the oscillations in F_V/F_M , dropped below a detection threshold significance of $p=0.05$ above random noise. A higher damping index shows a taxa and measurement condition that supports more persistent S-State synchronization of the PSII population, reflecting more successive photochemical charge separations before photochemical misses, or photochemical double hits, desynchronize the S-State cycle to a randomized distribution of S-States across the PSII population.

Predictions of the damping index across conditions were then generated using generalized additive modelling, using the tensor product smooth of the measurement temperature (°C) and the equivalent effective light level ($\mu\text{mol photons m}^{-2}\text{s}^{-1}$) set by flash spacing, for each strain; Table 4; Table 5. Only data from cultures growth at their respective near-optimal salinities were used for GAM, to limit influences of stress. All of the model fits explained over 50% of the variation in the response variable, damping index, except for *C. euryale*.

The polar diatom *Fragilariopsis cylindrus* (grown at either 0 or 6 °C) showed higher damping indices, and thus maintained S-State cycling longest, at higher effective light levels, with a monotonic decline in damping index with decreasing effective light. *Fragilariopsis cylindrus* showed a modest decrease in S-State damping index with increasing measurement temperature (Fig 5) from 0 to 10°C.

In contrast the temperate diatom, *Thalassiosira pseudonana* (grown at 22 °C), showed a strong decrease in damping index with increasing measurement temperature from 10 to 28°C, with sharp peak of ~ 12 for damping index at lower measurement temperatures and $\sim 0.4 \mu\text{mol photons m}^{-2}\text{s}^{-1}$, equivalent to a flash spacing of 2 s. The disparity between the polar and temperate diatoms was particularly evident above their respective growth temperatures, and when longer spacing between flashes produced lower effective light levels (Fig 5), where *Thalassiosira pseudonana* cultures did not retain significant 4-step oscillation in F_V/F_M indicative of synchronized S-State cycling, shown as blank areas on the contour plot.

The polar diatom achieved a higher damping index than did the temperate diatom around their respective growth temperatures. Under equivalent measurement conditions at 10 °C, *Fragilariopsis cylindrus* maintained synchronized S-State cycling over a wider range of equivalent light levels. *Fragilariopsis cylindrus* was our only analyzed comparison of a single strain grown at two temperatures; a peak damping index of ~ 11-12 was achieved by the cultures at both growth temperatures, but growth at 0°C somewhat lowered performance under the very lowest equivalent light levels (widest flash spacings), suggesting that PSII synchronization performance may respond to growth temperature. The similar peak damping indices between polar and temperate diatoms suggest similar intrinsic capacities for PSII performance, with manifestation of synchronized S-States across a wider range of conditions in the polar diatom resulting from the ability of the polar diatom to grow at low enough temperatures to manifest S-State synchronization.

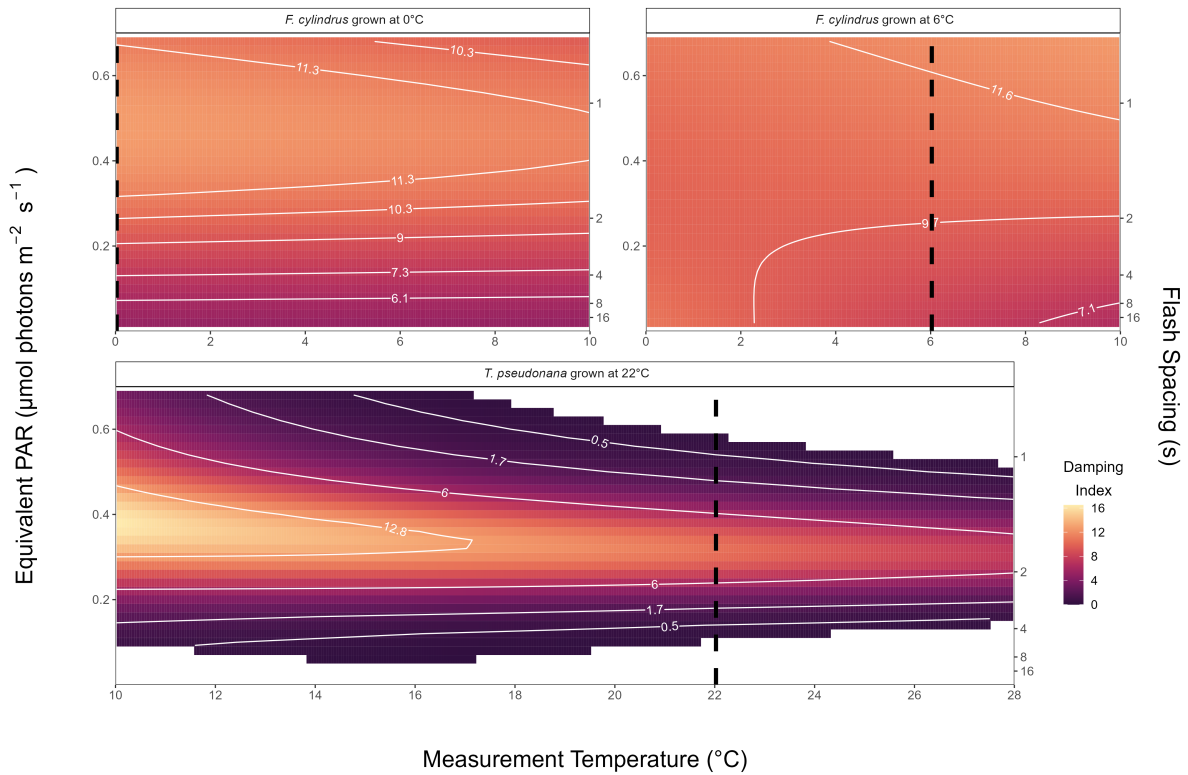


Fig 5: Generalized Additive Models (GAM) for polar and temperate diatoms, of consecutive flashes before damping of S-State induced oscillations in maximum quantum yield of photosystem II (F_V/F_M). GAM predicted by the measurement temperature ($^{\circ}\text{C}$) and the effective light level ($\mu\text{mol photons m}^{-2}\text{s}^{-1}$, estimated from flash spacings (s) and effective absorption cross-section of photosystem II, σ_{PSII} (A^2)). Dashed vertical lines represent the growth temperatures. Measurement temperatures varied across taxa but the temperature axis scalings are comparable across plots. Data for GAM was taken from cultures grown and measured near their respective optimal salinities.

Similarly, the GAM outputs of damping index varied across the green algae tested (Table 5). The polar, saline strains *Chlamydomonas* ICE-MDV, *Chlamydomonas malina* and *Chlamydomonas priscui* showed GAM predictions of maximum damping indices of ~ 10 -11 flashes, centred at lower measurement temperatures and the higher effective light levels (Fig 6). The polar freshwater *Chlamydomonas klinobasis* also showed maximum damping indices at lower measurement temperatures, and the higher effective light levels. Similarly to the temperate diatom, the freshwater temperate algae *Chlamydomonas reinhardtii* and

Chlorella vulgaris did not exhibit significant periodic oscillations in F_V/F_M at measurement temperatures near or above their growth temperature, nor under the longer flash spacings equivalent to lower light, shown by blank areas on the contour plots. The marine temperate *Chlamydomonas euryale* showed limited evidence for periodic oscillations, and thus did not generate a statistically significant GAM. *Chlamydomonas euryale* also showed F_V/F_M somewhat lower than other *Chlamydomonas* (Table 4), and might not have been in optimal growth (data not shown).

At their respective growth temperatures of 4°C for the polar green algae vs. 22-24°C for the temperate strains, the polar taxa showed significant S-State synchronization over much wider ranges of effective light, maintaining detectable synchronization even at the lowest effective light levels. Again, in parallel with the diatoms, the temperate green algae achieve peak damping indices similar to the polar taxa, so the performance of the polar green algae appears to result from their ability to grow at the lower temperatures where PSII manifests sustained synchronization, rather than more fundamental differences in PSII function.

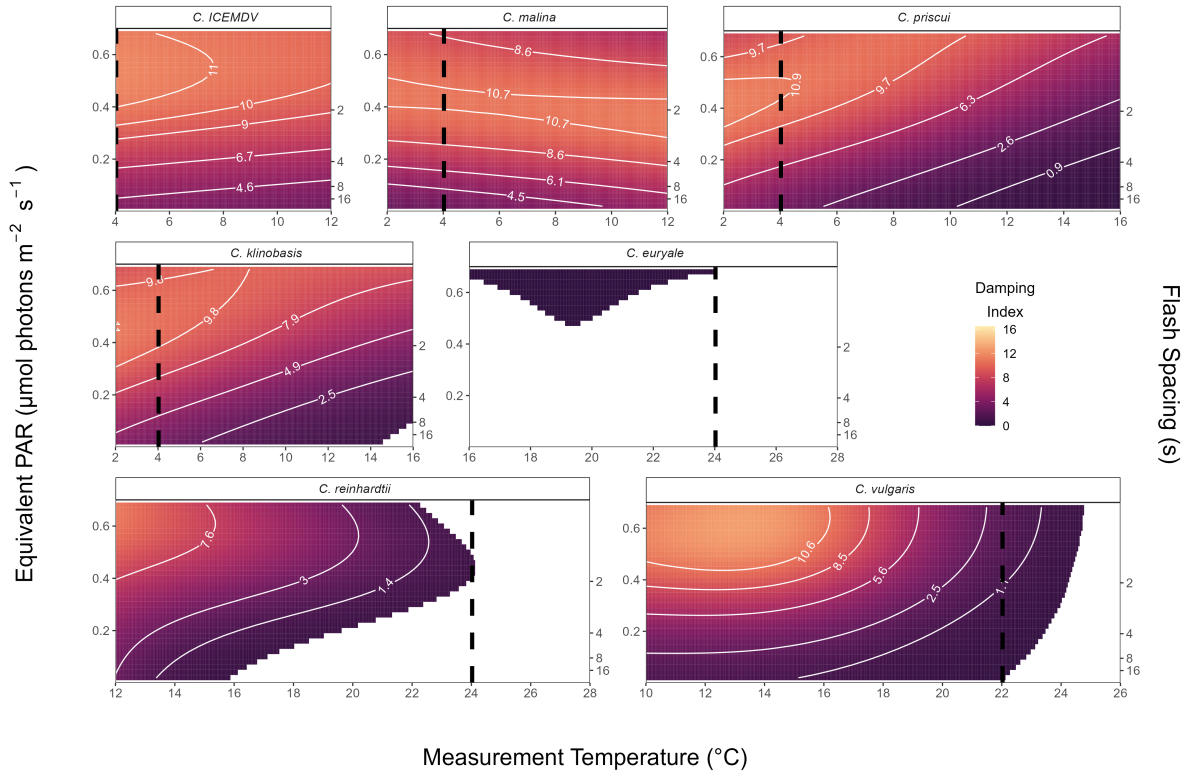


Fig 6: Generalized Additive Models (GAM) for polar and temperate green algae, of consecutive flashes before damping of S-State induced oscillations in maximum quantum yield of photosystem II (F_V/F_M). GAM predicted by the measurement temperature ($^{\circ}\text{C}$) and the effective light level ($\mu\text{mol photons m}^{-2}\text{s}^{-1}$, estimated from flash spacings (s) and effective absorption cross-section of photosystem II, σ_{PSII} , (A^2)). Dashed vertical lines represent the growth temperatures. Measurement temperatures varied across taxa but the temperature axis scalings are comparable across plots. Data for GAM was taken from cultures grown and measured near their respective optimal salinities.

The *Chlamydomonas* taxa we used can grow under a range of salinities. Furthermore, across several taxa of *Chlamydomonas* growth under higher salinities increases photosynthetic resilience during stress (Kalra et al., 2020; Kalra et al., 2023; Stahl-Rommel et al., 2022). Therefore, to test for effects of salinity acclimation upon the performance of PSII under extreme low light, we grew *Chlamydomonas* taxa across three growth-permissive salinities (Table 1) and then performed saturating flash trains over a range of flash spacings, and at measurement temperatures flanking the growth temperature for each

taxa. We then ran an ANOVA for damping indices, testing effects of taxa, effective equivalent light level (derived from the saturating flash spacing and σ_{PSII}), measurement temperature, and the growth concentration of NaCl, with interactions among all factors (Table 2). As expected, taxa, effective equivalent light level and measurement temperature all showed strongly significant influences on damping indices. NaCl concentration also showed a significant influence on damping indices when tested across taxa, with significant interaction between NaCl and effective equivalent light level.

Table 2: Analyses of Variation of damping indices, testing effects of *Chlamydomonas* taxa, effective equivalent light level (LightLevel_ue, $\mu\text{mol photons m}^{-2}\text{s}^{-1}$, estimated from flash spacings (s) and effective absorption cross-section of photosystem II, σ_{PSII} , (A^2)), measurement temperature (Temp_C), and the growth media concentration of NaCl (NaCl_mM), with interactions among all factors. Degrees of freedom (df); Sum of Squares of Residuals (sumsq); Mean of Sum of Squares of Residuals (meansq); F statistic (statistic); P Value (p.value)

term	df	sumsq	meansq	statistic	p.value
Strain	4	2128	532	78.5	0.00
LightLevel_ue	1	577	577	85.1	0.00
Temp_C	1	488	488	72.0	0.00
NaCl_mM	1	35	35	5.2	0.02
Strain:LightLevel_ue	4	369	92	13.6	0.00
Strain:Temp_C	4	203	51	7.5	0.00
LightLevel_ue:NaCl_mM	1	27	27	4.0	0.05
Strain:LightLevel_ue:Temp_C	4	74	18	2.7	0.03

We then ran the ANOVA for each taxa separately (Table 3). Within a taxa only *Chlamydomonas priscui* showed a significant effect of NaCl on damping indices, suggesting the statistical influence of NaCl detected across taxa derives from different taxa growing at different salinities and showing different responses of damping indices across taxa, rather than a direct effect of NaCl on damping indices in most taxa, except *Chlamydomonas priscui*.

Table 3: Analyses of Variation for each *Chlamydomonas* species, testing effects upon damping indices of effective equivalent light level (LightLevel_ue, $\mu\text{mol photons m}^{-2}\text{s}^{-1}$, estimated from flash spacings (s) and effective absorption cross-section of photosystem II, σ_{PSII} , (A^2)), measurement temperature (Temp_C), and the growth media concentration of NaCl (NaCl_mM), with interactions among all factors. Degrees of freedom (df); Sum of Squares of Residuals (sumsq); Mean of Sum of Squares of Residuals (meansq); F statistic (statistic); P Value (p.value)

Taxa	term	df	sumsq	meansq	statistic	p.value
<i>C. euryale</i>	LightLevel_ue	1	30	30	5.1	0.03
<i>C. klinobasis</i>	LightLevel_ue	1	550	550	77.5	0.00
<i>C. klinobasis</i>	Temp_C	1	137	137	19.2	0.00
<i>C. malina</i>	LightLevel_ue	1	140	140	14.5	0.00
<i>C. priscui</i>	LightLevel_ue	1	255	255	31.3	0.00
<i>C. priscui</i>	Temp_C	1	476	476	58.5	0.00
<i>C. priscui</i>	NaCl_mM	1	40	40	4.9	0.03
<i>C. reinhardtii</i>	Temp_C	1	18	18	5.2	0.03

Discussion

Across all study taxa oscillations in F_V/F_M , provoked by a series of single turnover saturating flashes, showed wavelet power at a period of four, that generally declined with increasing measurement temperatures and increasing spacing of flashes, equivalent to decreasing equivalent effective light. Thus, sustained synchronized S-State cycling of PSII decayed faster under higher measurement temperatures and lower equivalent effective light (Fig. 3), which we interpret as a faster accumulation of photochemical misses across the PSII population (de Wijn and van Gorkom, 2002) when arrivals of photons are widely spaced, and when higher temperature accelerates the probabilities of recombinations. While light absorption and excitation energy transfer within PSII are not temperature-sensitive, the redox reactions associated with downstream electron transport slow with decreasing temperature, (Hüner et al., 2023), while probabilities of charge recombinations (Ivanov et al., 2006; Han et al., 2022) decrease as ambient temperature falls below their activation energies. The temperature dependencies of recombination reactions may also shift through acclimatory mechanisms within a taxa, to regulate photosynthetic electron flow, S-State transitions, and energy partitioning (Hüner et al., 2023). In future, treatments with plastoquinone-analog inhibitors of PSII such as dichlorophenyl-dimethylurea (DCMU) might help dissect distinctions among responses across psychrophile and temperate taxa.

The desynchronization of S-State cycling among the PSII in a population indicates that sufficient photochemical misses, including charge recombinations, have taken place to create a PSII population with a randomized distribution of S-States. Thus, if this desynchronization occurs over more consecutive flashes, it signifies a decreased proportion

of PSII undergoing photochemical misses for each flash. By inference, PSII populations with decreased incidence of energetically wasteful misses, including charge recombinations, such as those under lower temperatures and higher effective light levels, are more efficient in their photosynthetic energy conversion (Rappaport et al., 2005), which becomes a critical factor for maintaining photosynthesis under extreme low light, with wide spacings between sequential excitations of each PSII.

These results are consistent with previous literature evaluating the responses of photochemical misses to light conditions. As light levels decline, there are longer intervals between successive PSII excitations, pushing fewer electrons through the electron transport chain (Keren et al., 1997). Consequently, the probability of energetically wasteful charge recombinations is higher over the longer intervals between excitations, leading to weaker maintenance of S-State cycling (Keren et al., 1997; de Wijn and van Gorkom, 2002). Furthermore, our findings are consistent with recombination reactions decreasing as measurement temperatures (Ivanov et al., 2006) fall below the activation temperatures of the recombinations. In contrast, in PSII-enriched membrane fractions isolated from spinach, the average miss probability of S-State transitions was highest at -10 °C and lowest at 10 °C (Han et al., 2022), although both these temperatures fall well below temperatures equivalent to the expected activation energies for recombinations.

Chlamydomonas sp. ICE-MDV and *Chlamydomonas priscui* originate from Lake Bonney in Antarctica, but previous work found differential responses to light among the taxa. Unlike the obligately shade-adapted *Chlamydomonas priscui* that is only present in the deep photic zone, *Chlamydomonas* sp. ICE-MDV has been isolated from various depths

within the lake (Li and Morgan-Kiss, 2019) including shallow and seasonally open waters (Sherwell et al., 2022) where it encounters a wider range of light conditions. In contrast with *Chlamydomonas priscui* that has a “locked” photophysiology, *Chlamydomonas* sp. ICE-MDV has retained the ability to balance excitation pressure through state transitions (Kalra et al., 2023), can modify its photosynthetic apparatus due to nutrient availability (Cook et al., 2019), exhibits phototactic motility in response to light signals (Poirier et al., 2023) and can successfully acclimate to a range of wavelengths and light levels (Poirier et al., 2025). Interestingly the GAM contour plots also show *Chlamydomonas* sp. ICE-MDV maintains higher PSII synchronization over a wider range of measurement temperatures and effective light levels, than does *Chlamydomonas priscui*. It is evident that physiological differences exist among polar species, likely as a result of life in comparably cold, but otherwise differing environments.

Nevertheless, we found strong, sustained periodicity of F_V/F_M as a common trait across the diverse polar diatoms and green algae (Figs. 3; 4; 5; 6). Further, the polar taxa exhibited significant 4-step F_V/F_M oscillations under a broader range of measurement conditions around their growth temperatures than did their temperate counterparts. These findings illustrate that polar phytoplankton species, measured at and around growth temperatures typical for their native environments, show higher capacity to maintain synchronized S-State cycling than do temperate taxa around their optimal growth temperatures. The inferred rates of photochemical misses, notably through charge recombinations, relative to productive photochemistry, increase under lower light and higher measurement temperatures (Keren et al., 1997; Ivanov et al., 2006). The most sustained PSII synchronizations are around 10-12 flashes for both temperate and polar

taxa, but temperate taxa only achieve this peak under narrow conditions. This suggests the suppression of photochemical misses across wider ranges of conditions in polar taxa is an outcome of growth at temperatures low enough for PSII stabilization to manifest, rather than a fundamental difference in the capacities of PSII between the polar and temperate taxa.

Sustained synchronized S-State cycling reflects suppression of energetically wasteful photochemical misses in the polar taxa, as measured at the low growth temperatures possible for polar taxa. Stable S-State cycling with minimal energy losses through charge recombination or other forms of electron cycling within PSII (Ananyev et al., 2017; Zournas et al., 2023) can support linear electron flow, sustaining ATP and NADPH production, and minimizing the risk of photodamage to the photosynthetic machinery (Rappaport et al., 2005; Keren et al., 1997). Our findings support our hypothesis that diverse polar phytoplankton suppress energetically wasteful photochemical misses, while growing under low temperature, thereby supporting their photosynthesis even under extremely low growth light. Sustaining PSII function even under widely spaced excitations appears as an emergent property supporting the productivity of polar phytoplankton under the ice during the polar night.

Salinities vary widely across the habitats of the tested *Chlamydomonas* taxa, and fluctuate within some habitats during ice formation and melting (Morgan-Kiss et al., 2016; Smoot and Hopcroft, 2017; Carton et al., 2015). Higher salinity may have photoprotective properties for the polar taxa of *Chlamydomonas* (Kalra et al., 2020; Kalra et al., 2023; Stahl-Rommel et al., 2022). NaCl concentration showed a co-variate influence on dampening

indices, when analyzed across all tested *Chlamydomonas* taxa, but when tested within taxa, NaCl only showed an influence within *Chlamydomonas priscui*. Out of all the *Chlamydomonas* species used here, *Chlamydomonas priscui* exhibits the widest range of permissive growth salinities (10 mM - 1.2 M). Salinity significantly affects photosystem partitioning and thylakoid organization in *Chlamydomonas priscui* (Szyszka-Mroz et al., 2019). Thus the detected influence of changing salinity on PSII synchronization in *Chlamydomonas priscui* is consistent with its photophysiological responses to changing salinity. It is important to note that faster desynchronization of S-State cycling does not indicate a loss nor instability of PSII function *per se*. Desynchronization rather reflects randomization of the S-States across the PSII population. Prolonged synchronization under low effective light levels is thus an indicator of the capacity of PSII to exploit widely spaced photons under low light, not the stability of PSII function itself.

Field results show that net photosynthetic accumulation is possible very early in the Arctic season under ice, at photon fluxes so low that they push the boundaries of our knowledge of photobiology (Randelhoff et al., 2020; Hoppe et al., 2024). With global warming, the underwater light climate is expected to change and likely increase the time window of light available under the ice during spring (Kristiansen et al., 2025; Levis et al., 2020), which would have an impact on the phenology of spring blooms as well as the growth dynamics between the main phytoplankton groups, which are mainly controlled by the availability of light during this season (Benoît-Gagné et al., 2024; Plassart et al., 2026).

Understanding and detecting mechanisms enabling polar phytoplankton to sustain slow, but significant, productivity under the ice in the winter can help predict the changing

dynamics of spring phytoplankton blooms, in the face of rapid warming (Ardyna and Arrigo, 2020). Beyond direct temperature changes, polar aquatic ecosystems are experiencing declines in sea ice extent and thickness, escalating freshwater inputs, acidification, and increased winds and storms (Ardyna and Arrigo, 2020; Cvetkovska et al., 2022). These pressures are altering the productivity and seasonal peaks of phytoplankton blooms (Ardyna and Arrigo, 2020). Maintaining photosynthetic energy conversion over winter (Morin et al., 2020) is crucial for the timing and speed of spring bloom initiation (Hancke et al., 2018). Our findings indicate that low growth temperature itself is a key factor enabling polar phytoplankton to exploit extreme low light. Taxa with the capacity for low temperature & low light growth then possess a competitive advantage to quickly initiate more rapid spring growth, giving them first access to the nutrients required to form an extensive bloom and in turn, exert bottom-up effects on changing polar ecosystems (Ardyna and Arrigo, 2020). We suggest that low temperatures may actually support phytoplankton survival of the polar night.

Supporting Information

Table 4: Single turnover saturating flash measurement conditions, with key growth conditions, equivalent effective light levels ($\mu\text{mol photons m}^{-2}\text{s}^{-1}$, estimated from flash spacings (s) and σ_{PSII}), mean effective absorption cross section of photosystem II, σ_{PSII} (A^2) and mean photochemical yield of photosystem II, F_V/F_M , for each measurement

Taxa	Flash Spacing (s)	Temp (°C)	Light	Growth Temp (°C)	NaCl (mM)	σ_{PSII}	F _V /F _M
<i>C. euryale</i>	1	16	0.71	24	30	238	0.47
<i>C. euryale</i>	1	16	0.74	24	70	224	0.48
<i>C. euryale</i>	1	16	0.76	24	500	222	0.44
<i>C. euryale</i>	1	20	0.70	24	30	242	0.45
<i>C. euryale</i>	1	20	0.71	24	70	239	0.46
<i>C. euryale</i>	1	20	0.75	24	500	231	0.45
<i>C. euryale</i>	1	24	0.72	24	30	232	0.43
<i>C. euryale</i>	1	24	0.75	24	70	223	0.62
<i>C. euryale</i>	1	24	0.76	24	500	224	0.55
<i>C. euryale</i>	1	28	0.69	24	30	248	0.43
<i>C. euryale</i>	1	28	0.71	24	70	238	0.44
<i>C. euryale</i>	1	28	0.72	24	500	234	0.42
<i>C. euryale</i>	2	16	0.35	24	30	234	0.46
<i>C. euryale</i>	2	16	0.37	24	500	224	0.44
<i>C. euryale</i>	2	20	0.34	24	30	241	0.45
<i>C. euryale</i>	2	20	0.35	24	70	241	0.46
<i>C. euryale</i>	2	20	0.36	24	500	234	0.45
<i>C. euryale</i>	2	24	0.37	24	70	228	0.62

Taxa	Flash Spacing (s)	Temp (°C)	Light	Growth Temp (°C)	NaCl (mM)	σ_{PSII}	F _V /F _M
<i>C. euryale</i>	2	24	0.37	24	500	225	0.54
<i>C. euryale</i>	2	24	0.37	24	30	227	0.42
<i>C. euryale</i>	2	28	0.32	24	30	262	0.43
<i>C. euryale</i>	2	28	0.35	24	70	239	0.43
<i>C. euryale</i>	2	28	0.35	24	500	236	0.42
<i>C. euryale</i>	4	16	0.17	24	30	245	0.47
<i>C. euryale</i>	4	16	0.18	24	70	232	0.48
<i>C. euryale</i>	4	16	0.18	24	500	229	0.46
<i>C. euryale</i>	4	20	0.17	24	30	248	0.45
<i>C. euryale</i>	4	20	0.17	24	70	246	0.46
<i>C. euryale</i>	4	20	0.18	24	500	235	0.45
<i>C. euryale</i>	4	24	0.18	24	30	235	0.43
<i>C. euryale</i>	4	24	0.18	24	70	226	0.63
<i>C. euryale</i>	4	24	0.18	24	500	232	0.54
<i>C. euryale</i>	4	28	0.18	24	70	244	0.44
<i>C. euryale</i>	4	28	0.18	24	500	238	0.42
<i>C. euryale</i>	8	16	0.08	24	30	254	0.47
<i>C. euryale</i>	8	16	0.09	24	70	238	0.48

Taxa	Flash Spacing (s)	Temp (°C)	Light	Growth Temp (°C)	NaCl (mM)	σ_{PSII}	F _V /F _M
<i>C. euryale</i>	8	16	0.09	24	500	233	0.46
<i>C. euryale</i>	8	20	0.08	24	70	250	0.46
<i>C. euryale</i>	8	20	0.09	24	30	247	0.45
<i>C. euryale</i>	8	20	0.09	24	500	244	0.45
<i>C. euryale</i>	8	24	0.09	24	70	242	0.64
<i>C. euryale</i>	8	24	0.09	24	500	225	0.55
<i>C. euryale</i>	8	24	0.09	24	30	230	0.43
<i>C. euryale</i>	8	28	0.08	24	30	255	0.46
<i>C. euryale</i>	8	28	0.09	24	70	237	0.46
<i>C. euryale</i>	8	28	0.09	24	500	240	0.44
<i>C. euryale</i>	16	16	0.04	24	30	239	0.47
<i>C. euryale</i>	16	16	0.04	24	70	237	0.48
<i>C. euryale</i>	16	16	0.05	24	500	236	0.46
<i>C. euryale</i>	16	20	0.04	24	70	245	0.47
<i>C. euryale</i>	16	20	0.04	24	500	243	0.46
<i>C. euryale</i>	16	20	0.04	24	30	242	0.47
<i>C. euryale</i>	16	24	0.05	24	70	240	0.65
<i>C. euryale</i>	16	24	0.05	24	500	237	0.57

Taxa	Flash Spacing (s)	Temp (°C)	Light	Growth Temp (°C)	NaCl (mM)	σ_{PSII}	F _V /F _M
<i>C. euryale</i>	16	24	0.05	24	30	226	0.44
<i>C. euryale</i>	16	28	0.04	24	30	248	0.47
<i>C. euryale</i>	16	28	0.05	24	500	232	0.46
<i>C. euryale</i>	16	28	0.05	24	70	233	0.48
<i>C. ICEMDV</i>	1	4	0.74	4	70	227	0.65
<i>C. ICEMDV</i>	1	4	0.74	4	70	224	0.61
<i>C. ICEMDV</i>	1	8	0.72	4	70	227	0.61
<i>C. ICEMDV</i>	1	8	0.72	4	70	239	0.60
<i>C. ICEMDV</i>	1	12	0.68	4	70	234	0.63
<i>C. ICEMDV</i>	1	12	0.68	4	70	258	0.63
<i>C. ICEMDV</i>	2	4	0.36	4	70	229	0.66
<i>C. ICEMDV</i>	2	4	0.36	4	70	229	0.61
<i>C. ICEMDV</i>	2	8	0.35	4	70	228	0.62
<i>C. ICEMDV</i>	2	8	0.35	4	70	241	0.60
<i>C. ICEMDV</i>	2	12	0.34	4	70	236	0.64
<i>C. ICEMDV</i>	2	12	0.34	4	70	261	0.64
<i>C. ICEMDV</i>	4	4	0.18	4	70	231	0.66
<i>C. ICEMDV</i>	4	4	0.18	4	70	232	0.61

Taxa	Flash Spacing (s)	Temp (°C)	Light	Growth Temp (°C)	NaCl (mM)	σ_{PSII}	F _V /F _M
<i>C. ICEMDV</i>	4	8	0.18	4	70	229	0.62
<i>C. ICEMDV</i>	4	8	0.18	4	70	244	0.60
<i>C. ICEMDV</i>	4	12	0.17	4	70	240	0.64
<i>C. ICEMDV</i>	4	12	0.17	4	70	264	0.64
<i>C. ICEMDV</i>	8	4	0.09	4	70	235	0.66
<i>C. ICEMDV</i>	8	4	0.09	4	70	233	0.61
<i>C. ICEMDV</i>	8	8	0.09	4	70	229	0.62
<i>C. ICEMDV</i>	8	8	0.09	4	70	248	0.60
<i>C. ICEMDV</i>	8	12	0.08	4	70	241	0.64
<i>C. ICEMDV</i>	8	12	0.08	4	70	264	0.64
<i>C. ICEMDV</i>	16	4	0.04	4	70	229	0.66
<i>C. ICEMDV</i>	16	4	0.04	4	70	242	0.61
<i>C. ICEMDV</i>	16	8	0.04	4	70	229	0.62
<i>C. ICEMDV</i>	16	8	0.04	4	70	250	0.61
<i>C. ICEMDV</i>	16	12	0.04	4	70	238	0.64
<i>C. ICEMDV</i>	16	12	0.04	4	70	261	0.65
<i>C. klinobasi s</i>	1	2	0.75	4	10	225	0.73

Taxa	Flash Spacing (s)	Temp (°C)	Light	Growth Temp (°C)	NaCl (mM)	σ_{PSII}	F _V /F _M
<i>C. klinobasi s</i>	1	2	0.75	4	0	224	0.73
<i>C. klinobasi s</i>	1	2	0.80	4	30	208	0.70
<i>C. klinobasi s</i>	1	4	0.74	4	0	226	0.78
<i>C. klinobasi s</i>	1	4	0.75	4	10	222	0.74
<i>C. klinobasi s</i>	1	4	0.80	4	30	208	0.71
<i>C. klinobasi s</i>	1	8	0.72	4	0	236	0.70
<i>C. klinobasi s</i>	1	8	0.73	4	10	228	0.72
<i>C. klinobasi s</i>	1	8	0.77	4	30	218	0.69
<i>C. klinobasi s</i>	1	12	0.70	4	0	241	0.70
<i>C. klinobasi s</i>	1	12	0.70	4	10	239	0.69
<i>C. klinobasi s</i>	1	12	0.74	4	30	224	0.67
<i>C. klinobasi s</i>	1	16	0.79	4	30	211	0.53
<i>C. klinobasi s</i>	2	2	0.38	4	10	220	0.73

Taxa	Flash Spacing (s)	Temp (°C)	Light	Growth Temp (°C)	NaCl (mM)	σ_{PSII}	F_V/F_M
<i>C. klinobasis</i>	2	2	0.38	4	0	220	0.73
<i>C. klinobasis</i>	2	2	0.41	4	30	204	0.71
<i>C. klinobasis</i>	2	4	0.38	4	0	221	0.78
<i>C. klinobasis</i>	2	4	0.38	4	10	220	0.74
<i>C. klinobasis</i>	2	4	0.40	4	30	206	0.71
<i>C. klinobasis</i>	2	8	0.36	4	0	233	0.70
<i>C. klinobasis</i>	2	8	0.37	4	10	228	0.71
<i>C. klinobasis</i>	2	8	0.39	4	30	215	0.69
<i>C. klinobasis</i>	4	2	0.19	4	0	220	0.73
<i>C. klinobasis</i>	4	2	0.19	4	10	220	0.73
<i>C. klinobasis</i>	4	2	0.20	4	30	207	0.71
<i>C. klinobasis</i>	4	4	0.19	4	0	222	0.77
<i>C. klinobasis</i>	4	4	0.19	4	10	222	0.74

Taxa	Flash Spacing (s)	Temp (°C)	Light	Growth Temp (°C)	NaCl (mM)	σ_{PSII}	F _V /F _M
<i>C. klinobasis</i>	4	4	0.20	4	30	208	0.71
<i>C. klinobasis</i>	4	8	0.18	4	0	231	0.70
<i>C. klinobasis</i>	4	8	0.18	4	10	229	0.71
<i>C. klinobasis</i>	4	8	0.19	4	30	218	0.69
<i>C. klinobasis</i>	4	12	0.18	4	10	238	0.69
<i>C. klinobasis</i>	4	12	0.18	4	0	238	0.70
<i>C. klinobasis</i>	4	12	0.18	4	30	228	0.67
<i>C. klinobasis</i>	4	16	0.19	4	30	219	0.53
<i>C. klinobasis</i>	8	2	0.09	4	10	220	0.73
<i>C. klinobasis</i>	8	2	0.10	4	0	220	0.72
<i>C. klinobasis</i>	8	2	0.10	4	30	209	0.71
<i>C. klinobasis</i>	8	4	0.09	4	10	226	0.74
<i>C. klinobasis</i>	8	4	0.09	4	0	224	0.77

Taxa	Flash Spacing (s)	Temp (°C)	Light	Growth Temp (°C)	NaCl (mM)	σ_{PSII}	F _V /F _M
<i>C. klinobasis</i>	8	4	0.10	4	30	214	0.70
<i>C. klinobasis</i>	8	8	0.09	4	0	234	0.69
<i>C. klinobasis</i>	8	8	0.09	4	10	234	0.70
<i>C. klinobasis</i>	8	8	0.09	4	30	226	0.69
<i>C. klinobasis</i>	8	12	0.09	4	10	244	0.67
<i>C. klinobasis</i>	8	12	0.09	4	0	236	0.69
<i>C. klinobasis</i>	8	12	0.09	4	30	234	0.67
<i>C. klinobasis</i>	8	16	0.09	4	30	221	0.53
<i>C. klinobasis</i>	16	2	0.05	4	10	225	0.72
<i>C. klinobasis</i>	16	2	0.05	4	0	222	0.72
<i>C. klinobasis</i>	16	2	0.05	4	30	213	0.70
<i>C. klinobasis</i>	16	4	0.05	4	10	227	0.73
<i>C. klinobasis</i>	16	4	0.05	4	0	225	0.76

Taxa	Flash Spacing (s)	Temp (°C)	Light	Growth Temp (°C)	NaCl (mM)	σ_{PSII}	F _V /F _M
<i>C. klinobasis</i>	16	4	0.05	4	30	218	0.70
<i>C. klinobasis</i>	16	8	0.04	4	10	237	0.69
<i>C. klinobasis</i>	16	8	0.04	4	0	233	0.69
<i>C. klinobasis</i>	16	8	0.05	4	30	227	0.69
<i>C. klinobasis</i>	16	12	0.04	4	10	240	0.68
<i>C. klinobasis</i>	16	12	0.04	4	30	236	0.68
<i>C. klinobasis</i>	16	12	0.04	4	0	235	0.69
<i>C. klinobasis</i>	16	16	0.05	4	30	221	0.53
<i>C. malina</i>	1	2	0.64	4	500	258	0.57
<i>C. malina</i>	1	2	0.74	4	70	222	0.63
<i>C. malina</i>	1	2	0.78	4	30	211	0.69
<i>C. malina</i>	1	4	0.65	4	70	253	0.66
<i>C. malina</i>	1	4	0.65	4	70	259	0.62
<i>C. malina</i>	1	4	0.66	4	500	254	0.56
<i>C. malina</i>	1	4	0.72	4	30	229	0.70
<i>C. malina</i>	1	4	0.74	4	70	225	0.72
<i>C. malina</i>	1	8	0.59	4	500	282	0.55
<i>C. malina</i>	1	8	0.66	4	70	244	0.62
<i>C. malina</i>	1	8	0.66	4	70	263	0.60

Taxa	Flash Spacing (s)	Temp (°C)	Light	Growth Temp (°C)	NaCl (mM)	σ_{PSII}	F _V /F _M
<i>C. malina</i>	1	8	0.70	4	30	242	0.65
<i>C. malina</i>	1	8	0.70	4	70	239	0.68
<i>C. malina</i>	1	12	0.57	4	500	295	0.53
<i>C. malina</i>	1	12	0.60	4	70	259	0.65
<i>C. malina</i>	1	12	0.60	4	70	296	0.64
<i>C. malina</i>	1	12	0.69	4	30	242	0.68
<i>C. malina</i>	1	12	0.70	4	70	239	0.64
<i>C. malina</i>	2	2	0.32	4	500	256	0.57
<i>C. malina</i>	2	2	0.38	4	70	222	0.64
<i>C. malina</i>	2	2	0.40	4	30	211	0.69
<i>C. malina</i>	2	4	0.32	4	500	262	0.57
<i>C. malina</i>	2	4	0.32	4	70	254	0.67
<i>C. malina</i>	2	4	0.32	4	70	261	0.62
<i>C. malina</i>	2	4	0.37	4	30	225	0.70
<i>C. malina</i>	2	4	0.37	4	70	222	0.72
<i>C. malina</i>	2	8	0.30	4	500	278	0.55
<i>C. malina</i>	2	8	0.32	4	70	251	0.63
<i>C. malina</i>	2	8	0.32	4	70	265	0.61
<i>C. malina</i>	2	8	0.35	4	30	237	0.65
<i>C. malina</i>	2	8	0.36	4	70	237	0.69
<i>C. malina</i>	2	12	0.30	4	70	263	0.65
<i>C. malina</i>	2	12	0.30	4	70	298	0.65
<i>C. malina</i>	4	2	0.16	4	500	255	0.57
<i>C. malina</i>	4	2	0.19	4	70	213	0.64
<i>C. malina</i>	4	2	0.20	4	30	210	0.69
<i>C. malina</i>	4	4	0.16	4	70	254	0.66
<i>C. malina</i>	4	4	0.16	4	70	262	0.62
<i>C. malina</i>	4	4	0.16	4	500	254	0.57

Taxa	Flash Spacing (s)	Temp (°C)	Light	Growth Temp (°C)	NaCl (mM)	σ_{PSII}	F_v/F_m
<i>C. malina</i>	4	4	0.19	4	30	221	0.70
<i>C. malina</i>	4	4	0.19	4	70	224	0.71
<i>C. malina</i>	4	8	0.15	4	500	278	0.55
<i>C. malina</i>	4	8	0.16	4	70	255	0.64
<i>C. malina</i>	4	8	0.16	4	70	265	0.61
<i>C. malina</i>	4	8	0.18	4	70	236	0.70
<i>C. malina</i>	4	8	0.18	4	30	235	0.65
<i>C. malina</i>	4	12	0.15	4	500	287	0.53
<i>C. malina</i>	4	12	0.15	4	70	265	0.65
<i>C. malina</i>	4	12	0.15	4	70	300	0.65
<i>C. malina</i>	4	12	0.17	4	70	246	0.64
<i>C. malina</i>	4	12	0.17	4	30	238	0.67
<i>C. malina</i>	8	2	0.08	4	500	252	0.56
<i>C. malina</i>	8	2	0.10	4	30	208	0.69
<i>C. malina</i>	8	2	0.10	4	70	204	0.63
<i>C. malina</i>	8	4	0.08	4	70	256	0.67
<i>C. malina</i>	8	4	0.08	4	70	263	0.63
<i>C. malina</i>	8	4	0.08	4	500	254	0.57
<i>C. malina</i>	8	4	0.09	4	30	228	0.70
<i>C. malina</i>	8	4	0.09	4	70	225	0.71
<i>C. malina</i>	8	8	0.07	4	500	286	0.54
<i>C. malina</i>	8	8	0.08	4	70	263	0.61
<i>C. malina</i>	8	8	0.09	4	30	239	0.65
<i>C. malina</i>	8	8	0.09	4	70	239	0.69
<i>C. malina</i>	8	12	0.07	4	500	288	0.52
<i>C. malina</i>	8	12	0.07	4	70	266	0.65
<i>C. malina</i>	8	12	0.07	4	70	301	0.65
<i>C. malina</i>	8	12	0.08	4	30	253	0.67

Taxa	Flash Spacing (s)	Temp (°C)	Light	Growth Temp (°C)	NaCl (mM)	σ_{PSII}	F _V /F _M
<i>C. malina</i>	8	12	0.09	4	70	243	0.65
<i>C. malina</i>	16	2	0.05	4	30	210	0.68
<i>C. malina</i>	16	2	0.05	4	70	210	0.63
<i>C. malina</i>	16	4	0.04	4	70	255	0.67
<i>C. malina</i>	16	4	0.04	4	70	266	0.65
<i>C. malina</i>	16	4	0.04	4	500	267	0.56
<i>C. malina</i>	16	4	0.04	4	30	232	0.69
<i>C. malina</i>	16	4	0.05	4	70	231	0.71
<i>C. malina</i>	16	8	0.04	4	500	290	0.53
<i>C. malina</i>	16	8	0.04	4	70	246	0.65
<i>C. malina</i>	16	8	0.04	4	70	259	0.63
<i>C. malina</i>	16	8	0.04	4	30	246	0.65
<i>C. malina</i>	16	8	0.04	4	70	245	0.69
<i>C. malina</i>	16	12	0.03	4	500	305	0.52
<i>C. malina</i>	16	12	0.04	4	70	267	0.65
<i>C. malina</i>	16	12	0.04	4	70	300	0.64
<i>C. malina</i>	16	12	0.04	4	30	260	0.67
<i>C. malina</i>	16	12	0.04	4	70	248	0.65
<i>C. priscui</i>	1	2	0.68	4	500	245	0.59
<i>C. priscui</i>	1	2	0.79	4	70	213	0.65
<i>C. priscui</i>	1	2	0.82	4	30	203	0.69
<i>C. priscui</i>	1	4	0.71	4	500	238	0.59
<i>C. priscui</i>	1	4	0.77	4	70	209	0.64
<i>C. priscui</i>	1	4	0.77	4	70	213	0.62
<i>C. priscui</i>	1	4	0.77	4	700	224	0.45
<i>C. priscui</i>	1	4	0.78	4	70	212	0.64
<i>C. priscui</i>	1	4	0.81	4	30	204	0.70
<i>C. priscui</i>	1	8	0.65	4	500	259	0.58

Taxa	Flash Spacing (s)	Temp (°C)	Light	Growth Temp (°C)	NaCl (mM)	σ_{PSII}	F _V /F _M
<i>C. priscui</i>	1	8	0.70	4	70	240	0.65
<i>C. priscui</i>	1	8	0.70	4	70	229	0.61
<i>C. priscui</i>	1	8	0.70	4	700	249	0.45
<i>C. priscui</i>	1	8	0.76	4	70	219	0.65
<i>C. priscui</i>	1	8	0.76	4	30	220	0.65
<i>C. priscui</i>	1	12	0.65	4	70	234	0.63
<i>C. priscui</i>	1	12	0.65	4	70	263	0.63
<i>C. priscui</i>	1	12	0.65	4	700	269	0.43
<i>C. priscui</i>	1	12	0.67	4	500	251	0.48
<i>C. priscui</i>	1	12	0.79	4	30	213	0.64
<i>C. priscui</i>	1	12	0.80	4	70	209	0.71
<i>C. priscui</i>	1	16	0.65	4	500	257	0.47
<i>C. priscui</i>	1	16	0.67	4	70	249	0.45
<i>C. priscui</i>	1	16	0.79	4	30	210	0.52
<i>C. priscui</i>	2	2	0.35	4	500	238	0.60
<i>C. priscui</i>	2	2	0.40	4	70	202	0.65
<i>C. priscui</i>	2	2	0.42	4	30	200	0.69
<i>C. priscui</i>	2	4	0.34	4	500	250	0.58
<i>C. priscui</i>	2	4	0.38	4	70	208	0.64
<i>C. priscui</i>	2	4	0.38	4	70	215	0.63
<i>C. priscui</i>	2	4	0.38	4	700	225	0.46
<i>C. priscui</i>	2	4	0.40	4	70	212	0.64
<i>C. priscui</i>	2	4	0.41	4	30	206	0.70
<i>C. priscui</i>	2	8	0.33	4	500	252	0.58
<i>C. priscui</i>	2	8	0.35	4	70	231	0.64
<i>C. priscui</i>	2	8	0.35	4	70	231	0.61
<i>C. priscui</i>	2	8	0.35	4	700	250	0.45
<i>C. priscui</i>	2	8	0.38	4	30	221	0.65

Taxa	Flash Spacing (s)	Temp (°C)	Light	Growth Temp (°C)	NaCl (mM)	σ_{PSII}	F _V /F _M
<i>C. priscui</i>	2	8	0.38	4	70	216	0.66
<i>C. priscui</i>	2	12	0.32	4	70	259	0.55
<i>C. priscui</i>	2	12	0.33	4	70	235	0.63
<i>C. priscui</i>	2	12	0.33	4	70	263	0.64
<i>C. priscui</i>	2	12	0.33	4	700	268	0.43
<i>C. priscui</i>	2	12	0.35	4	30	239	0.62
<i>C. priscui</i>	2	12	0.37	4	500	227	0.56
<i>C. priscui</i>	2	16	0.31	4	500	275	0.53
<i>C. priscui</i>	2	16	0.36	4	70	235	0.58
<i>C. priscui</i>	2	16	0.39	4	70	215	0.53
<i>C. priscui</i>	2	16	0.39	4	30	211	0.51
<i>C. priscui</i>	2	16	0.51	4	500	162	0.54
<i>C. priscui</i>	4	2	0.18	4	500	237	0.60
<i>C. priscui</i>	4	2	0.20	4	70	207	0.65
<i>C. priscui</i>	4	2	0.21	4	30	199	0.69
<i>C. priscui</i>	4	4	0.18	4	500	229	0.60
<i>C. priscui</i>	4	4	0.19	4	70	206	0.64
<i>C. priscui</i>	4	4	0.19	4	70	215	0.63
<i>C. priscui</i>	4	4	0.19	4	700	227	0.46
<i>C. priscui</i>	4	4	0.20	4	70	211	0.64
<i>C. priscui</i>	4	4	0.20	4	30	206	0.70
<i>C. priscui</i>	4	8	0.16	4	500	247	0.58
<i>C. priscui</i>	4	8	0.18	4	70	231	0.64
<i>C. priscui</i>	4	8	0.18	4	70	231	0.61
<i>C. priscui</i>	4	8	0.18	4	700	251	0.45
<i>C. priscui</i>	4	8	0.19	4	30	225	0.65
<i>C. priscui</i>	4	8	0.19	4	70	217	0.66
<i>C. priscui</i>	4	12	0.16	4	70	236	0.62

Taxa	Flash Spacing (s)	Temp (°C)	Light	Growth Temp (°C)	NaCl (mM)	σ_{PSII}	F _V /F _M
<i>C. priscui</i>	4	12	0.16	4	70	263	0.64
<i>C. priscui</i>	4	12	0.16	4	700	268	0.43
<i>C. priscui</i>	4	12	0.17	4	500	248	0.47
<i>C. priscui</i>	4	12	0.19	4	30	218	0.63
<i>C. priscui</i>	4	12	0.20	4	70	214	0.70
<i>C. priscui</i>	4	16	0.16	4	500	257	0.47
<i>C. priscui</i>	4	16	0.17	4	70	254	0.46
<i>C. priscui</i>	4	16	0.19	4	30	216	0.53
<i>C. priscui</i>	8	2	0.09	4	500	231	0.60
<i>C. priscui</i>	8	2	0.10	4	70	205	0.65
<i>C. priscui</i>	8	2	0.10	4	30	203	0.69
<i>C. priscui</i>	8	4	0.09	4	500	232	0.59
<i>C. priscui</i>	8	4	0.10	4	70	208	0.64
<i>C. priscui</i>	8	4	0.10	4	70	215	0.63
<i>C. priscui</i>	8	4	0.10	4	700	230	0.47
<i>C. priscui</i>	8	4	0.10	4	30	209	0.70
<i>C. priscui</i>	8	4	0.10	4	70	208	0.64
<i>C. priscui</i>	8	8	0.08	4	500	254	0.59
<i>C. priscui</i>	8	8	0.09	4	70	232	0.64
<i>C. priscui</i>	8	8	0.09	4	70	232	0.61
<i>C. priscui</i>	8	8	0.09	4	700	251	0.45
<i>C. priscui</i>	8	8	0.09	4	30	229	0.64
<i>C. priscui</i>	8	8	0.10	4	70	219	0.66
<i>C. priscui</i>	8	12	0.08	4	70	234	0.62
<i>C. priscui</i>	8	12	0.08	4	70	263	0.63
<i>C. priscui</i>	8	12	0.08	4	700	267	0.43
<i>C. priscui</i>	8	12	0.08	4	500	252	0.47
<i>C. priscui</i>	8	12	0.09	4	30	224	0.62

Taxa	Flash Spacing (s)	Temp (°C)	Light	Growth Temp (°C)	NaCl (mM)	σ_{PSII}	F _V /F _M
<i>C. priscui</i>	8	12	0.10	4	70	225	0.69
<i>C. priscui</i>	8	16	0.08	4	500	254	0.47
<i>C. priscui</i>	8	16	0.08	4	70	252	0.46
<i>C. priscui</i>	8	16	0.09	4	30	223	0.52
<i>C. priscui</i>	16	2	0.05	4	500	223	0.61
<i>C. priscui</i>	16	2	0.05	4	70	204	0.64
<i>C. priscui</i>	16	2	0.05	4	30	206	0.69
<i>C. priscui</i>	16	4	0.05	4	70	220	0.63
<i>C. priscui</i>	16	4	0.05	4	700	232	0.47
<i>C. priscui</i>	16	4	0.05	4	500	224	0.61
<i>C. priscui</i>	16	4	0.05	4	30	214	0.69
<i>C. priscui</i>	16	4	0.05	4	70	215	0.64
<i>C. priscui</i>	16	8	0.04	4	500	251	0.58
<i>C. priscui</i>	16	8	0.04	4	70	231	0.64
<i>C. priscui</i>	16	8	0.04	4	70	222	0.63
<i>C. priscui</i>	16	8	0.04	4	700	253	0.46
<i>C. priscui</i>	16	8	0.04	4	30	236	0.64
<i>C. priscui</i>	16	8	0.04	4	70	232	0.65
<i>C. priscui</i>	16	12	0.04	4	500	249	0.47
<i>C. priscui</i>	16	12	0.04	4	70	226	0.61
<i>C. priscui</i>	16	12	0.04	4	70	257	0.63
<i>C. priscui</i>	16	12	0.04	4	700	261	0.44
<i>C. priscui</i>	16	12	0.05	4	70	232	0.70
<i>C. priscui</i>	16	12	0.05	4	30	228	0.64
<i>C. priscui</i>	16	16	0.04	4	500	245	0.47
<i>C. priscui</i>	16	16	0.04	4	70	245	0.46
<i>C. priscui</i>	16	16	0.05	4	30	224	0.52
<i>C. priscui</i>	NA	4	NA	4	70	205	0.64

Taxa	Flash Spacing (s)	Temp (°C)	Light	Growth Temp (°C)	NaCl (mM)	σ_{PSII}	F_V/F_M
<i>C. reinhardt ii</i>	1	12	0.83	24	0	199	0.62
<i>C. reinhardt ii</i>	1	12	0.83	24	0	203	0.62
<i>C. reinhardt ii</i>	1	12	1.00	24	10	164	0.83
<i>C. reinhardt ii</i>	1	12	1.01	24	30	164	0.81
<i>C. reinhardt ii</i>	1	12	1.03	24	0	163	0.80
<i>C. reinhardt ii</i>	1	16	0.78	24	0	222	0.62
<i>C. reinhardt ii</i>	1	16	0.78	24	0	209	0.61
<i>C. reinhardt ii</i>	1	16	0.96	24	0	161	0.58
<i>C. reinhardt ii</i>	1	16	1.06	24	10	157	0.58
<i>C. reinhardt ii</i>	1	16	1.07	24	30	155	0.58
<i>C. reinhardt ii</i>	1	20	0.81	24	0	216	0.59
<i>C. reinhardt ii</i>	1	20	0.81	24	0	200	0.57
<i>C. reinhardt ii</i>	1	20	1.03	24	0	164	0.56

Taxa	Flash Spacing (s)	Temp (°C)	Light	Growth Temp (°C)	NaCl (mM)	σ_{PSII}	F _V /F _M
<i>C. reinhardt ii</i>	1	20	1.04	24	10	162	0.57
<i>C. reinhardt ii</i>	1	20	1.05	24	30	159	0.56
<i>C. reinhardt ii</i>	1	24	0.84	24	0	197	0.55
<i>C. reinhardt ii</i>	1	24	0.84	24	0	202	0.55
<i>C. reinhardt ii</i>	1	24	0.92	24	10	167	0.56
<i>C. reinhardt ii</i>	1	24	1.02	24	30	163	0.54
<i>C. reinhardt ii</i>	1	24	1.03	24	0	161	0.55
<i>C. reinhardt ii</i>	1	28	0.96	24	30	173	0.50
<i>C. reinhardt ii</i>	1	28	1.00	24	0	168	0.53
<i>C. reinhardt ii</i>	1	28	1.02	24	10	162	0.61
<i>C. reinhardt ii</i>	1	28	1.02	24	30	162	0.62
<i>C. reinhardt ii</i>	2	12	0.41	24	0	203	0.62
<i>C. reinhardt ii</i>	2	12	0.41	24	0	204	0.63

Taxa	Flash Spacing (s)	Temp (°C)	Light	Growth Temp (°C)	NaCl (mM)	σ_{PSII}	F _V /F _M
<i>C. reinhardt ii</i>	2	16	0.38	24	0	226	0.62
<i>C. reinhardt ii</i>	2	16	0.38	24	0	212	0.61
<i>C. reinhardt ii</i>	2	16	0.45	24	30	172	0.56
<i>C. reinhardt ii</i>	2	16	0.46	24	10	169	0.56
<i>C. reinhardt ii</i>	2	16	0.51	24	0	162	0.55
<i>C. reinhardt ii</i>	2	20	0.40	24	0	220	0.60
<i>C. reinhardt ii</i>	2	20	0.40	24	0	200	0.58
<i>C. reinhardt ii</i>	2	20	0.49	24	30	172	0.53
<i>C. reinhardt ii</i>	2	20	0.49	24	10	169	0.54
<i>C. reinhardt ii</i>	2	20	0.51	24	0	161	0.53
<i>C. reinhardt ii</i>	2	24	0.41	24	0	202	0.56
<i>C. reinhardt ii</i>	2	24	0.41	24	0	206	0.56
<i>C. reinhardt ii</i>	2	24	0.49	24	10	171	0.53

Taxa	Flash Spacing (s)	Temp (°C)	Light	Growth Temp (°C)	NaCl (mM)	σ_{PSII}	F _V /F _M
<i>C. reinhardt ii</i>	2	24	0.50	24	30	170	0.52
<i>C. reinhardt ii</i>	2	24	0.50	24	0	166	0.53
<i>C. reinhardt ii</i>	2	28	0.48	24	30	178	0.50
<i>C. reinhardt ii</i>	2	28	0.49	24	10	170	0.51
<i>C. reinhardt ii</i>	2	28	0.50	24	0	168	0.49
<i>C. reinhardt ii</i>	4	12	0.20	24	0	206	0.62
<i>C. reinhardt ii</i>	4	12	0.20	24	0	209	0.63
<i>C. reinhardt ii</i>	4	12	0.25	24	10	169	0.84
<i>C. reinhardt ii</i>	4	12	0.25	24	30	166	0.82
<i>C. reinhardt ii</i>	4	12	0.26	24	0	162	0.82
<i>C. reinhardt ii</i>	4	16	0.19	24	0	229	0.63
<i>C. reinhardt ii</i>	4	16	0.19	24	0	219	0.62
<i>C. reinhardt ii</i>	4	16	0.26	24	0	164	0.57

Taxa	Flash Spacing (s)	Temp (°C)	Light	Growth Temp (°C)	NaCl (mM)	σ_{PSII}	F _V /F _M
<i>C. reinhardt ii</i>	4	16	0.27	24	10	158	0.58
<i>C. reinhardt ii</i>	4	16	0.27	24	30	155	0.57
<i>C. reinhardt ii</i>	4	20	0.20	24	0	227	0.60
<i>C. reinhardt ii</i>	4	20	0.20	24	0	198	0.58
<i>C. reinhardt ii</i>	4	20	0.24	24	0	165	0.55
<i>C. reinhardt ii</i>	4	20	0.25	24	10	168	0.57
<i>C. reinhardt ii</i>	4	20	0.26	24	30	159	0.56
<i>C. reinhardt ii</i>	4	24	0.20	24	0	202	0.56
<i>C. reinhardt ii</i>	4	24	0.20	24	0	211	0.57
<i>C. reinhardt ii</i>	4	24	0.23	24	0	170	0.55
<i>C. reinhardt ii</i>	4	24	0.24	24	10	176	0.57
<i>C. reinhardt ii</i>	4	24	0.25	24	30	166	0.54
<i>C. reinhardt ii</i>	4	28	0.25	24	10	171	0.53

Taxa	Flash Spacing (s)	Temp (°C)	Light	Growth Temp (°C)	NaCl (mM)	σ_{PSII}	F _V /F _M
<i>C. reinhardtii</i>	4	28	0.25	24	30	167	0.64
<i>C. reinhardtii</i>	4	28	0.25	24	0	166	0.53
<i>C. reinhardtii</i>	8	12	0.10	24	0	210	0.63
<i>C. reinhardtii</i>	8	12	0.10	24	0	213	0.63
<i>C. reinhardtii</i>	8	12	0.12	24	10	169	0.87
<i>C. reinhardtii</i>	8	12	0.12	24	30	167	0.83
<i>C. reinhardtii</i>	8	12	0.13	24	0	167	0.86
<i>C. reinhardtii</i>	8	16	0.09	24	0	232	0.63
<i>C. reinhardtii</i>	8	16	0.09	24	0	222	0.62
<i>C. reinhardtii</i>	8	16	0.12	24	0	164	0.57
<i>C. reinhardtii</i>	8	16	0.13	24	10	164	0.58
<i>C. reinhardtii</i>	8	16	0.13	24	30	157	0.57
<i>C. reinhardtii</i>	8	20	0.10	24	0	226	0.61

Taxa	Flash Spacing (s)	Temp (°C)	Light	Growth Temp (°C)	NaCl (mM)	σ_{PSII}	F _V /F _M
<i>C. reinhardtii</i>	8	20	0.10	24	0	195	0.59
<i>C. reinhardtii</i>	8	20	0.12	24	10	170	0.57
<i>C. reinhardtii</i>	8	20	0.12	24	0	171	0.56
<i>C. reinhardtii</i>	8	20	0.13	24	30	162	0.56
<i>C. reinhardtii</i>	8	24	0.10	24	0	197	0.57
<i>C. reinhardtii</i>	8	24	0.10	24	0	204	0.58
<i>C. reinhardtii</i>	8	24	0.12	24	10	171	0.56
<i>C. reinhardtii</i>	8	24	0.13	24	30	166	0.55
<i>C. reinhardtii</i>	8	24	0.13	24	0	167	0.54
<i>C. reinhardtii</i>	8	28	0.12	24	10	174	0.54
<i>C. reinhardtii</i>	8	28	0.12	24	30	169	0.62
<i>C. reinhardtii</i>	8	28	0.13	24	0	167	0.54
<i>C. reinhardtii</i>	16	12	0.05	24	0	214	0.64

Taxa	Flash Spacing (s)	Temp (°C)	Light	Growth Temp (°C)	NaCl (mM)	σ_{PSII}	F _V /F _M
<i>C. reinhardt ii</i>	16	12	0.05	24	0	216	0.64
<i>C. reinhardt ii</i>	16	12	0.06	24	0	159	0.98
<i>C. reinhardt ii</i>	16	12	0.06	24	30	170	0.84
<i>C. reinhardt ii</i>	16	12	0.06	24	10	155	1.00
<i>C. reinhardt ii</i>	16	16	0.05	24	0	233	0.64
<i>C. reinhardt ii</i>	16	16	0.05	24	0	224	0.63
<i>C. reinhardt ii</i>	16	16	0.06	24	10	173	0.58
<i>C. reinhardt ii</i>	16	16	0.06	24	0	169	0.56
<i>C. reinhardt ii</i>	16	16	0.07	24	30	158	0.57
<i>C. reinhardt ii</i>	16	20	0.05	24	0	215	0.62
<i>C. reinhardt ii</i>	16	20	0.05	24	0	192	0.61
<i>C. reinhardt ii</i>	16	20	0.06	24	0	171	0.56
<i>C. reinhardt ii</i>	16	20	0.06	24	10	167	0.58

Taxa	Flash Spacing (s)	Temp (°C)	Light	Growth Temp (°C)	NaCl (mM)	σ_{PSII}	F _V /F _M
<i>C. reinhardtii</i>	16	20	0.06	24	30	162	0.57
<i>C. reinhardtii</i>	16	24	0.05	24	0	184	0.58
<i>C. reinhardtii</i>	16	24	0.05	24	0	202	0.61
<i>C. reinhardtii</i>	16	24	0.06	24	0	172	0.54
<i>C. reinhardtii</i>	16	24	0.06	24	10	168	0.57
<i>C. reinhardtii</i>	16	24	0.06	24	30	165	0.57
<i>C. reinhardtii</i>	16	28	0.06	24	10	171	0.55
<i>C. reinhardtii</i>	16	28	0.06	24	0	174	0.54
<i>C. reinhardtii</i>	16	28	0.06	24	30	169	0.64
<i>F. cylindrus</i>	1	0	0.60	0	500	327	0.60
<i>F. cylindrus</i>	1	0	0.60	0	500	210	0.55
<i>F. cylindrus</i>	1	0	0.66	6	500	308	0.62
<i>F. cylindrus</i>	1	0	0.66	6	500	199	0.57
<i>F. cylindrus</i>	1	10	0.49	6	500	342	0.61

Taxa	Flash Spacing (s)	Temp (°C)	Light	Growth Temp (°C)	NaCl (mM)	σ_{PSII}	F_V/F_M
<i>F. cylindrus</i>	1	10	0.49	6	500	338	0.62
<i>F. cylindrus</i>	1	10	0.55	0	500	375	0.58
<i>F. cylindrus</i>	1	10	0.55	0	500	236	0.53
<i>F. cylindrus</i>	1	2	0.61	0	500	342	0.55
<i>F. cylindrus</i>	1	2	0.61	0	500	203	0.54
<i>F. cylindrus</i>	1	2	0.65	6	500	307	0.63
<i>F. cylindrus</i>	1	2	0.65	6	500	202	0.57
<i>F. cylindrus</i>	1	6	0.60	6	500	348	0.58
<i>F. cylindrus</i>	1	6	0.60	6	500	215	0.57
<i>F. cylindrus</i>	1	6	0.77	0	500	214	0.54
<i>F. cylindrus</i>	16	0	0.04	0	500	341	0.62
<i>F. cylindrus</i>	16	0	0.04	0	500	217	0.58
<i>F. cylindrus</i>	16	0	0.04	6	500	317	0.65
<i>F. cylindrus</i>	16	0	0.04	6	500	204	0.60
<i>F. cylindrus</i>	16	10	0.03	6	500	357	0.61
<i>F. cylindrus</i>	16	10	0.03	6	500	346	0.62
<i>F. cylindrus</i>	16	10	0.03	0	500	388	0.58
<i>F. cylindrus</i>	16	10	0.03	0	500	245	0.55

Taxa	Flash Spacing (s)	Temp (°C)	Light	Growth Temp (°C)	NaCl (mM)	σ_{PSII}	F _V /F _M
<i>F. cylindrus</i>	16	2	0.03	0	500	344	0.56
<i>F. cylindrus</i>	16	2	0.03	0	500	346	0.55
<i>F. cylindrus</i>	16	2	0.03	0	500	210	0.56
<i>F. cylindrus</i>	16	2	0.04	6	500	322	0.64
<i>F. cylindrus</i>	16	2	0.04	6	500	214	0.58
<i>F. cylindrus</i>	16	6	0.04	6	500	356	0.59
<i>F. cylindrus</i>	16	6	0.04	6	500	228	0.59
<i>F. cylindrus</i>	16	6	0.05	0	500	223	0.56
<i>F. cylindrus</i>	2	0	0.30	0	500	329	0.60
<i>F. cylindrus</i>	2	0	0.30	0	500	211	0.56
<i>F. cylindrus</i>	2	0	0.33	6	500	307	0.63
<i>F. cylindrus</i>	2	0	0.33	6	500	198	0.58
<i>F. cylindrus</i>	2	10	0.24	6	500	344	0.62
<i>F. cylindrus</i>	2	10	0.24	6	500	339	0.62
<i>F. cylindrus</i>	2	10	0.27	0	500	376	0.58
<i>F. cylindrus</i>	2	10	0.27	0	500	238	0.54
<i>F. cylindrus</i>	2	2	0.31	0	500	338	0.55
<i>F. cylindrus</i>	2	2	0.31	0	500	203	0.55

Taxa	Flash Spacing (s)	Temp (°C)	Light	Growth Temp (°C)	NaCl (mM)	σ_{PSII}	F_V/F_M
<i>F. cylindrus</i>	2	2	0.32	6	500	311	0.64
<i>F. cylindrus</i>	2	2	0.32	6	500	202	0.57
<i>F. cylindrus</i>	2	6	0.30	6	500	347	0.58
<i>F. cylindrus</i>	2	6	0.30	6	500	215	0.57
<i>F. cylindrus</i>	2	6	0.38	0	500	215	0.54
<i>F. cylindrus</i>	4	0	0.15	0	500	331	0.61
<i>F. cylindrus</i>	4	0	0.15	0	500	213	0.56
<i>F. cylindrus</i>	4	0	0.16	6	500	308	0.64
<i>F. cylindrus</i>	4	0	0.16	6	500	199	0.58
<i>F. cylindrus</i>	4	10	0.12	6	500	347	0.62
<i>F. cylindrus</i>	4	10	0.12	6	500	341	0.63
<i>F. cylindrus</i>	4	10	0.13	0	500	379	0.59
<i>F. cylindrus</i>	4	10	0.13	0	500	241	0.54
<i>F. cylindrus</i>	4	2	0.15	0	500	333	0.56
<i>F. cylindrus</i>	4	2	0.15	0	500	206	0.55
<i>F. cylindrus</i>	4	2	0.16	6	500	308	0.64
<i>F. cylindrus</i>	4	2	0.16	6	500	203	0.58
<i>F. cylindrus</i>	4	6	0.15	6	500	349	0.59

Taxa	Flash Spacing (s)	Temp (°C)	Light	Growth Temp (°C)	NaCl (mM)	σ_{PSII}	F_V/F_M
<i>F. cylindrus</i>	4	6	0.15	6	500	217	0.58
<i>F. cylindrus</i>	4	6	0.19	0	500	217	0.55
<i>F. cylindrus</i>	8	0	0.07	0	500	337	0.61
<i>F. cylindrus</i>	8	0	0.07	0	500	215	0.57
<i>F. cylindrus</i>	8	0	0.08	6	500	312	0.64
<i>F. cylindrus</i>	8	0	0.08	6	500	201	0.59
<i>F. cylindrus</i>	8	10	0.06	6	500	352	0.62
<i>F. cylindrus</i>	8	10	0.06	6	500	343	0.63
<i>F. cylindrus</i>	8	10	0.07	0	500	384	0.58
<i>F. cylindrus</i>	8	10	0.07	0	500	245	0.55
<i>F. cylindrus</i>	8	2	0.07	0	500	349	0.55
<i>F. cylindrus</i>	8	2	0.07	0	500	209	0.56
<i>F. cylindrus</i>	8	2	0.08	6	500	310	0.64
<i>F. cylindrus</i>	8	2	0.08	6	500	207	0.59
<i>F. cylindrus</i>	8	6	0.07	6	500	352	0.59
<i>F. cylindrus</i>	8	6	0.07	6	500	221	0.58
<i>F. cylindrus</i>	8	6	0.09	0	500	222	0.55
<i>C. vulgaris</i>	1	10	0.90	22	0	203	0.67

Taxa	Flash Spacing (s)	Temp (°C)	Light	Growth Temp (°C)	NaCl (mM)	σ_{PSII}	F _V /F _M
<i>C. vulgaris</i>	1	10	0.90	22	0	178	0.71
<i>C. vulgaris</i>	1	10	0.90	22	0	181	0.64
<i>C. vulgaris</i>	1	14	0.85	22	0	211	0.66
<i>C. vulgaris</i>	1	14	0.85	22	0	184	0.71
<i>C. vulgaris</i>	1	14	0.85	22	0	199	0.67
<i>C. vulgaris</i>	1	18	0.90	22	0	214	0.65
<i>C. vulgaris</i>	1	18	0.90	22	0	185	0.70
<i>C. vulgaris</i>	1	18	0.90	22	0	165	0.65
<i>C. vulgaris</i>	1	22	0.85	22	0	222	0.65
<i>C. vulgaris</i>	1	22	0.85	22	0	197	0.71
<i>C. vulgaris</i>	1	22	0.85	22	0	171	0.64
<i>C. vulgaris</i>	1	26	0.81	22	0	194	0.62
<i>C. vulgaris</i>	1	26	0.81	22	0	209	0.63
<i>C. vulgaris</i>	16	10	0.05	22	0	209	0.68
<i>C. vulgaris</i>	16	10	0.05	22	0	191	0.72
<i>C. vulgaris</i>	16	10	0.05	22	0	202	0.66
<i>C. vulgaris</i>	16	14	0.05	22	0	213	0.67
<i>C. vulgaris</i>	16	14	0.05	22	0	200	0.73

Taxa	Flash Spacing (s)	Temp (°C)	Light	Growth Temp (°C)	NaCl (mM)	σ_{PSII}	F _V /F _M
<i>C. vulgaris</i>	16	14	0.05	22	0	209	0.66
<i>C. vulgaris</i>	16	18	0.05	22	0	218	0.66
<i>C. vulgaris</i>	16	18	0.05	22	0	202	0.73
<i>C. vulgaris</i>	16	18	0.05	22	0	183	0.66
<i>C. vulgaris</i>	16	22	0.05	22	0	222	0.66
<i>C. vulgaris</i>	16	22	0.05	22	0	202	0.73
<i>C. vulgaris</i>	16	22	0.05	22	0	187	0.66
<i>C. vulgaris</i>	16	26	0.05	22	0	200	0.74
<i>C. vulgaris</i>	16	26	0.05	22	0	195	0.64
<i>C. vulgaris</i>	16	26	0.05	22	0	215	0.67
<i>C. vulgaris</i>	2	10	0.44	22	0	203	0.67
<i>C. vulgaris</i>	2	10	0.44	22	0	180	0.72
<i>C. vulgaris</i>	2	10	0.44	22	0	184	0.65
<i>C. vulgaris</i>	2	14	0.42	22	0	211	0.67
<i>C. vulgaris</i>	2	14	0.42	22	0	186	0.71
<i>C. vulgaris</i>	2	14	0.42	22	0	200	0.67
<i>C. vulgaris</i>	2	18	0.44	22	0	217	0.66
<i>C. vulgaris</i>	2	18	0.44	22	0	190	0.70

Taxa	Flash Spacing (s)	Temp (°C)	Light	Growth Temp (°C)	NaCl (mM)	σ_{PSII}	F _V /F _M
<i>C. vulgaris</i>	2	18	0.44	22	0	165	0.65
<i>C. vulgaris</i>	2	22	0.42	22	0	224	0.65
<i>C. vulgaris</i>	2	22	0.42	22	0	201	0.71
<i>C. vulgaris</i>	2	22	0.42	22	0	178	0.65
<i>C. vulgaris</i>	2	26	0.41	22	0	204	0.69
<i>C. vulgaris</i>	2	26	0.41	22	0	196	0.62
<i>C. vulgaris</i>	2	26	0.41	22	0	210	0.63
<i>C. vulgaris</i>	4	10	0.22	22	0	204	0.67
<i>C. vulgaris</i>	4	10	0.22	22	0	183	0.72
<i>C. vulgaris</i>	4	10	0.22	22	0	187	0.66
<i>C. vulgaris</i>	4	14	0.21	22	0	212	0.66
<i>C. vulgaris</i>	4	14	0.21	22	0	190	0.72
<i>C. vulgaris</i>	4	14	0.21	22	0	202	0.67
<i>C. vulgaris</i>	4	18	0.21	22	0	218	0.66
<i>C. vulgaris</i>	4	18	0.21	22	0	198	0.71
<i>C. vulgaris</i>	4	18	0.21	22	0	172	0.66
<i>C. vulgaris</i>	4	22	0.21	22	0	225	0.64
<i>C. vulgaris</i>	4	22	0.21	22	0	202	0.70

Taxa	Flash Spacing (s)	Temp (°C)	Light	Growth Temp (°C)	NaCl (mM)	σ_{PSII}	F _V /F _M
<i>C. vulgaris</i>	4	22	0.21	22	0	182	0.65
<i>C. vulgaris</i>	4	26	0.20	22	0	208	0.70
<i>C. vulgaris</i>	4	26	0.20	22	0	198	0.62
<i>C. vulgaris</i>	4	26	0.20	22	0	217	0.63
<i>C. vulgaris</i>	8	10	0.11	22	0	206	0.67
<i>C. vulgaris</i>	8	10	0.11	22	0	187	0.72
<i>C. vulgaris</i>	8	10	0.11	22	0	194	0.66
<i>C. vulgaris</i>	8	14	0.10	22	0	211	0.66
<i>C. vulgaris</i>	8	14	0.10	22	0	195	0.72
<i>C. vulgaris</i>	8	14	0.10	22	0	207	0.67
<i>C. vulgaris</i>	8	18	0.11	22	0	218	0.65
<i>C. vulgaris</i>	8	18	0.11	22	0	203	0.72
<i>C. vulgaris</i>	8	18	0.11	22	0	179	0.66
<i>C. vulgaris</i>	8	22	0.10	22	0	227	0.64
<i>C. vulgaris</i>	8	22	0.10	22	0	205	0.71
<i>C. vulgaris</i>	8	22	0.10	22	0	186	0.65
<i>C. vulgaris</i>	8	26	0.10	22	0	206	0.71
<i>C. vulgaris</i>	8	26	0.10	22	0	197	0.62

Taxa	Flash Spacing (s)	Temp (°C)	Light	Growth Temp (°C)	NaCl (mM)	σ_{PSII}	F _V /F _M
<i>C. vulgaris</i>	8	26	0.10	22	0	219	0.65
<i>T. pseudona na</i>	1	10	0.53	22	0	282	0.58
<i>T. pseudona na</i>	1	10	0.53	22	0	305	0.71
<i>T. pseudona na</i>	1	10	0.53	22	0	348	0.67
<i>T. pseudona na</i>	1	14	0.44	22	0	228	0.61
<i>T. pseudona na</i>	1	14	0.44	22	0	611	0.65
<i>T. pseudona na</i>	1	14	0.44	22	0	354	0.65
<i>T. pseudona na</i>	1	14	0.44	22	0	343	0.61
<i>T. pseudona na</i>	1	18	0.42	22	0	306	0.56
<i>T. pseudona na</i>	1	18	0.42	22	0	639	0.65
<i>T. pseudona na</i>	1	18	0.42	22	0	330	0.60
<i>T. pseudona na</i>	1	18	0.42	22	0	338	0.60
<i>T. pseudona na</i>	1	20	0.39	22	0	645	0.64

Taxa	Flash Spacing (s)	Temp (°C)	Light	Growth Temp (°C)	NaCl (mM)	σ_{PSII}	F_V/F_M
<i>T. pseudona na</i>	1	20	0.39	22	0	384	0.67
<i>T. pseudona na</i>	1	20	0.39	22	0	353	0.58
<i>T. pseudona na</i>	1	20	0.39	22	0	336	0.59
<i>T. pseudona na</i>	1	22	0.52	22	0	371	0.77
<i>T. pseudona na</i>	1	22	0.52	22	0	247	0.59
<i>T. pseudona na</i>	1	22	0.52	22	0	356	0.58
<i>T. pseudona na</i>	1	22	0.52	22	0	315	0.58
<i>T. pseudona na</i>	1	24	0.38	22	0	655	0.63
<i>T. pseudona na</i>	1	24	0.38	22	0	387	0.59
<i>T. pseudona na</i>	1	24	0.38	22	0	360	0.57
<i>T. pseudona na</i>	1	24	0.38	22	0	342	0.57
<i>T. pseudona na</i>	1	28	0.42	22	0	500	0.53
<i>T. pseudona na</i>	1	28	0.42	22	0	391	0.57

Taxa	Flash Spacing (s)	Temp (°C)	Light	Growth Temp (°C)	NaCl (mM)	σ_{PSII}	F _V /F _M
<i>T. pseudonana</i>	1	28	0.42	22	0	353	0.55
<i>T. pseudonana</i>	1	28	0.42	22	0	346	0.55
<i>T. pseudonana</i>	16	10	0.03	22	0	301	0.70
<i>T. pseudonana</i>	16	10	0.03	22	0	353	0.66
<i>T. pseudonana</i>	16	10	0.03	22	0	336	0.61
<i>T. pseudonana</i>	16	10	0.03	22	0	319	0.63
<i>T. pseudonana</i>	16	14	0.03	22	0	621	0.65
<i>T. pseudonana</i>	16	14	0.03	22	0	357	0.65
<i>T. pseudonana</i>	16	14	0.03	22	0	346	0.61
<i>T. pseudonana</i>	16	14	0.03	22	0	330	0.62
<i>T. pseudonana</i>	16	18	0.03	22	0	624	0.65
<i>T. pseudonana</i>	16	18	0.03	22	0	317	0.61
<i>T. pseudonana</i>	16	18	0.03	22	0	352	0.60

Taxa	Flash Spacing (s)	Temp (°C)	Light	Growth Temp (°C)	NaCl (mM)	σ_{PSII}	F _V /F _M
<i>T. pseudonana</i>	16	18	0.03	22	0	330	0.61
<i>T. pseudonana</i>	16	20	0.02	22	0	638	0.64
<i>T. pseudonana</i>	16	20	0.02	22	0	374	0.62
<i>T. pseudonana</i>	16	20	0.02	22	0	353	0.59
<i>T. pseudonana</i>	16	20	0.02	22	0	334	0.61
<i>T. pseudonana</i>	16	22	0.03	22	0	356	0.59
<i>T. pseudonana</i>	16	22	0.03	22	0	335	0.60
<i>T. pseudonana</i>	16	24	0.03	22	0	543	0.62
<i>T. pseudonana</i>	16	24	0.03	22	0	375	0.62
<i>T. pseudonana</i>	16	24	0.03	22	0	356	0.58
<i>T. pseudonana</i>	16	24	0.03	22	0	332	0.59
<i>T. pseudonana</i>	16	28	0.02	22	0	662	0.60
<i>T. pseudonana</i>	16	28	0.02	22	0	370	0.60

Taxa	Flash Spacing (s)	Temp (°C)	Light	Growth Temp (°C)	NaCl (mM)	σ_{PSII}	F _V /F _M
<i>T. pseudonana</i>	16	28	0.02	22	0	351	0.57
<i>T. pseudonana</i>	16	28	0.02	22	0	328	0.58
<i>T. pseudonana</i>	2	10	0.27	22	0	289	0.58
<i>T. pseudonana</i>	2	10	0.27	22	0	308	0.71
<i>T. pseudonana</i>	2	10	0.27	22	0	349	0.67
<i>T. pseudonana</i>	2	14	0.21	22	0	229	0.61
<i>T. pseudonana</i>	2	14	0.21	22	0	633	0.65
<i>T. pseudonana</i>	2	14	0.21	22	0	356	0.65
<i>T. pseudonana</i>	2	14	0.21	22	0	347	0.60
<i>T. pseudonana</i>	2	18	0.21	22	0	305	0.56
<i>T. pseudonana</i>	2	18	0.21	22	0	633	0.65
<i>T. pseudonana</i>	2	18	0.21	22	0	329	0.60
<i>T. pseudonana</i>	2	18	0.21	22	0	338	0.60

Taxa	Flash Spacing (s)	Temp (°C)	Light	Growth Temp (°C)	NaCl (mM)	σ_{PSII}	F _V /F _M
<i>T. pseudonana</i>	2	20	0.19	22	0	643	0.64
<i>T. pseudonana</i>	2	20	0.19	22	0	386	0.67
<i>T. pseudonana</i>	2	20	0.19	22	0	356	0.58
<i>T. pseudonana</i>	2	20	0.19	22	0	340	0.59
<i>T. pseudonana</i>	2	22	0.26	22	0	246	0.59
<i>T. pseudonana</i>	2	22	0.26	22	0	361	0.57
<i>T. pseudonana</i>	2	22	0.26	22	0	343	0.58
<i>T. pseudonana</i>	2	24	0.19	22	0	654	0.63
<i>T. pseudonana</i>	2	24	0.19	22	0	389	0.59
<i>T. pseudonana</i>	2	24	0.19	22	0	364	0.57
<i>T. pseudonana</i>	2	24	0.19	22	0	344	0.57
<i>T. pseudonana</i>	2	28	0.19	22	0	669	0.60
<i>T. pseudonana</i>	2	28	0.19	22	0	394	0.57

Taxa	Flash Spacing (s)	Temp (°C)	Light	Growth Temp (°C)	NaCl (mM)	σ_{PSII}	F _V /F _M
<i>T. pseudonana</i>	2	28	0.19	22	0	365	0.54
<i>T. pseudonana</i>	2	28	0.19	22	0	346	0.55
<i>T. pseudonana</i>	4	10	0.13	22	0	279	0.58
<i>T. pseudonana</i>	4	10	0.13	22	0	300	0.70
<i>T. pseudonana</i>	4	10	0.13	22	0	349	0.67
<i>T. pseudonana</i>	4	14	0.11	22	0	229	0.61
<i>T. pseudonana</i>	4	14	0.11	22	0	625	0.65
<i>T. pseudonana</i>	4	14	0.11	22	0	357	0.65
<i>T. pseudonana</i>	4	14	0.11	22	0	347	0.61
<i>T. pseudonana</i>	4	18	0.10	22	0	304	0.57
<i>T. pseudonana</i>	4	18	0.10	22	0	630	0.65
<i>T. pseudonana</i>	4	18	0.10	22	0	328	0.60
<i>T. pseudonana</i>	4	18	0.10	22	0	339	0.60

Taxa	Flash Spacing (s)	Temp (°C)	Light	Growth Temp (°C)	NaCl (mM)	σ_{PSII}	F _V /F _M
<i>T. pseudona na</i>	4	20	0.10	22	0	641	0.64
<i>T. pseudona na</i>	4	20	0.10	22	0	386	0.67
<i>T. pseudona na</i>	4	20	0.10	22	0	358	0.58
<i>T. pseudona na</i>	4	20	0.10	22	0	339	0.59
<i>T. pseudona na</i>	4	22	0.13	22	0	368	0.76
<i>T. pseudona na</i>	4	22	0.13	22	0	248	0.59
<i>T. pseudona na</i>	4	22	0.13	22	0	362	0.57
<i>T. pseudona na</i>	4	22	0.13	22	0	343	0.58
<i>T. pseudona na</i>	4	24	0.10	22	0	624	0.63
<i>T. pseudona na</i>	4	24	0.10	22	0	387	0.59
<i>T. pseudona na</i>	4	24	0.10	22	0	365	0.57
<i>T. pseudona na</i>	4	24	0.10	22	0	343	0.57
<i>T. pseudona na</i>	4	28	0.09	22	0	662	0.60

Taxa	Flash Spacing (s)	Temp (°C)	Light	Growth Temp (°C)	NaCl (mM)	σ_{PSII}	F _V /F _M
<i>T. pseudona na</i>	4	28	0.09	22	0	389	0.57
<i>T. pseudona na</i>	4	28	0.09	22	0	364	0.54
<i>T. pseudona na</i>	4	28	0.09	22	0	349	0.55
<i>T. pseudona na</i>	8	10	0.07	22	0	292	0.58
<i>T. pseudona na</i>	8	10	0.07	22	0	302	0.70
<i>T. pseudona na</i>	8	10	0.07	22	0	350	0.66
<i>T. pseudona na</i>	8	14	0.05	22	0	229	0.61
<i>T. pseudona na</i>	8	14	0.05	22	0	619	0.65
<i>T. pseudona na</i>	8	14	0.05	22	0	356	0.65
<i>T. pseudona na</i>	8	14	0.05	22	0	348	0.60
<i>T. pseudona na</i>	8	18	0.05	22	0	302	0.57
<i>T. pseudona na</i>	8	18	0.05	22	0	624	0.65
<i>T. pseudona na</i>	8	18	0.05	22	0	327	0.60

Taxa	Flash Spacing (s)	Temp (°C)	Light	Growth Temp (°C)	NaCl (mM)	σ_{PSII}	F _V /F _M
<i>T. pseudonana</i>	8	18	0.05	22	0	339	0.61
<i>T. pseudonana</i>	8	20	0.05	22	0	636	0.64
<i>T. pseudonana</i>	8	20	0.05	22	0	383	0.61
<i>T. pseudonana</i>	8	20	0.05	22	0	357	0.58
<i>T. pseudonana</i>	8	20	0.05	22	0	339	0.60
<i>T. pseudonana</i>	8	22	0.06	22	0	365	0.76
<i>T. pseudonana</i>	8	22	0.06	22	0	245	0.60
<i>T. pseudonana</i>	8	22	0.06	22	0	361	0.58
<i>T. pseudonana</i>	8	22	0.06	22	0	340	0.59
<i>T. pseudonana</i>	8	24	0.05	22	0	603	0.63
<i>T. pseudonana</i>	8	24	0.05	22	0	385	0.60
<i>T. pseudonana</i>	8	24	0.05	22	0	363	0.57
<i>T. pseudonana</i>	8	24	0.05	22	0	340	0.58

Taxa	Flash Spacing (s)	Temp (°C)	Light	Growth Temp (°C)	NaCl (mM)	σ_{PSII}	F_V/F_M
<i>T. pseudonana</i>	8	28	0.05	22	0	662	0.60
<i>T. pseudonana</i>	8	28	0.05	22	0	385	0.58
<i>T. pseudonana</i>	8	28	0.05	22	0	358	0.55
<i>T. pseudonana</i>	8	28	0.05	22	0	342	0.56

Table 5: Summary statistics by phytoplankton strain of Generalized Additive Models (GAM) using the restricted maximum likelihood (REML) method to model the response of the damping of S-State induced chlorophyll fluorescence oscillations to the predictors of measurement temperature (°C) and the effective light level ($\mu\text{mol photons m}^{-2}\text{s}^{-1}$). GAM were fit at the optimal growth salinity for each strain, and separate models were fit for strains with multiple growth temperatures. The GAM terms indicate main effect smooth (s) and interaction-only tensor product (ti); the edf (effective degrees of freedom) of the smooth; ref.df (reference degrees of freedom); statistic and p.value

Strain	term	edf	ref.df	statistic	p.value
<i>F. cylindrus</i>	s(LightLevel_ue)	1.8290	1.9708	4.7142	0.0318
<i>F. cylindrus</i>	s(Temp_C)	1.0000	1.0000	0.0556	0.8169
<i>F. cylindrus</i>	ti(Temp_C,LightLevel_ue)	1.0000	1.0001	0.0321	0.8603
<i>F. cylindrus</i>	s(LightLevel_ue)	1.1050	1.1959	3.2671	0.0689
<i>F. cylindrus</i>	s(Temp_C)	1.0000	1.0001	2.3553	0.1457
<i>F. cylindrus</i>	ti(Temp_C,LightLevel_ue)	1.4570	1.7021	2.8169	0.1782
<i>C. vulgaris</i>	s(LightLevel_ue)	1.9428	1.9967	8.4123	0.0021
<i>C. vulgaris</i>	s(Temp_C)	1.8934	1.9886	4.7064	0.0222
<i>C. vulgaris</i>	ti(Temp_C,LightLevel_ue)	1.0001	1.0001	0.7103	0.4099
<i>T. pseudonana</i>	s(LightLevel_ue)	1.9550	1.9926	10.2515	0.0004
<i>T. pseudonana</i>	s(Temp_C)	1.0005	1.0008	2.8345	0.1029
<i>T. pseudonana</i>	ti(Temp_C,LightLevel_ue)	2.1743	2.6305	3.9571	0.0375
<i>C. euryale</i>	s(LightLevel_ue)	1.0000	1.0000	188.8093	0.0000
<i>C. euryale</i>	s(Temp_C)	1.9633	1.0000	537.2389	0.0000
<i>C. euryale</i>	ti(Temp_C,LightLevel_ue)	1.0000	1.0000	42.5569	0.0000
<i>C. ICEMDV</i>	s(LightLevel_ue)	1.7363	1.9305	3.3306	0.0673
<i>C. ICEMDV</i>	s(Temp_C)	1.0000	1.0000	0.3048	0.5930
<i>C. ICEMDV</i>	ti(Temp_C,LightLevel_ue)	1.0000	1.0000	0.0293	0.8675
<i>C. klinobasis</i>	s(LightLevel_ue)	1.8486	1.9770	5.8087	0.0126
<i>C. klinobasis</i>	s(Temp_C)	1.2533	1.4424	2.0661	0.1485
<i>C. klinobasis</i>	ti(Temp_C,LightLevel_ue)	1.0000	1.0001	2.7576	0.1190

Strain	term	edf	ref.df	statistic	p.value
<i>C. malina</i>	s(LightLevel_ue)	1.8390	1.9741	3.7353	0.0574
<i>C. malina</i>	s(Temp_C)	1.0000	1.0000	0.5369	0.4768
<i>C. malina</i>	ti(Temp_C,LightLevel_ue)	1.0001	1.0001	0.9886	0.3382
<i>C. priscui</i>	s(LightLevel_ue)	1.9037	1.9906	11.0715	0.0002
<i>C. priscui</i>	s(Temp_C)	1.7671	1.9456	7.2106	0.0024
<i>C. priscui</i>	ti(Temp_C,LightLevel_ue)	1.0002	1.0004	7.7825	0.0085
<i>C. reinhardtii</i>	s(LightLevel_ue)	1.8658	1.9339	2.5493	0.1074
<i>C. reinhardtii</i>	s(Temp_C)	1.0001	1.0001	6.5384	0.0147
<i>C. reinhardtii</i>	ti(Temp_C,LightLevel_ue)	2.1942	2.6491	1.0762	0.4848

Strain: *F. cylindrus*, Growth Temperature 0 °C, Salinity 500 mM

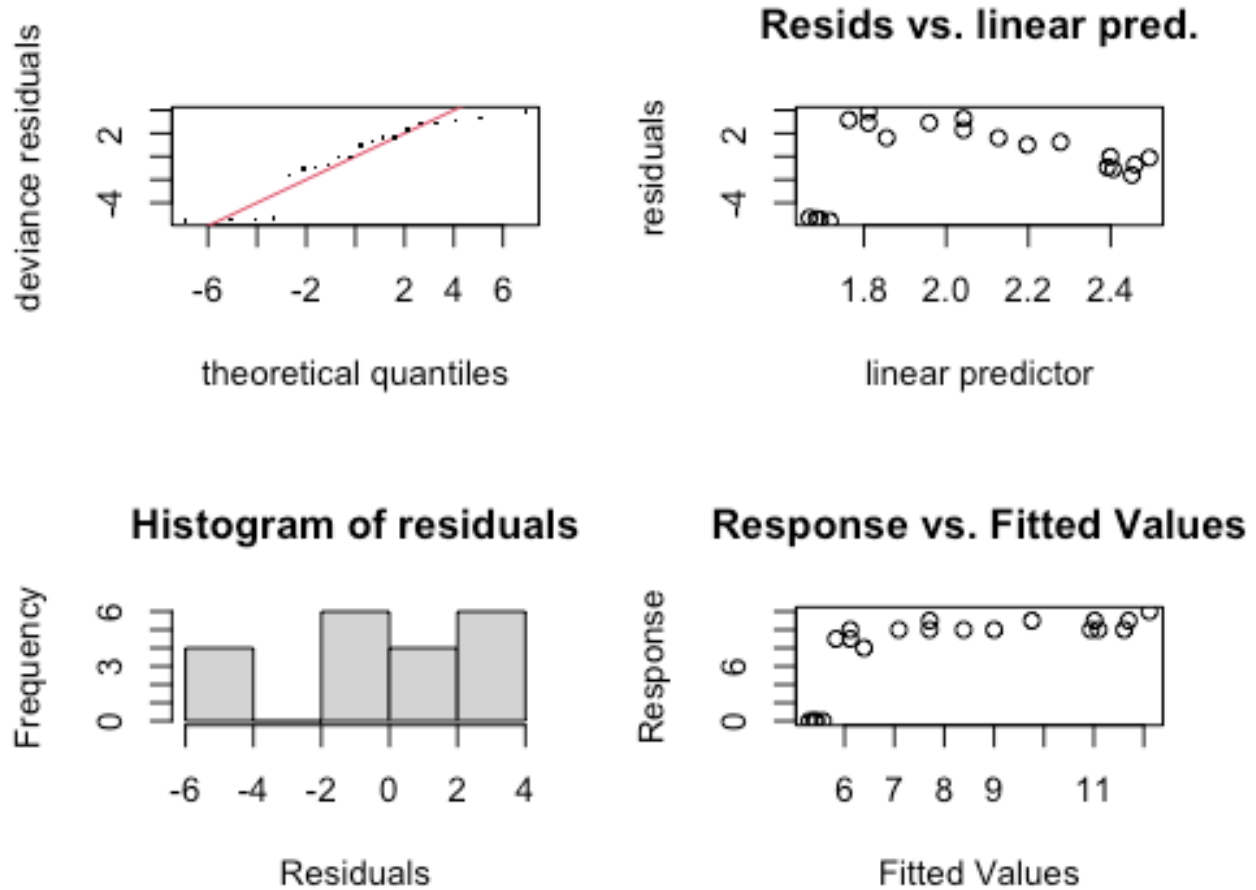


Fig 7: Generalized Additive Model (GAM) diagnostic outputs from `mgcv::gam.check()` for polar and temperate diatoms, of consecutive flashes before damping of S-State induced chlorophyll fluorescence oscillations. GAM predicted by the temperature (°C) imposed during measurements and the effective light level ($\mu\text{mol photons m}^{-2}\text{s}^{-1}$, estimated from flash spacings (s) and effective absorption cross-section of photosystem II, σ_{PSII} (A^2)).

```
Method: REML    Optimizer: outer newton
full convergence after 12 iterations.
Gradient range [-1.148855e-05,2.458705e-06]
(score 56.65425 & scale 12.43105).
Hessian positive definite, eigenvalue range [4.616448e-06,8.015894].
Model rank = 9 / 9
```

Basis dimension (k) checking results. Low p-value (k-index<1) may indicate that k is too low, especially if edf is close to k'.

	k'	edf	k-index	p-value	
s(LightLevel_ue)	2.00	1.83	0.26	<2e-16	***
s(Temp_C)	2.00	1.00	0.81	0.14	
ti(Temp_C,LightLevel_ue)	4.00	1.00	0.55	<2e-16	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Strain: *F. cylindrus*, Growth Temperature 6 °C, Salinity 500 mM

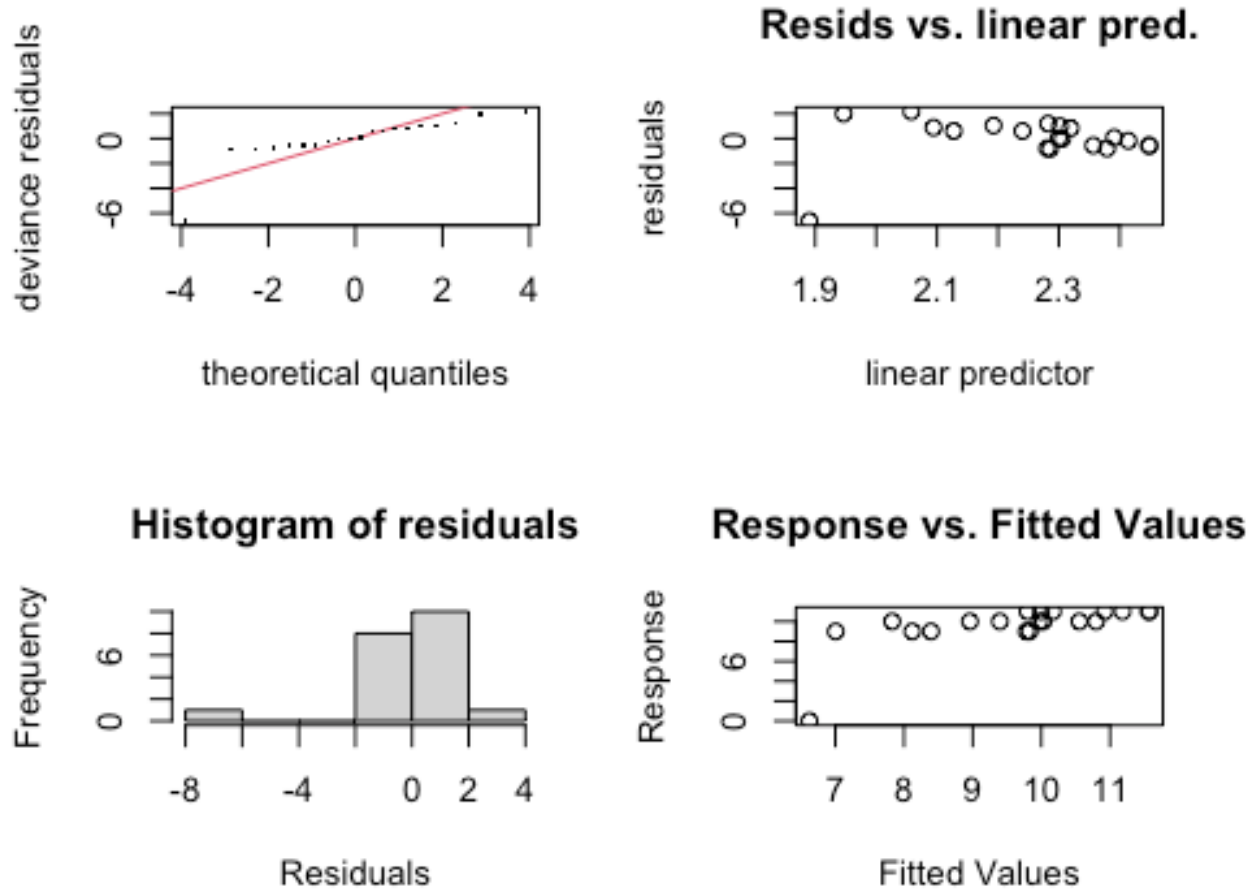


Fig 8: Generalized Additive Model (GAM) diagnostic outputs from `mgcv::gam.check()` for polar and temperate diatoms, of consecutive flashes before damping of S-State induced chlorophyll fluorescence oscillations. GAM predicted by the temperature (°C) imposed during measurements and the effective light level ($\mu\text{mol photons m}^{-2}\text{s}^{-1}$, estimated from flash spacings (s) and effective absorption cross-section of photosystem II, σ_{PSII} (A^2)).

```
Method: REML   Optimizer: outer newton
full convergence after 12 iterations.
Gradient range [-1.045605e-05,1.128619e-05]
(score 47.79832 & scale 3.965611).
Hessian positive definite, eigenvalue range [6.856764e-06,8.007953].
Model rank = 9 / 9
```

Basis dimension (k) checking results. Low p-value ($k\text{-index} < 1$) may indicate that k is too low, especially if edf is close to k'.

	k'	edf	k-index	p-value	
s(LightLevel_ue)	2.00	1.10	0.76	0.06	.
s(Temp_C)	2.00	1.00	0.93	0.28	
ti(Temp_C,LightLevel_ue)	4.00	1.46	0.89	0.12	

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Strain: *C. vulgaris*, Growth Temperature 22 °C, Salinity 0 mM

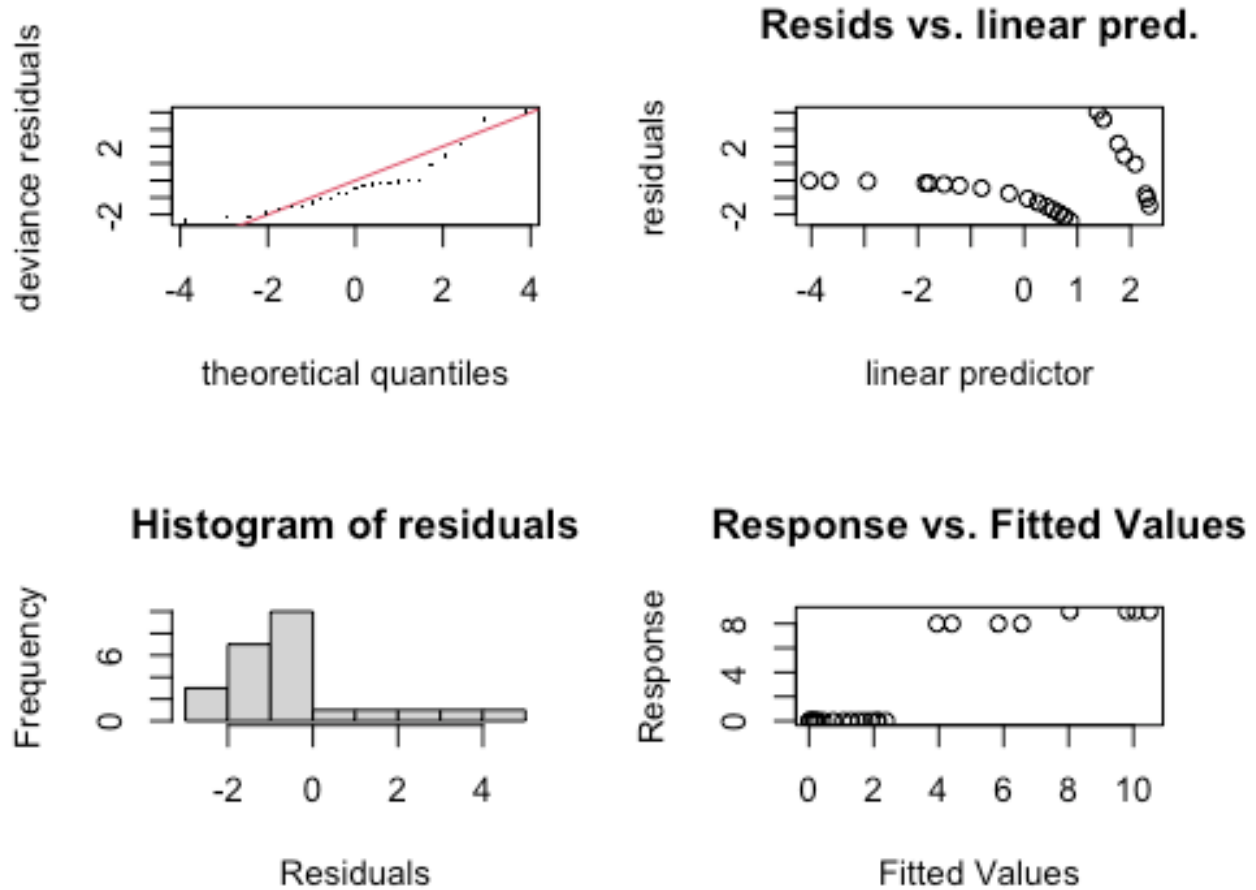


Fig 9: Generalized Additive Model (GAM) diagnostic outputs from `mgcv::gam.check()` for polar and temperate diatoms, of consecutive flashes before damping of S-State induced chlorophyll fluorescence oscillations. GAM predicted by the temperature (°C) imposed during measurements and the effective light level ($\mu\text{mol photons m}^{-2}\text{s}^{-1}$, estimated from flash spacings (s) and effective absorption cross-section of photosystem II, σ_{PSII} (A^2)).

```
Method: REML    Optimizer: outer newton
full convergence after 12 iterations.
Gradient range [-1.964407e-05,6.40554e-06]
(score 55.32928 & scale 3.571869).
Hessian positive definite, eigenvalue range [1.200439e-05,10.5307].
Model rank = 9 / 9
```

Basis dimension (k) checking results. Low p-value (k-index<1) may indicate that k is too low, especially if edf is close to k'.

	k'	edf	k-index	p-value
s(LightLevel_ue)	2.00	1.94	1.23	0.89
s(Temp_C)	2.00	1.89	1.37	0.99
ti(Temp_C,LightLevel_ue)	4.00	1.00	0.97	0.41

Strain: *T. pseudonana*, Growth Temperature 22 °C, Salinity 0 mM

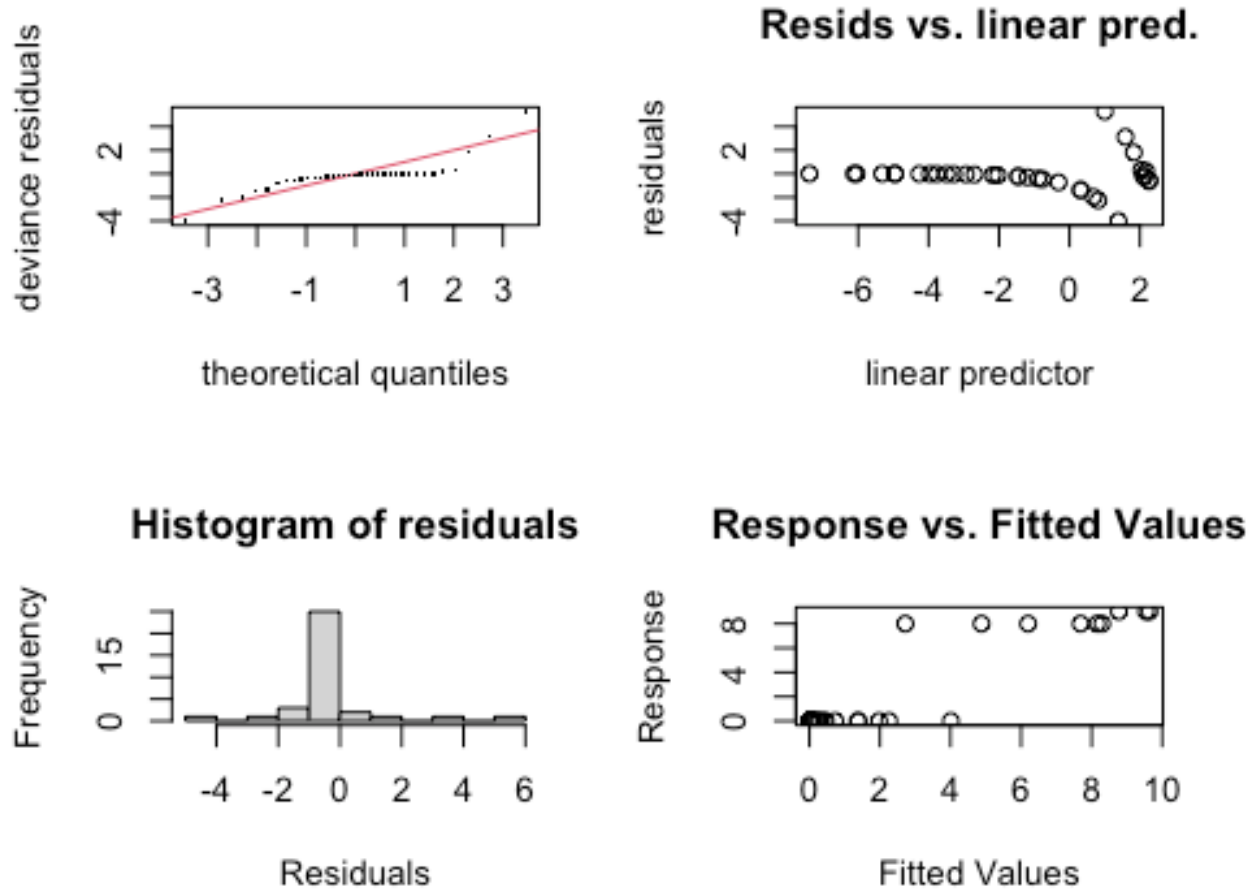


Fig 10: Generalized Additive Model (GAM) diagnostic outputs from `mgcv::gam.check()` for polar and temperate diatoms, of consecutive flashes before damping of S-State induced chlorophyll fluorescence oscillations. GAM predicted by the temperature (°C) imposed during measurements and the effective light level ($\mu\text{mol photons m}^{-2}\text{s}^{-1}$, estimated from flash spacings (s) and effective absorption cross-section of photosystem II, σ_{PSII} (A^2)).

```
Method: REML    Optimizer: outer newton
full convergence after 11 iterations.
Gradient range [-6.831923e-05,0.0001971058]
(score 71.16439 & scale 2.490609).
Hessian positive definite, eigenvalue range [5.281984e-06,15.52403].
Model rank = 9 / 9
```

Basis dimension (k) checking results. Low p-value (k-index<1) may indicate that k is too low, especially if edf is close to k'.

	k'	edf	k-index	p-value
s(LightLevel_ue)	2.00	1.96	1.60	1.00
s(Temp_C)	2.00	1.00	1.02	0.47
ti(Temp_C,LightLevel_ue)	4.00	2.17	1.48	0.99

Strain: *C. euryale*, Growth Temperature 24 °C, Salinity 500 mM

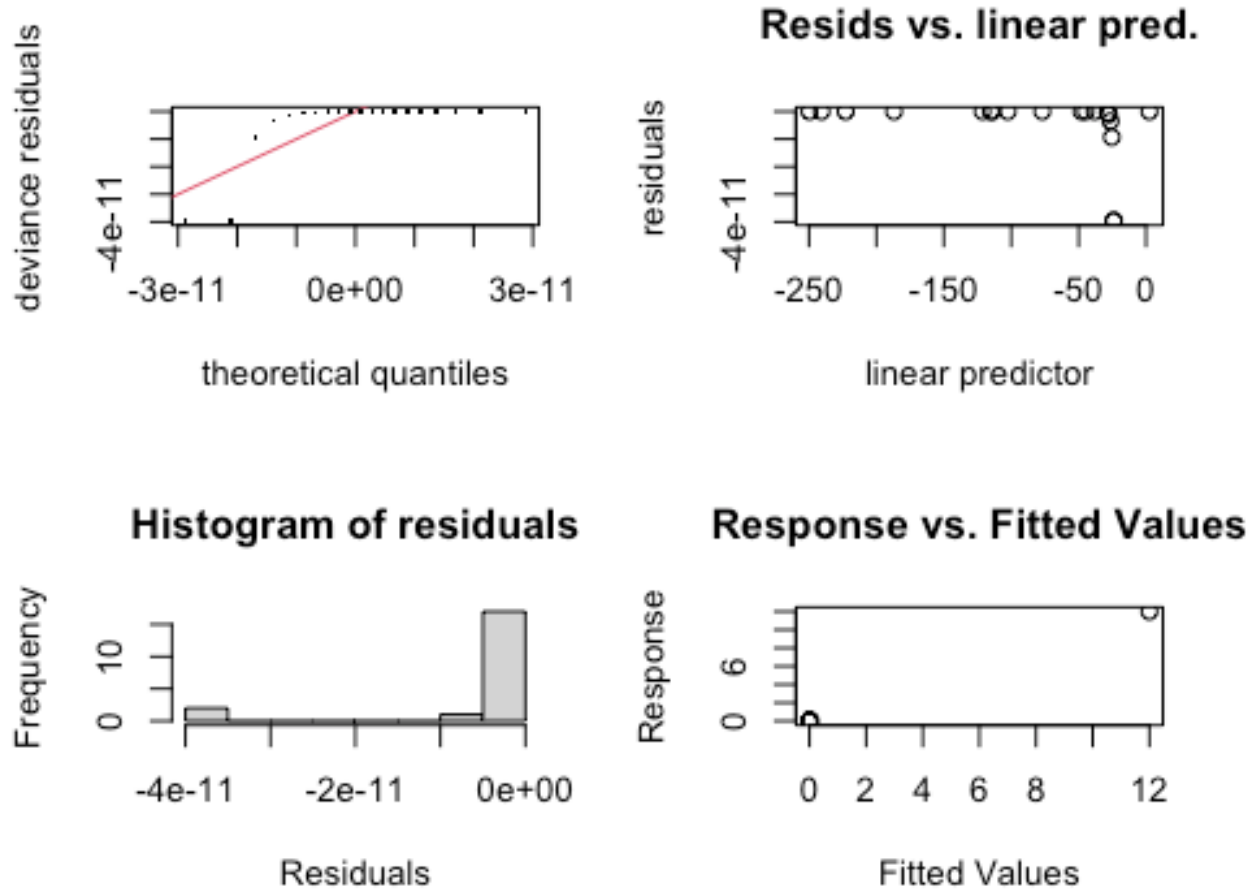


Fig 11: Generalized Additive Model (GAM) diagnostic outputs from `mgcv::gam.check()` for polar and temperate diatoms, of consecutive flashes before damping of S-State induced chlorophyll fluorescence oscillations. GAM predicted by the temperature (°C) imposed during measurements and the effective light level ($\mu\text{mol photons m}^{-2}\text{s}^{-1}$, estimated from flash spacings (s) and effective absorption cross-section of photosystem II, σ_{PSII} (A^2)).

```
Method: REML   Optimizer: outer newton
step failed after 1 iteration.
Gradient range [-1.61541e-06,9.174523]
(score -420.8737 & scale 2.151999e-22).
eigenvalue range [-9224757,7.998127].
Model rank = 9 / 9
```

Basis dimension (k) checking results. Low p-value (k-index<1) may indicate that k is too low, especially if edf is close to k'.

	k'	edf	k-index	p-value	
s(LightLevel_ue)	2.00	1.00	0.29	0.005	**
s(Temp_C)	2.00	1.96	0.75	0.120	
ti(Temp_C,LightLevel_ue)	4.00	1.00	0.67	0.055	.

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Strain: *C. ICEMDV*, Growth Temperature 4 °C, Salinity 500 mM

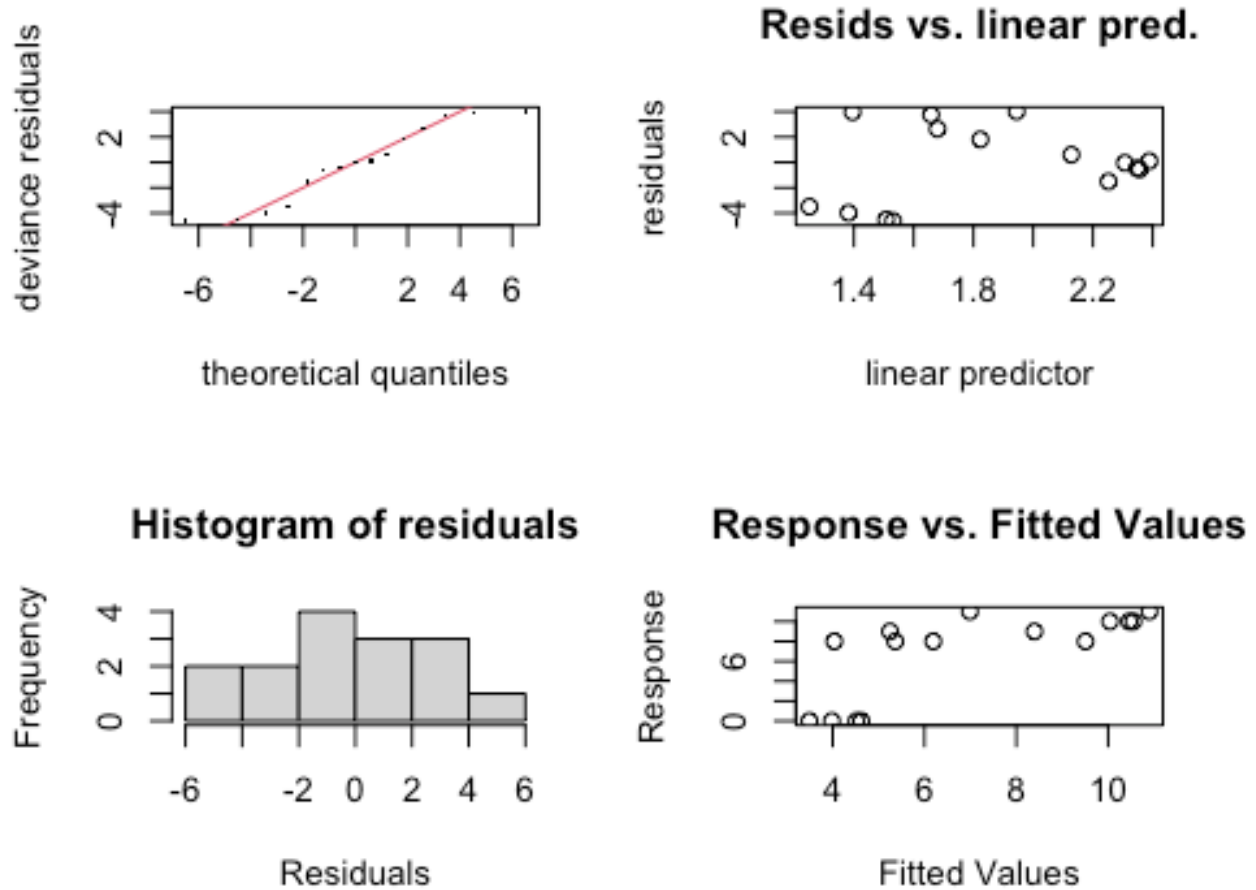


Fig 12: Generalized Additive Model (GAM) diagnostic outputs from `mgcv::gam.check()` for polar and temperate diatoms, of consecutive flashes before damping of S-State induced chlorophyll fluorescence oscillations. GAM predicted by the temperature (°C) imposed during measurements and the effective light level ($\mu\text{mol photons m}^{-2}\text{s}^{-1}$, estimated from flash spacings (s) and effective absorption cross-section of photosystem II, σ_{PSII} (A^2)).

```
Method: REML   Optimizer: outer newton
full convergence after 13 iterations.
Gradient range [-4.513978e-06,3.114671e-06]
(score 42.28655 & scale 12.57865).
Hessian positive definite, eigenvalue range [1.968747e-06,5.520196].
Model rank = 9 / 9
```

Basis dimension (k) checking results. Low p-value (k-index<1) may indicate that k is too low, especially if edf is close to k'.

	k'	edf	k-index	p-value
s(LightLevel_ue)	2.00	1.74	1.00	0.42
s(Temp_C)	2.00	1.00	0.99	0.35
ti(Temp_C,LightLevel_ue)	4.00	1.00	1.02	0.41

Strain: *C. klinobasis*, Growth Temperature 4 °C, Salinity 0 mM

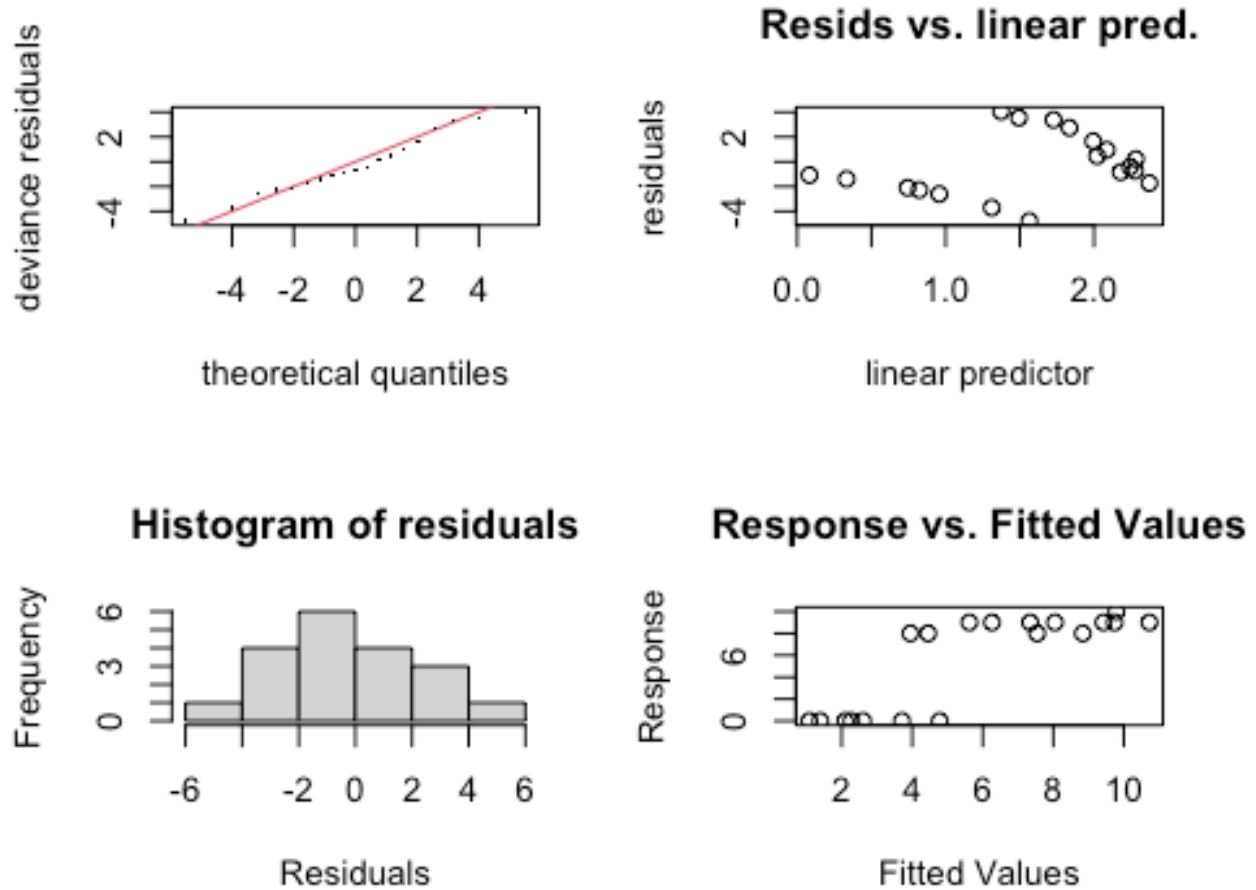


Fig 13: Generalized Additive Model (GAM) diagnostic outputs from `mgcv::gam.check()` for polar and temperate diatoms, of consecutive flashes before damping of S-State induced chlorophyll fluorescence oscillations. GAM predicted by the temperature (°C) imposed during measurements and the effective light level ($\mu\text{mol photons m}^{-2}\text{s}^{-1}$, estimated from flash spacings (s) and effective absorption cross-section of photosystem II, σ_{PSII} (Å^2)).

```
Method: REML    Optimizer: outer newton
full convergence after 10 iterations.
Gradient range [-1.145248e-05,7.291372e-06]
(score 49.70235 & scale 8.081872).
Hessian positive definite, eigenvalue range [3.948816e-06,7.521732].
Model rank = 9 / 9
```

Basis dimension (k) checking results. Low p-value (k-index<1) may indicate that k is too low, especially if edf is close to k'.

	k'	edf	k-index	p-value	
s(LightLevel_ue)	2.00	1.85	0.70	0.09	.
s(Temp_C)	2.00	1.25	1.14	0.63	
ti(Temp_C,LightLevel_ue)	4.00	1.00	0.77	0.04	*

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Strain: *C. malina*, Growth Temperature 4 °C, Salinity 500 mM

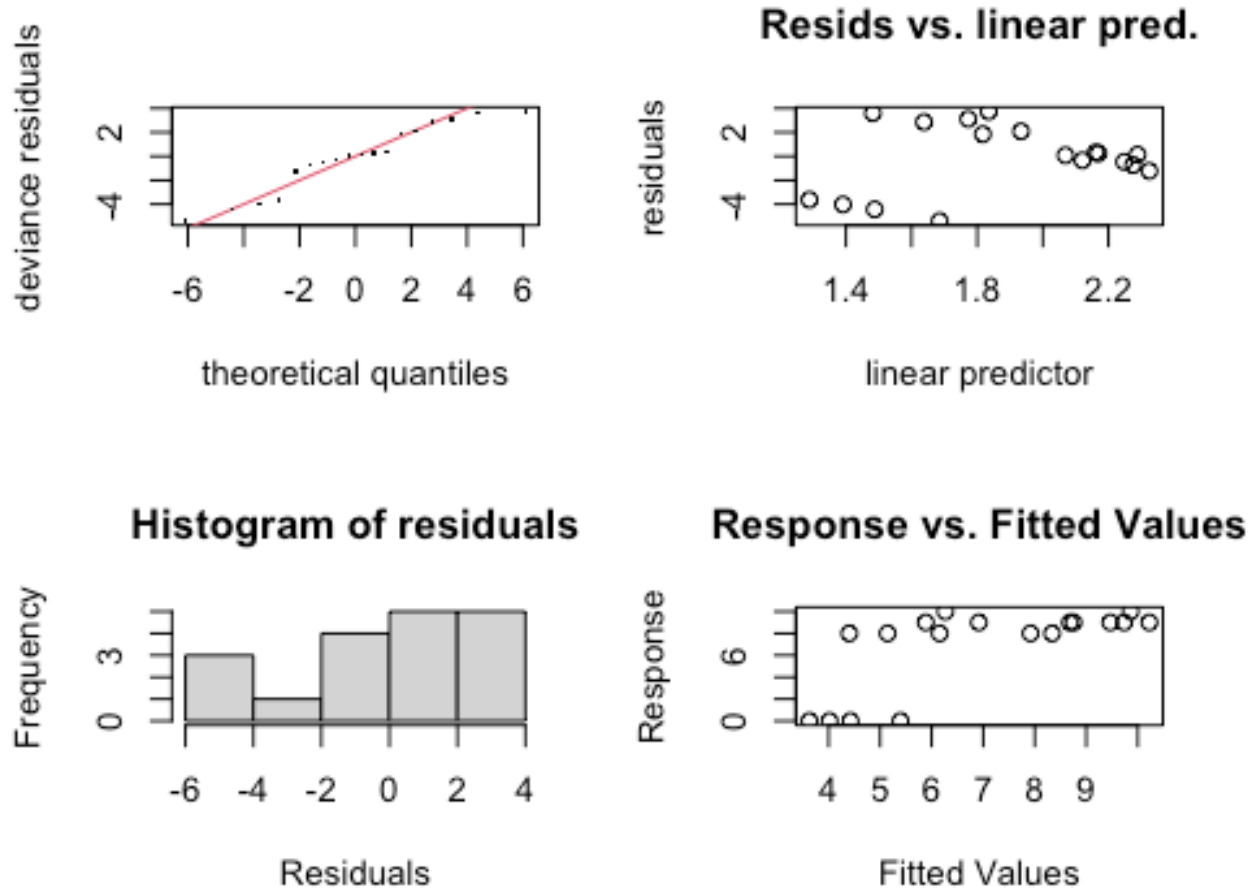


Fig 14: Generalized Additive Model (GAM) diagnostic outputs from `mgcv::gam.check()` for polar and temperate diatoms, of consecutive flashes before damping of S-State induced chlorophyll fluorescence oscillations. GAM predicted by the temperature (°C) imposed during measurements and the effective light level ($\mu\text{mol photons m}^{-2}\text{s}^{-1}$, estimated from flash spacings (s) and effective absorption cross-section of photosystem II, σ_{PSII} (A^2)).

```
Method: REML    Optimizer: outer newton
full convergence after 8 iterations.
Gradient range [-2.055142e-05,1.125567e-05]
(score 49.36194 & scale 10.11107).
Hessian positive definite, eigenvalue range [9.922912e-07,7.021343].
Model rank = 9 / 9
```

Basis dimension (k) checking results. Low p-value (k-index<1) may indicate that k is too low, especially if edf is close to k'.

	k'	edf	k-index	p-value
s(LightLevel_ue)	2.00	1.84	1.04	0.43
s(Temp_C)	2.00	1.00	1.06	0.51
ti(Temp_C,LightLevel_ue)	4.00	1.00	1.04	0.43

Strain: *C. priscui*, Growth Temperature 4 °C, Salinity 500 mM

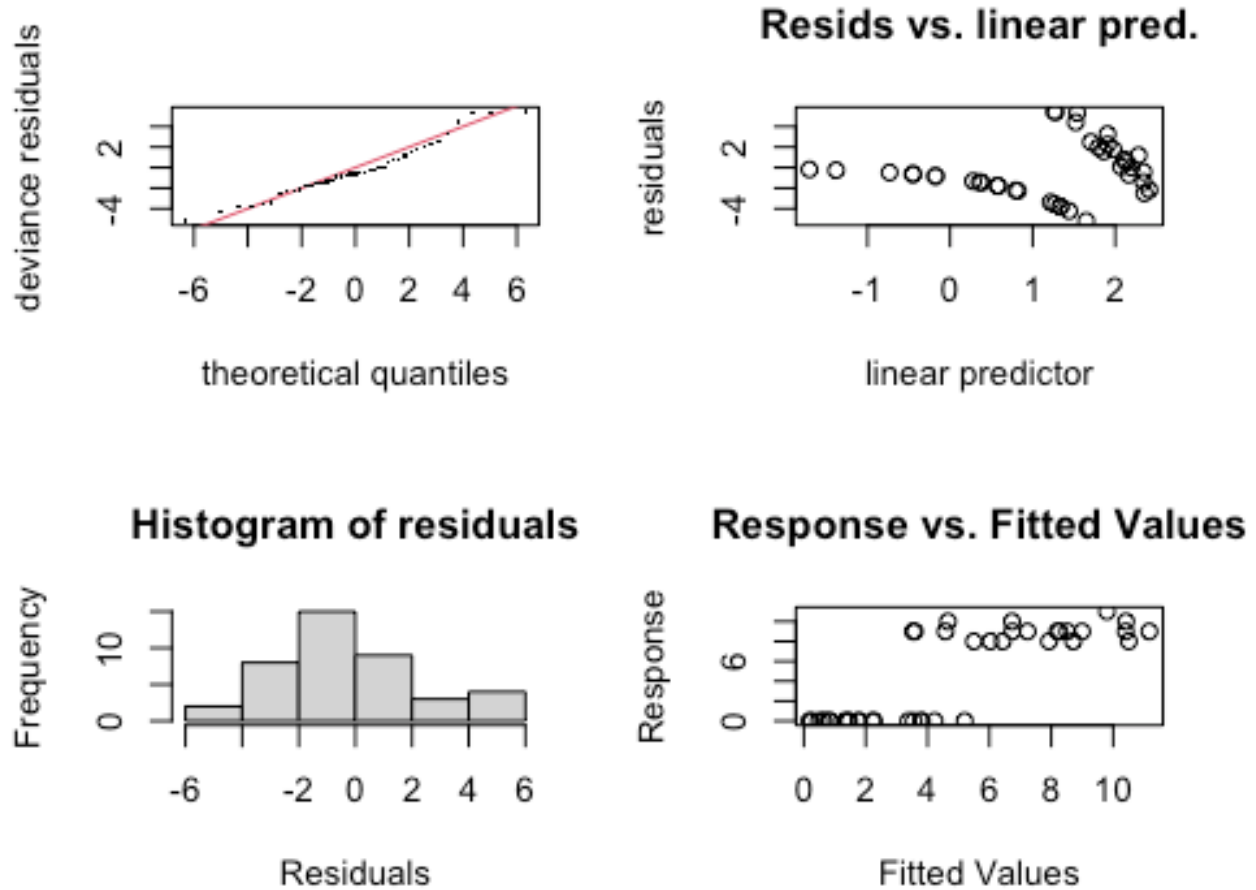


Fig 15: Generalized Additive Model (GAM) diagnostic outputs from `mgcv::gam.check()` for polar and temperate diatoms, of consecutive flashes before damping of S-State induced chlorophyll fluorescence oscillations. GAM predicted by the temperature (°C) imposed during measurements and the effective light level ($\mu\text{mol photons m}^{-2}\text{s}^{-1}$, estimated from flash spacings (s) and effective absorption cross-section of photosystem II, σ_{PSII} (A^2)).

```
Method: REML    Optimizer: outer newton
full convergence after 12 iterations.
Gradient range [-3.720374e-05,8.668272e-05]
(score 104.6995 & scale 7.872017).
Hessian positive definite, eigenvalue range [2.611616e-05,18.51441].
Model rank = 9 / 9
```

Basis dimension (k) checking results. Low p-value (k-index<1) may indicate that k is too low, especially if edf is close to k'.

	k'	edf	k-index	p-value	
s(LightLevel_ue)	2.00	1.90	0.72	0.015	*
s(Temp_C)	2.00	1.77	1.06	0.690	
ti(Temp_C,LightLevel_ue)	4.00	1.00	0.62	<2e-16	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Strain: *C. reinhardtii*, Growth Temperature 24 °C, Salinity 0 mM

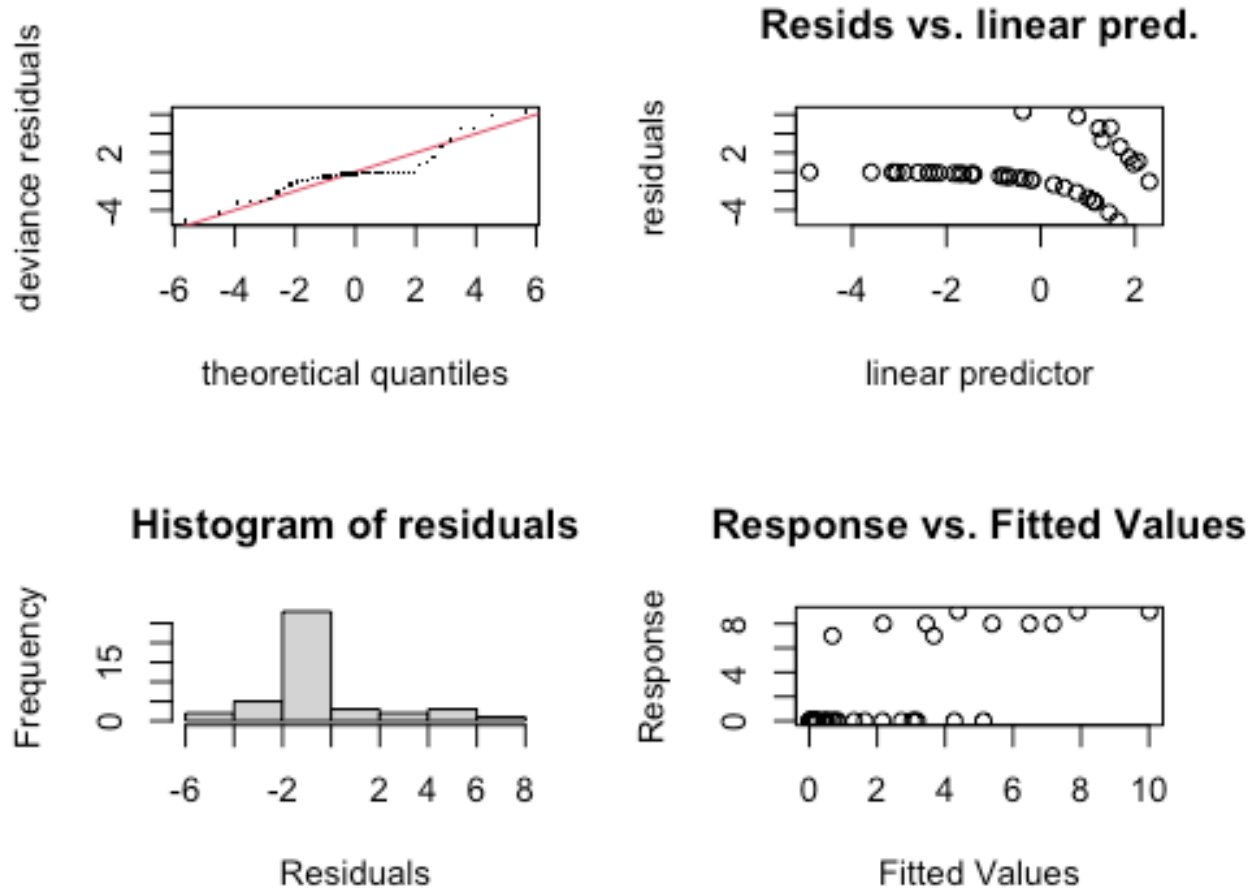


Fig 16: Generalized Additive Model (GAM) diagnostic outputs from `mgcv::gam.check()` for polar and temperate diatoms, of consecutive flashes before damping of S-State induced chlorophyll fluorescence oscillations. GAM predicted by the temperature (°C) imposed during measurements and the effective light level ($\mu\text{mol photons m}^{-2}\text{s}^{-1}$, estimated from flash spacings (s) and effective absorption cross-section of photosystem II, σ_{PSII} (A^2)).

```
Method: REML    Optimizer: outer newton
full convergence after 9 iterations.
Gradient range [-6.765617e-05,3.899604e-05]
(score 103.853 & scale 6.13007).
Hessian positive definite, eigenvalue range [5.51875e-05,20.0096].
Model rank = 9 / 9
```

Basis dimension (k) checking results. Low p-value (k-index<1) may indicate that k is too low, especially if edf is close to k'.

	k'	edf	k-index	p-value
s(LightLevel_ue)	2.00	1.87	1.01	0.47
s(Temp_C)	2.00	1.00	1.27	0.94
ti(Temp_C,LightLevel_ue)	4.00	2.19	0.85	0.07 .

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

References

- Ananyev, G., Gates, C., Kaplan, A., and Dismukes, G. C. (2017). Photosystem II-cyclic electron flow powers exceptional photoprotection and record growth in the microalga *Chlorella Ohadii*. *Biochimica et Biophysica Acta (BBA) - Bioenergetics* 1858, 873–883. doi:[10.1016/j.bbabbio.2017.07.001](https://doi.org/10.1016/j.bbabbio.2017.07.001).
- Ardyna, M., and Arrigo, K. R. (2020). Phytoplankton dynamics in a changing Arctic Ocean. *Nature Climate Change* 10, 892–903. doi:[10.1038/s41558-020-0905-y](https://doi.org/10.1038/s41558-020-0905-y).
- Balzano, S., Gourvil, P., Siano, R., Chanoine, M., Marie, D., Lessard, S., Sarno, D., and Vaulot, D. (2012). Diversity of cultured photosynthetic flagellates in the northeast Pacific and Arctic Oceans in summer. *Biogeosciences* 9, 4553–4571. doi:[10.5194/bg-9-4553-2012](https://doi.org/10.5194/bg-9-4553-2012).
- Bayer-Giraldi, M., Weikusat, I., Besir, H., and Dieckmann, G. (2011). Characterization of an antifreeze protein from the polar diatom *Fragilariopsis cylindrus* and its relevance in sea ice. *Cryobiology* 63, 210–219. doi:[10.1016/j.cryobiol.2011.08.006](https://doi.org/10.1016/j.cryobiol.2011.08.006).
- Benoît-Gagné, M., Dutkiewicz, S., Deschepper, I., Dufresne, C., Dumont, D., Larouche, R., Mémery, L., Olivier, G., and Maps, F. (2024). Exploring controls on the timing of the phytoplankton bloom in western Baffin Bay, Canadian Arctic. *Elementa: Science of the Anthropocene* 12, 00008. doi:[10.1525/elementa.2024.00008](https://doi.org/10.1525/elementa.2024.00008).
- Bryan, J. (2023). *googlesheets4: Access google sheets using the sheets API V4*. Available at: <https://googlesheets4.tidyverse.org>.
- Burch, T. A., Adams, W. W., Degrenne, B. L. S., Englert, C. H., Mines, B. R., Nash, P. C., Boone, E. C., and Demmig-Adams, B. (2015). Environmental manipulation of growth and energy carrier release from freshwater and marine *Chlamydomonas* species. *Journal of Applied Phycology* 27, 1127–1136. doi:[10.1007/s10811-014-0433-0](https://doi.org/10.1007/s10811-014-0433-0).
- Campitelli, E. (2021). *metR: Tools for easier analysis of meteorological fields*. doi:[10.5281/zenodo.2593516](https://doi.org/10.5281/zenodo.2593516).
- Canuel, E., Vaz, C., Matias, W. G., and Dewez, D. (2021). Interaction Effect of EDTA, Salinity, and Oxide Nanoparticles on Alga *Chlamydomonas reinhardtii* and *Chlamydomonas euryale*. *Plants* 10, 2118. doi:[10.3390/plants10102118](https://doi.org/10.3390/plants10102118).
- Carrasco Flores, D., Fricke, M., Wesp, V., Desirò, D., Kniewasser, A., Hölzer, M., Marz, M., and Mittag, M. (2021). A marine *Chlamydomonas* sp. Emerging as an algal model. *Journal of Phycology* 57, 54–69. doi:[10.1111/jpy.13083](https://doi.org/10.1111/jpy.13083).
- Carton, J. A., Ding, Y., and Arrigo, K. R. (2015). The seasonal cycle of the Arctic Ocean under climate change. *Geophysical Research Letters* 42, 7681–7686. doi:[10.1002/2015GL064514](https://doi.org/10.1002/2015GL064514).
- Cazelles, B., Chavez, M., Berteaux, D., Ménard, F., Vik, J. O., Jenouvrier, S., and Stenseth, N. C. (2008). Wavelet analysis of ecological time series. *Oecologia* 156, 287–304. doi:[10.1007/s00442-008-0993-2](https://doi.org/10.1007/s00442-008-0993-2).
- CCMP219 NCMA at Bigelow Laboratory.

- Cefarelli, A. O., Ferrario, M. E., Almandoz, G. O., Atencio, A. G., Akselman, R., and Vernet, M. (2010). Diversity of the diatom genus *Fragilariopsis* in the Argentine Sea and Antarctic waters: Morphology, distribution and abundance. *Polar Biology* 33. doi:[10.1007/s00300-010-0794-z](https://doi.org/10.1007/s00300-010-0794-z).
- Cohen, J. H., Berge, J., Moline, M. A., Johnsen, G., and Zolich, A. P. (2020). "Light in the Polar Night," in *POLAR NIGHT Marine Ecology*, eds. J. Berge, G. Johnsen, and J. H. Cohen (Cham: Springer International Publishing), 37–66. doi:[10.1007/978-3-030-33208-2_3](https://doi.org/10.1007/978-3-030-33208-2_3).
- Cook, G., Teufel, A., Kalra, I., Li, W., Wang, X., Priscu, J., and Morgan-Kiss, R. (2019). The Antarctic psychrophiles *Chlamydomonas* spp. UW0241 and ICEMDV exhibit differential restructuring of photosystem I in response to iron. *Photosynthesis Research* 141, 209–228. doi:[10.1007/s11120-019-00621-0](https://doi.org/10.1007/s11120-019-00621-0).
- Croteau, D., Jensen, E., Wilhelm, C., and Bailleul, B. (2024). "Comparing Diatom Photosynthesis with the Green Lineage," in *Diatom Photosynthesis* (John Wiley & Sons, Ltd), 1–44. doi:[10.1002/9781119842156.ch1](https://doi.org/10.1002/9781119842156.ch1).
- Cvetkovska, M., Vakulenko, G., Smith, D. R., Zhang, X., and Hüner, N. P. A. (2022). Temperature stress in psychrophilic green microalgae: Minireview. *Physiologia Plantarum* 174, e13811. doi:[10.1111/ppl.13811](https://doi.org/10.1111/ppl.13811).
- Dau, H., and Haumann, M. (2007). Time-resolved X-ray spectroscopy leads to an extension of the classical S-state cycle model of photosynthetic oxygen evolution. *Photosynthesis Research* 92, 327–343. doi:[10.1007/s11120-007-9141-9](https://doi.org/10.1007/s11120-007-9141-9).
- de Wijn, R., and van Gorkom, H. J. (2002). S-state dependence of the miss probability in Photosystem II. *Photosynthesis Research* 72, 217–222. doi:[10.1023/A:1016128632704](https://doi.org/10.1023/A:1016128632704).
- Falkowski, P. G. (1994). The role of phytoplankton photosynthesis in global biogeochemical cycles. *Photosynthesis Research* 39, 235–258. doi:[10.1007/BF00014586](https://doi.org/10.1007/BF00014586).
- Falkowski, P., Fenchel, T., and Delong, E. (2008). The Microbial Engines That Drive Earth's Biogeochemical Cycles. *Science* 320, 1034–1039. doi:[10.1126/science.1153213](https://doi.org/10.1126/science.1153213).
- Fasiolo, M., Nedellec, R., Goude, Y., and Wood, S. N. (2020). Scalable Visualization Methods for Modern Generalized Additive Models. *Journal of Computational and Graphical Statistics* 29, 78–86. doi:[10.1080/10618600.2019.1629942](https://doi.org/10.1080/10618600.2019.1629942).
- Fiala, M., and Oriol, L. (1990). Light-temperature interactions on the growth of Antarctic diatoms. *Polar Biology* 10, 629–636. doi:[10.1007/BF00239374](https://doi.org/10.1007/BF00239374).
- Gates, C., Ananyev, G., and Dismukes, G. C. (2020). Realtime kinetics of the light driven steps of photosynthetic water oxidation in living organisms by "stroboscopic" fluorometry. *Biochimica et Biophysica Acta (BBA) - Bioenergetics* 1861, 148212. doi:[10.1016/j.bbabi.2020.148212](https://doi.org/10.1016/j.bbabi.2020.148212).
- Halekoh, U., and Højsgaard, S. (2025). *doBy: Groupwise statistics, LSmeans, linear estimates, utilities*. Available at: <https://github.com/hojsgaard/doBy>.

- Han, G., Chernev, P., Styring, S., Messinger, J., and Mamedov, F. (2022). Molecular basis for turnover inefficiencies (misses) during water oxidation in photosystem II. *Chemical Science* 13, 8667–8678. doi:[10.1039/D2SC00854H](https://doi.org/10.1039/D2SC00854H).
- Han, G., Ho, F. M., Havelius, K. G. V., Morvaridi, S. F., Mamedov, F., and Styring, S. (2008). Direct quantification of the four individual S states in Photosystem II using EPR spectroscopy. *Biochimica et Biophysica Acta (BBA) - Bioenergetics* 1777, 496–503. doi:[10.1016/j.bbabi.2008.03.007](https://doi.org/10.1016/j.bbabi.2008.03.007).
- Han, G., Mamedov, F., and Styring, S. (2012). Misses during water oxidation in photosystem II are S state-dependent. *The Journal of Biological Chemistry* 287, 13422–13429. doi:[10.1074/jbc.M112.342543](https://doi.org/10.1074/jbc.M112.342543).
- Hancke, K., Lund-Hansen, L. C., Lamare, M. L., Højlund Pedersen, S., King, M. D., Andersen, P., and Sorrell, B. K. (2018). Extreme Low Light Requirement for Algae Growth Underneath Sea Ice: A Case Study From Station Nord, NE Greenland. *Journal of Geophysical Research: Oceans* 123, 985–1000. doi:[10.1002/2017JC013263](https://doi.org/10.1002/2017JC013263).
- Hoppe, C. J. M., Fuchs, N., Notz, D., Anderson, P., Assmy, P., Berge, J., Bratbak, G., Guillou, G., Kraberg, A., Larsen, A., et al. (2024). Photosynthetic light requirement near the theoretical minimum detected in Arctic microalgae. *Nature Communications* 15, 7385. doi:[10.1038/s41467-024-51636-8](https://doi.org/10.1038/s41467-024-51636-8).
- Hoppmann, M., Kuznetsov, I., Fang, Y.-C., and Rabe, B. (2022). Mesoscale observations of temperature and salinity in the Arctic Transpolar Drift: A high-resolution dataset from the MOSAiC Distributed Network. *Earth System Science Data* 14, 4901–4921. doi:[10.5194/essd-14-4901-2022](https://doi.org/10.5194/essd-14-4901-2022).
- Hulatt, C. J., Berecz, O., Egeland, E. S., Wijffels, R. H., and Kiron, V. (2017). Polar snow algae as a valuable source of lipids? *Bioresource Technology* 235, 338–347. doi:[10.1016/j.biortech.2017.03.130](https://doi.org/10.1016/j.biortech.2017.03.130).
- Hüner, N. P. A., Szyszka-Mroz, B., Ivanov, A. G., Kata, V., Lye, H., and Smith, D. R. (2023). Photosynthetic adaptation and multicellularity in the Antarctic psychrophile, *Chlamydomonas priscuii*. *Algal Research* 74, 103220. doi:[10.1016/j.algal.2023.103220](https://doi.org/10.1016/j.algal.2023.103220).
- Ibarbalz, F. M., Henry, N., Mahé, F., Ardyna, M., Zingone, A., Scalco, E., Lovejoy, C., Lombard, F., Jaillon, O., Iudicone, D., et al. (2023). Pan-Arctic plankton community structure and its global connectivity. *Elementa: Science of the Anthropocene* 11, 00060. doi:[10.1525/elementa.2022.00060](https://doi.org/10.1525/elementa.2022.00060).
- Ivanov, A. G., Sane, P. V., Krol, M., Gray, G. R., Balseris, A., Savitch, L. V., Öquist, G., and Hüner, N. P. A. (2006). Acclimation to temperature and irradiance modulates PSII charge recombination. *FEBS Letters* 580, 2797–2802. doi:[10.1016/j.febslet.2006.04.018](https://doi.org/10.1016/j.febslet.2006.04.018).
- Ivanov, A. G., Sane, P., Hurry, V., Król, M., Sveshnikov, D., Huner, N. P. A., and Öquist, G. (2003). Low-temperature modulation of the redox properties of the acceptor side of photosystem II: Photoprotection through reaction centre quenching of excess energy. *Physiologia Plantarum* 119, 376–383. doi:[10.1034/j.1399-3054.2003.00225.x](https://doi.org/10.1034/j.1399-3054.2003.00225.x).

Kalra, I., Wang, X., Cvetkovska, M., Jeong, J., McHargue, W., Zhang, R., Hüner, N., Yuan, J. S., and Morgan-Kiss, R. (2020). Chlamydomonas sp. UWO 241 Exhibits High Cyclic Electron Flow and Rewired Metabolism under High Salinity1 [OPEN]. *Plant Physiology* 183, 588–601. doi:[10.1104/pp.19.01280](https://doi.org/10.1104/pp.19.01280).

Kalra, I., Wang, X., Zhang, R., and Morgan-Kiss, R. (2023). High salt-induced PSI-supercomplex is associated with high CEF and attenuation of state transitions. *Photosynthesis Research* 157, 65–84. doi:[10.1007/s11120-023-01032-y](https://doi.org/10.1007/s11120-023-01032-y).

Kang, S.-H., and Fryxell, G. A. (1992). Fragilariopsis cylindrus (Grunow) Krieger: The most abundant diatom in water column assemblages of Antarctic marginal ice-edge zones. *Polar Biology* 12, 609–627. doi:[10.1007/BF00236984](https://doi.org/10.1007/BF00236984).

Keren, N., Berg, A., van Kan, P. J. M., Levanon, H., and Ohad, I. (1997). Mechanism of photosystem II photoinactivation and D1 protein degradation at low light: The role of back electron flow. *Proceedings of the National Academy of Sciences* 94, 1579–1584. doi:[10.1073/pnas.94.4.1579](https://doi.org/10.1073/pnas.94.4.1579).

Kirk, J. T. O. (2011). *Light and Photosynthesis in Aquatic Ecosystems*. 3rd ed. Cambridge, UK: Cambridge University Press.

Kok, B., Forbush, B., and McGLOIN, M. (1970). Cooperation of Charges in Photosynthetic O₂ Evolution—I. A Linear Four Step Mechanism. *Photochemistry and Photobiology* 11, 457–475. doi:[10.1111/j.1751-1097.1970.tb06017.x](https://doi.org/10.1111/j.1751-1097.1970.tb06017.x).

Kolber, Z. S. (1998). Measurements of variable chlorophyll fluorescence using fast repetition rate techniques: Defining methodology and experimental protocols. *Biochimica et Biophysica Acta* 1367, 19. doi:[10.1016/S0005-2728\(98\)00135-2](https://doi.org/10.1016/S0005-2728(98)00135-2).

Kvernvik, A. C., Hoppe, C. J. M., Lawrenz, E., Prášil, O., Greenacre, M., Wiktor, J. M., and Leu, E. (2018). Fast reactivation of photosynthesis in arctic phytoplankton during the polar Night1. *Journal of Phycology* 54, 461–470. doi:[10.1111/jpy.12750](https://doi.org/10.1111/jpy.12750).

Laloi, C., Przybyla, D., and Apel, K. (2006). A genetic approach towards elucidating the biological activity of different reactive oxygen species in Arabidopsis thaliana. *Journal of Experimental Botany* 57, 1719–1724. doi:[10.1093/jxb/erj183](https://doi.org/10.1093/jxb/erj183).

Leu, E., Mundy, C. J., Assmy, P., Campbell, K., Gabrielsen, T. M., Gosselin, M., Juul-Pedersen, T., and Gradinger, R. (2015). Arctic spring awakening – Steering principles behind the phenology of vernal ice algal blooms. *Progress in Oceanography* 139, 151–170. doi:[10.1016/j.pocean.2015.07.012](https://doi.org/10.1016/j.pocean.2015.07.012).

Leyva, L. A., Bashan, Y., Mendoza, A., and de-Bashan, L. E. (2014). Accumulation fatty acids of in Chlorella vulgaris under heterotrophic conditions in relation to activity of acetyl-CoA carboxylase, temperature, and co-immobilization with Azospirillum brasilense. *Naturwissenschaften* 101, 819–830. doi:[10.1007/s00114-014-1223-x](https://doi.org/10.1007/s00114-014-1223-x).

Li, J., Wu, X., and Wang, X. (2025). An Arctic Ocean Thermohaline Dataset. *Scientific Data* 12, 1607. doi:[10.1038/s41597-025-05855-3](https://doi.org/10.1038/s41597-025-05855-3).

- Li, W., and Morgan-Kiss, R. M. (2019). Influence of Environmental Drivers and Potential Interactions on the Distribution of Microbial Communities From Three Permanently Stratified Antarctic Lakes. *Frontiers in Microbiology* 10. doi:[10.3389/fmicb.2019.01067](https://doi.org/10.3389/fmicb.2019.01067).
- Li, W., Podar, M., and Morgan-Kiss, R. M. (2016). Ultrastructural and Single-Cell-Level Characterization Reveals Metabolic Versatility in a Microbial Eukaryote Community from an Ice-Covered Antarctic Lake. *Applied and Environmental Microbiology* 82, 3659–3670. doi:[10.1128/AEM.00478-16](https://doi.org/10.1128/AEM.00478-16).
- Lyon, B. R., and Mock, T. (2014). Polar Microalgae: New Approaches towards Understanding Adaptations to an Extreme and Changing Environment. *Biology* 3, 56–80. doi:[10.3390/biology3010056](https://doi.org/10.3390/biology3010056).
- McMinn, A., Ryan, K., and Gademann, R. (2003). Diurnal changes in photosynthesis of Antarctic fast ice algal communities determined by pulse amplitude modulation fluorometry. *Marine Biology* 143, 359–367. doi:[10.1007/s00227-003-1052-5](https://doi.org/10.1007/s00227-003-1052-5).
- Mock, T., Otilar, R. P., Strauss, J., McMullan, M., Paajanen, P., Schmutz, J., Salamov, A., Sanges, R., Toseland, A., Ward, B. J., et al. (2017). Evolutionary genomics of the cold-adapted diatom *Fragilariopsis cylindrus*. *Nature* 541, 536–540. doi:[10.1038/nature20803](https://doi.org/10.1038/nature20803).
- Morales-Sánchez, D., Schulze, P. S. C., Kiron, V., and Wijffels, R. H. (2020). Temperature-Dependent Lipid Accumulation in the Polar Marine Microalga *Chlamydomonas malina* RCC2488. *Frontiers in Plant Science* 11. doi:[10.3389/fpls.2020.619064](https://doi.org/10.3389/fpls.2020.619064).
- Morgan-Kiss, R. M., Lizotte, M. P., Kong, W., and Priscu, J. C. (2016). Photoadaptation to the polar night by phytoplankton in a permanently ice-covered Antarctic lake. *Limnology and Oceanography* 61, 3–13. doi:[10.1002/lno.10107](https://doi.org/10.1002/lno.10107).
- Morin, P.-I., Lacour, T., Grondin, P.-L., Bruyant, F., Ferland, J., Forget, M.-H., Massicotte, P., Donaher, N., Campbell, D. A., Lavaud, J., et al. (2020). Response of the sea-ice diatom *Fragilariopsis cylindrus* to simulated polar night darkness and return to light. *Limnology and Oceanography* 65, 1041–1060. doi:[10.1002/lno.11368](https://doi.org/10.1002/lno.11368).
- Mukhopadhyay, S., Mandal, S. K., Bhaduri, S., and Armstrong, W. H. (2004). Manganese Clusters with Relevance to Photosystem II. *Chemical Reviews* 104, 3981–4026. doi:[10.1021/cr0206014](https://doi.org/10.1021/cr0206014).
- Otte, A., Winder, J. C., Deng, L., Schmutz, J., Jenkins, J., Grigoriev, I. V., Hopes, A., and Mock, T. (2023). The diatom *Fragilariopsis cylindrus*: A model alga to understand cold-adapted life. *Journal of Phycology* 59, 301–306. doi:[10.1111/jpy.13325](https://doi.org/10.1111/jpy.13325).
- Pierella Karlusich, J. J., Ibarbalz, F. M., and Bowler, C. (2020). Phytoplankton in the Tara Ocean. *Annual Review of Marine Science* 12, 233–265. doi:[10.1146/annurev-marine-010419-010706](https://doi.org/10.1146/annurev-marine-010419-010706).
- Pinilla, J., and Negrín, M. (2021). Non-Parametric Generalized Additive Models as a Tool for Evaluating Policy Interventions. *Mathematics* 9, 299. doi:[10.3390/math9040299](https://doi.org/10.3390/math9040299).

- Plassart, A., Joli, N., Sivaram, S., Guérin, S., Bruyant, F., Forget, M.-H., Bowler, C., and Babin, M. (2026). Paradoxical energetics in the polar diatom *Fragilariopsis cylindrus* exposed to extreme low light. 2026.01.30.702734. doi:[10.64898/2026.01.30.702734](https://doi.org/10.64898/2026.01.30.702734).
- Poirier, M. C., Fugard, K., and Cvetkovska, M. (2025). Light quality affects chlorophyll biosynthesis and photosynthetic performance in Antarctic *Chlamydomonas*. *Photosynthesis Research* 163, 9. doi:[10.1007/s11120-024-01127-0](https://doi.org/10.1007/s11120-024-01127-0).
- Poirier, M., Osmers, P., Wilkins, K., Morgan-Kiss, R. M., and Cvetkovska, M. (2023). Aberrant light sensing and motility in the green alga *Chlamydomonas priscuii* from the ice-covered Antarctic Lake Bonney. *Plant Signaling & Behavior* 18, 2184588. doi:[10.1080/15592324.2023.2184588](https://doi.org/10.1080/15592324.2023.2184588).
- Posit team (2025). *RStudio: Integrated development environment for R*. Boston, MA: Posit Software, PBC.
- Poulsen, N., and Kröger, N. (2023). *Thalassiosira pseudonana* (*Cyclotella nana*) (Hustedt) Hasle et Heimdal (Bacillariophyceae): A genetically tractable model organism for studying diatom biology, including biological silica formation. *Journal of Phycology* 59, 809–817. doi:[10.1111/jpy.13362](https://doi.org/10.1111/jpy.13362).
- Quigg, A., Beardall, J., and Wydrzynski, T. (2003). Photoacclimation involves modulation of the photosynthetic oxygen-evolving reactions in *Dunaliella tertiolecta* and *Phaeodactylum tricornutum*. *Functional Plant Biology* 30, 301. doi:[10.1071/FP02140](https://doi.org/10.1071/FP02140).
- R Core Team (2023). *R: A language and environment for statistical computing*. Vienna, Austria: R Foundation for Statistical Computing.
- Randelhoff, A., Lacour, L., Marec, C., Leymarie, E., Lagunas, J., Xing, X., Darnis, G., Penkerch, C., Sampei, M., Fortier, L., et al. (2020). Arctic mid-winter phytoplankton growth revealed by autonomous profilers. *Science Advances* 6, eabc2678. doi:[10.1126/sciadv.abc2678](https://doi.org/10.1126/sciadv.abc2678).
- Rappaport, F., Cuni, A., Xiong, L., Sayre, R., and Lavergne, J. (2005). Charge Recombination and Thermoluminescence in Photosystem II. *Biophysical Journal* 88, 1948–1958. doi:[10.1529/biophysj.104.050237](https://doi.org/10.1529/biophysj.104.050237).
- Rappaport, F., Guergova-Kuras, M., Nixon, P. J., Diner, B. A., and Lavergne, J. (2002). Kinetics and Pathways of Charge Recombination in Photosystem II. *Biochemistry* 41, 8518–8527. doi:[10.1021/bi025725p](https://doi.org/10.1021/bi025725p).
- Raven, J. A., Kübler, J. E., and Beardall, J. (2000). Put out the light, and then put out the light. *Journal of the Marine Biological Association of the United Kingdom* 80, 1–25. doi:[10.1017/S0025315499001526](https://doi.org/10.1017/S0025315499001526).
- Robinson, D. H., Arrigo, K. R., Iturriaga, R., and Sullivan, C. W. (1995). Microalgal Light-Harvesting in Extreme Low-Light Environments in Mcmurdo Sound, Antarctica. *Journal of Phycology* 31, 508–520. doi:[10.1111/j.1529-8817.1995.tb02544.x](https://doi.org/10.1111/j.1529-8817.1995.tb02544.x).

- Robinson, D., Hayes, A., and Couch, S. (2025). *Broom: Convert statistical objects into tidy tibbles*. Available at: <https://broom.tidymodels.org/>.
- Roesch, A., and Schmidbauer, H. (2018). *WaveletComp: Computational wavelet analysis*.
- Rutherford, A. W., Osyczka, A., and Rappaport, F. (2012). Back-reactions, short-circuits, leaks and other energy wasteful reactions in biological electron transfer: Redox tuning to survive life in O₂. *FEBS Letters* 586, 603–616. doi:[10.1016/j.febslet.2011.12.039](https://doi.org/10.1016/j.febslet.2011.12.039).
- Sane, P. V., Ivanov, A. G., Hurry, V., Huner, N. P. A., and Öquist, G. (2003). Changes in the Redox Potential of Primary and Secondary Electron-Accepting Quinones in Photosystem II Confer Increased Resistance to Photoinhibition in Low-Temperature-Acclimated Arabidopsis. *Plant Physiology* 132, 2144–2151. doi:[10.1104/pp.103.022939](https://doi.org/10.1104/pp.103.022939).
- Sane, P. V., Ivanov, A. G., Sveshnikov, D., Huner, N. P. A., and Oquist, G. (2002). A Transient Exchange of the Photosystem II Reaction Center Protein D1:1 with D1:2 during Low Temperature Stress of *Synechococcus* sp. PCC 7942 in the Light Lowers the Redox Potential of QB*. *Journal of Biological Chemistry* 277, 32739–32745. doi:[10.1074/jbc.M200444200](https://doi.org/10.1074/jbc.M200444200).
- Sasso, S., Stibor, H., Mittag, M., and Grossman, A. R. (2018). From molecular manipulation of domesticated *Chlamydomonas reinhardtii* to survival in nature. *eLife* 7, e39233. doi:[10.7554/eLife.39233](https://doi.org/10.7554/eLife.39233).
- Schoenig, D., Maynard, L., Brice, M.-H., Cazelles, K., Braga, P. H. P., Gongora, E., Sherin, L., Pedersen, E., Taranu, Z., Dussault, C. F., et al. (2023). *Workshop 8: Generalized additive models in R*. Québec Centre for Biodiversity Science.
- Schuback, N., Tortell, P. D., Berman-Frank, I., Campbell, D. A., Ciotti, A., Courtecuisse, E., Erickson, Z. K., Fujiki, T., Halsey, K., Hickman, A. E., et al. (2021). Single-Turnover Variable Chlorophyll Fluorescence as a Tool for Assessing Phytoplankton Photosynthesis and Primary Productivity: Opportunities, Caveats and Recommendations. *Frontiers in Marine Science* 8. doi:[10.3389/fmars.2021.690607](https://doi.org/10.3389/fmars.2021.690607).
- Shen, J.-R. (2021). “Photosynthesis Photosystem II: Protein Components, Structure and Electron Transfer,” in *Encyclopedia of Biological Chemistry III (Third Edition)*, ed. J. Jez (Oxford: Elsevier), 215–228. doi:[10.1016/B978-0-12-819460-7.00012-8](https://doi.org/10.1016/B978-0-12-819460-7.00012-8).
- Sherwell, S., Kalra, I., Li, W., McKnight, D. M., Priscu, J. C., and Morgan-Kiss, R. M. (2022). Antarctic lake phytoplankton and bacteria from near-surface waters exhibit high sensitivity to climate-driven disturbance. *Environmental Microbiology* 24, 6017–6032. doi:[10.1111/1462-2920.16113](https://doi.org/10.1111/1462-2920.16113).
- Shevela, D., Kern, J. F., Govindjee, G., and Messinger, J. (2023). Solar energy conversion by photosystem II: Principles and structures. *Photosynthesis Research* 156, 279–307. doi:[10.1007/s11120-022-00991-y](https://doi.org/10.1007/s11120-022-00991-y).
- Smoot, C. A., and Hopcroft, R. R. (2017). Depth-stratified community structure of Beaufort Sea slope zooplankton and its relations to water masses. *Journal of Plankton Research* 39, 79–91. doi:[10.1093/plankt/fbw087](https://doi.org/10.1093/plankt/fbw087).

Spinu, V., Grolemond, G., and Wickham, H. (2024). *Lubridate: Make dealing with dates a little easier*. Available at: <https://lubridate.tidyverse.org>.

Stahl-Rommel, S., Kalra, I., D'Silva, S., Hahn, M. M., Popson, D., Cvetkovska, M., and Morgan-Kiss, R. M. (2022). Cyclic electron flow (CEF) and ascorbate pathway activity provide constitutive photoprotection for the photopsychrophile, *Chlamydomonas* sp. UWO 241 (renamed *Chlamydomonas priscuii*). *Photosynthesis Research* 151, 235–250. doi:[10.1007/s11120-021-00877-5](https://doi.org/10.1007/s11120-021-00877-5).

Szyszkka-Mroz, B., Cvetkovska, M., Ivanov, A. G., Smith, D. R., Possmayer, M., Maxwell, D. P., and Hüner, N. P. A. (2019). Cold-Adapted Protein Kinases and Thylakoid Remodeling Impact Energy Distribution in an Antarctic Psychrophile¹. *Plant Physiology* 180, 1291–1309. doi:[10.1104/pp.19.00411](https://doi.org/10.1104/pp.19.00411).

Theis, F. J., and Meyer-Bäse, A. (2010). “Spectral Transformations,” in *Biomedical Signal Analysis : Contemporary Methods and Applications* The MIT Press Series. (MIT Press), 42.

van den Brand, T. (2025). *Ggh4x: Hacks for 'Ggplot2'*.

Vass, I. (2011). Role of charge recombination processes in photodamage and photoprotection of the photosystem II complex. *Physiologia Plantarum* 142, 6–16. doi:[10.1111/j.1399-3054.2011.01454.x](https://doi.org/10.1111/j.1399-3054.2011.01454.x).

Vass, I., and Cser, K. (2009). Janus-faced charge recombinations in photosystem II photoinhibition. *Trends in Plant Science* 14, 200–205. doi:[10.1016/j.tplants.2009.01.009](https://doi.org/10.1016/j.tplants.2009.01.009).

Vinyard, D. J., Ananyev, G. M., and Charles Dismukes, G. (2013). Photosystem II: The Reaction Center of Oxygenic Photosynthesis. *Annual Review of Biochemistry* 82, 577–606. doi:[10.1146/annurev-biochem-070511-100425](https://doi.org/10.1146/annurev-biochem-070511-100425).

von Quillfeldt, C. H. (2004). The diatom *Fragilariopsis cylindrus* and its potential as an indicator species for cold water rather than for sea ice. *Vie et Milieu / Life & Environment*, 137–143.

Wickham, H. (2016). *Ggplot2: Elegant graphics for data analysis*. Springer-Verlag New York.

Wickham, H. (2023). *Tidyverse: Easily install and load the tidyverse*. Available at: <https://tidyverse.tidyverse.org>.

Wiel, J. B. V., D. Mikulicz, J., R. Boysen, M., Hashemi, N., Kalgren, P., M. Nauman, L., J. Baetzold, S., G. Powell, G., He, Q., and Nastaran Hashemi, N. (2017). Characterization of *Chlorella vulgaris* and *Chlorella protothecoides* using multi-pixel photon counters in a 3D focusing optofluidic system. *RSC Advances* 7, 4402–4408. doi:[10.1039/C6RA25837A](https://doi.org/10.1039/C6RA25837A).

Wood, S. (2025). *Mgcv: Mixed GAM computation vehicle with automatic smoothness estimation*. Available at: <https://CRAN.R-project.org/package=mgcv>.

Wood, S. N. (2011). Fast stable restricted maximum likelihood and marginal likelihood estimation of semiparametric generalized linear models. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)* 73, 3–36. doi:[10.1111/j.1467-9868.2010.00749.x](https://doi.org/10.1111/j.1467-9868.2010.00749.x).

Wood, S. N. (2017). *Generalized Additive Models: An introduction with R*. 2nd ed. New York: Chapman and Hall/CRC.

Xie, B., Bishop, S., Stessman, D., Wright, D., Spalding, M. H., and Halverson, L. J. (2013). *Chlamydomonas reinhardtii* thermal tolerance enhancement mediated by a mutualistic interaction with vitamin B12-producing bacteria. *The ISME Journal* 7, 1544–1555. doi:[10.1038/ismej.2013.43](https://doi.org/10.1038/ismej.2013.43).

Xie, Y. (2025). *Knitr: A general-purpose package for dynamic report generation in r*. Available at: <https://yihui.org/knitr/>.

Xu, K., Lavaud, J., Perkins, R., Austen, E., Bonnanfant, M., and Campbell, D. A. (2018). Phytoplankton σ PSII and Excitation Dissipation; Implications for Estimates of Primary Productivity. *Frontiers in Marine Science* 5. doi:[10.3389/fmars.2018.00281](https://doi.org/10.3389/fmars.2018.00281).

Zaharieva, I., and Dau, H. (2019). Energetics and Kinetics of S-State Transitions Monitored by Delayed Chlorophyll Fluorescence. *Frontiers in Plant Science* 10.

Zhang, Z., Qu, C., Zhang, K., He, Y., Zhao, X., Yang, L., Zheng, Z., Ma, X., Wang, X., Wang, W., et al. (2020). Adaptation to Extreme Antarctic Environments Revealed by the Genome of a Sea Ice Green Alga. *Current Biology* 30, 3330–3341.e7. doi:[10.1016/j.cub.2020.06.029](https://doi.org/10.1016/j.cub.2020.06.029).

Zhu, H. (2024). *kableExtra: Construct complex table with kable and pipe syntax*. Available at: <http://haozhu233.github.io/kableExtra/>.

Zournas, A., Mani, K., and Dismukes, G. C. (2023). Cyclic electron flow around photosystem II in silico: How it works and functions in vivo. *Photosynthesis Research* 156, 129–145. doi:[10.1007/s11120-023-00997-0](https://doi.org/10.1007/s11120-023-00997-0).